Using Hand Gestures for Multimodal Interactions: Foundations, Methods, and Emerging Challenges

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Overview

1. Introduction: Hand Gestures in Multimodal Interaction 40 minutes – hazem — andrea
2. Hand Gesture Recognition: Methods, Benchmarks, and Applications — 80 minutes - marco
3. Hand Motion Generation: Recent Advances and Challenges — 80 minutes - Omar
4. Discussion and Open Challenges — 20 minutes

1 - INTRODUCTION: HAND GESTURES IN MULTIMODAL INTERACTION

Overview

- Hand Gestures: Foundations and Meaning
- Gesture Types and Application Domains
- Hand Tracking and Sensing Technologies
- Hand Gesture Representations
- Challenges in Gesture-Based Interaction

Understanding Human Motion

Human motion is a rich signal: it reveals what people do, what they intend, and how they feel.

Capturing it lets machines understand behavior and interact naturally with us.

Interaction
HCI, XR, robots

Health care
Falls, rehab, gait

Safety
Monitoring, surveillance

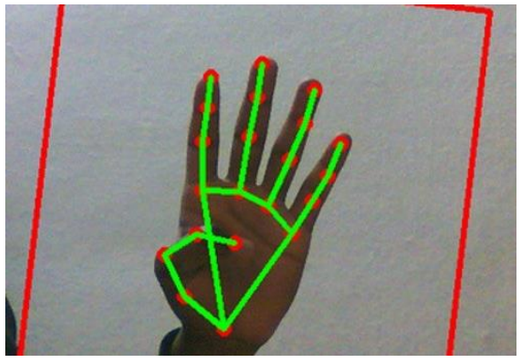
Activity → **Action** → **Gesture** → **Fine-grained gesture**



A sequence of actions with objects and a goal: cooking, tidying up



A single goal-directed body movement: raising the arms, sitting down



A movement meant to communicate, mostly with the hands: waving, pointing



Subtle finger articulation: pinching, playing the piano, handshapes

From coarse to fine: whole body → hands → fingers, from minutes to fractions of a second, and increasingly hard to capture.

Why Hands Matter

Hands carry meaning and action

Express intention, emotion, and emphasis

Enable manipulation, tool use, and action

AI systems increasingly interact with humans

They can see, speak, reason, and respond

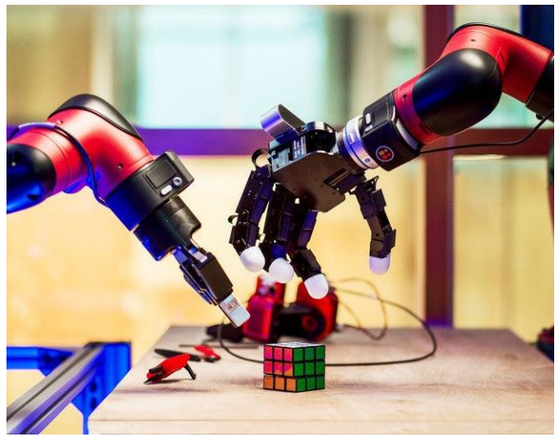
Next step: physical and embodied interaction



Natural HCI & gesture-based interfaces



VR/AR & embodied avatars



Robotics & teleoperation



Assistive & rehabilitation systems

Gestures

- A gesture is a movement of the body that contains information (Kurtenbach & Hulteen, 1990).
- Normally used as a communication channel by humans
  - Independently
  - Combined with other channels (e.g., speech)



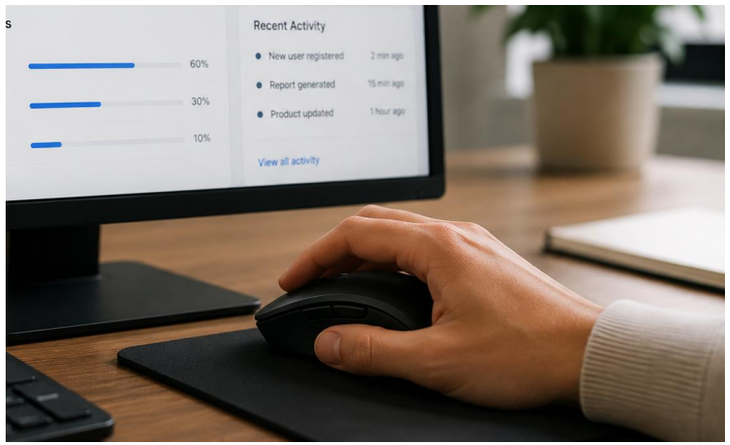


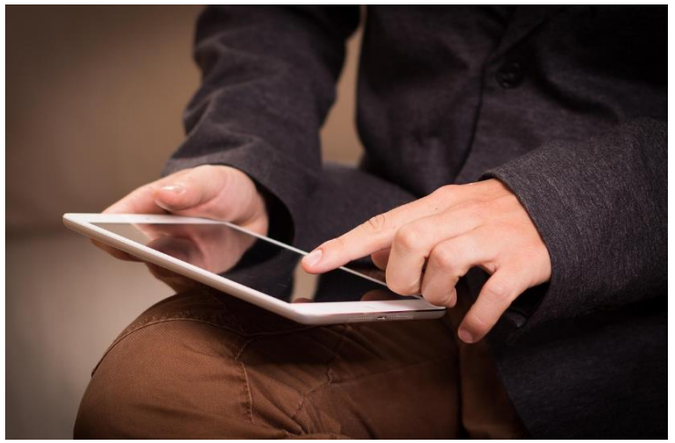


(6) *Che vuoi*

Gestures in interfaces

- Given this definition, gestures are quite common in existing interfaces
  - Touchscreen gestures
  - Controls with mouse, joystick or other devices can be considered gesture-based

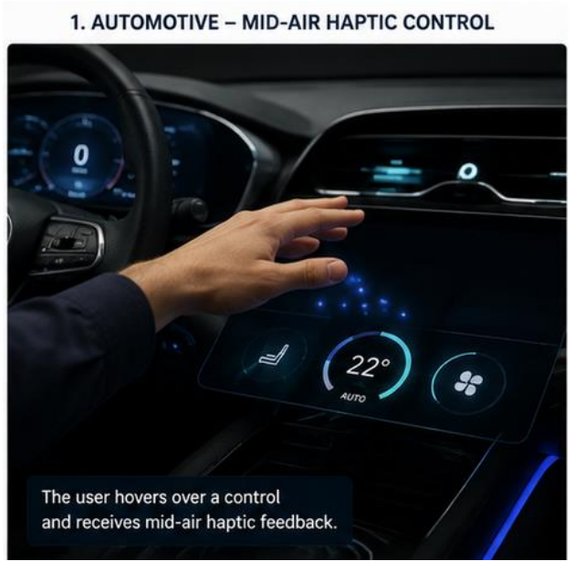






Mid-air gestures

- When we think of modern «gesture-based interface» we think something different: mid-air gestures used to control systems and communicate





Feedback and Guidance

Mid-air gestures have no physical affordance: users need to know what the system saw and what they can do.



Mid-air haptic feedback in a car interface

Visual

Hand avatar, highlighted targets, progress indicators while a gesture is performed

Haptic

Ultrasound mid-air haptics or vibrotactile wearables to confirm contact and selection

Auditory

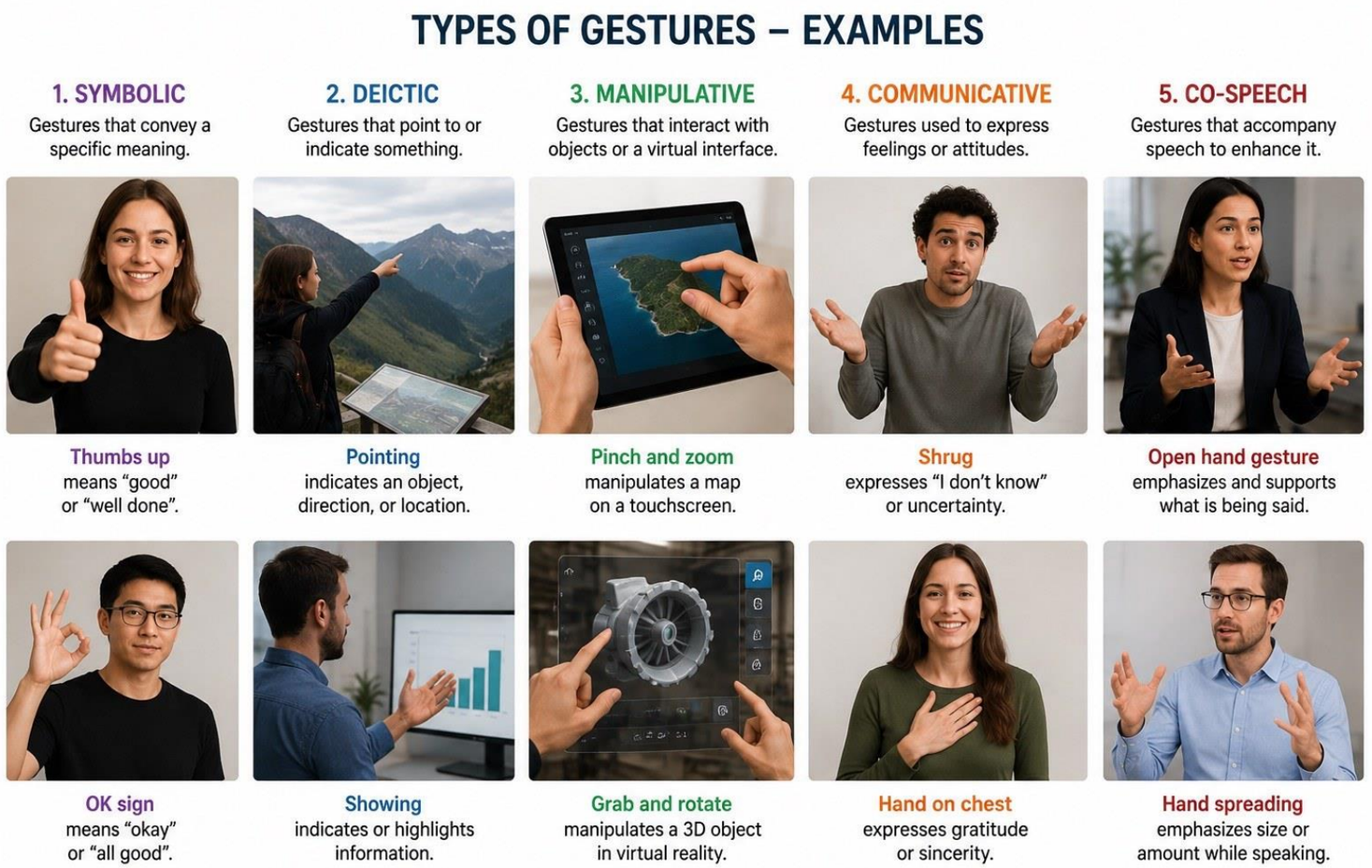
Short sounds confirming activation, completion, or rejection

Feedforward

Show which gestures are available and how to perform them, before the action

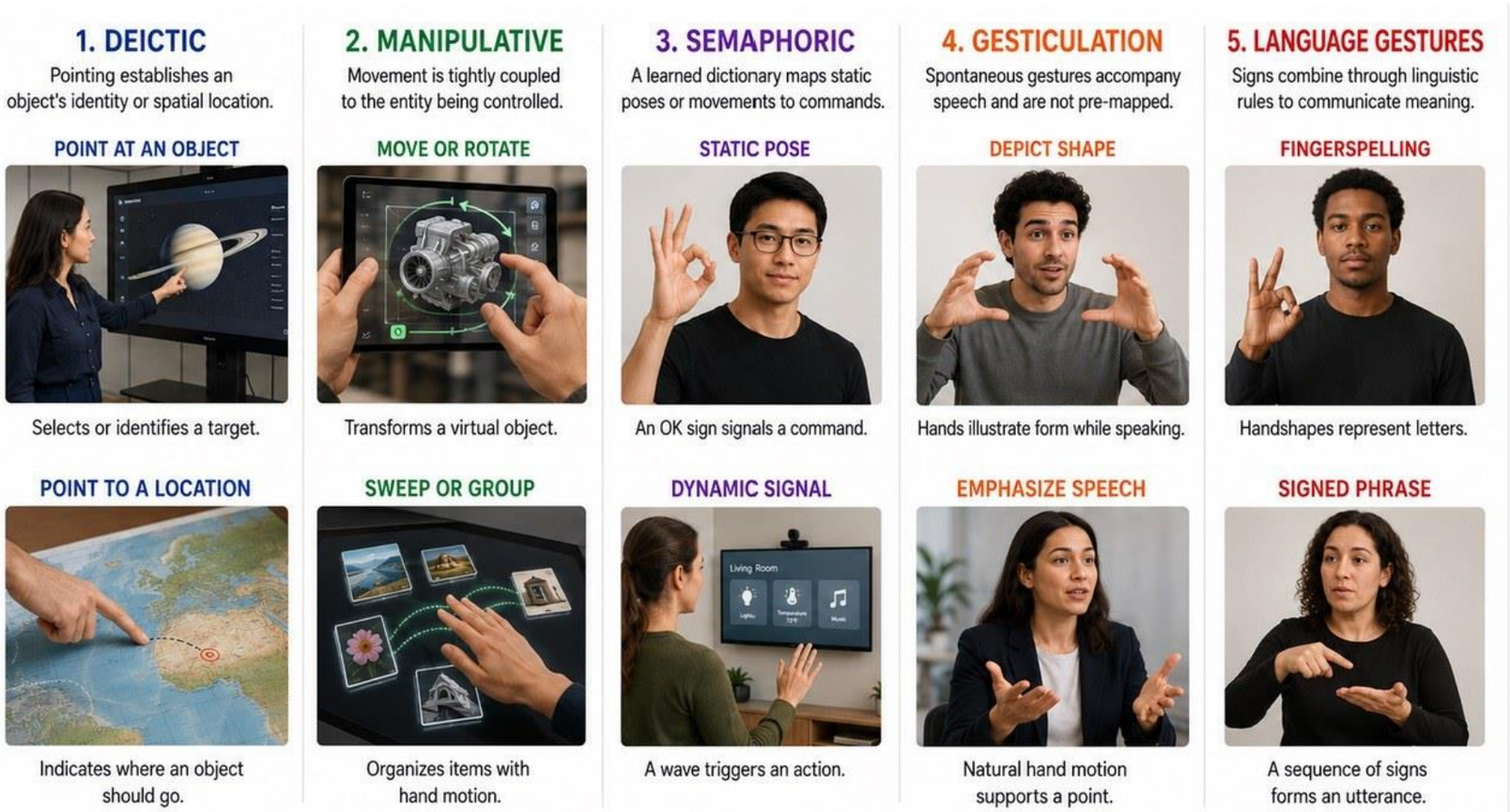
Gesture Types

- Different types, and different taxonomies
- AI-based analysis is useful for different kinds of gestures
- Interaction design mainly looks for sets of symbolic gestures useful for control (single-modal or multimodal)



Gesture Types: Another Taxonomy

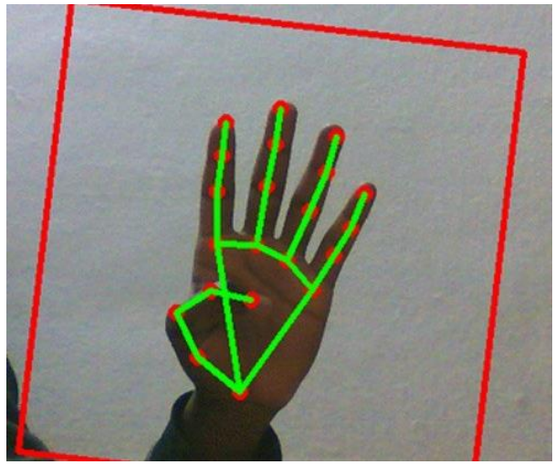
- Karam & schraefel (2005): deictic, manipulative, semaphoric, gesticulation, language gestures
- Taxonomies differ in what they emphasize: function, form, or relation to speech
- For control, interaction design mostly needs semaphoric gestures (single-modal or multimodal)

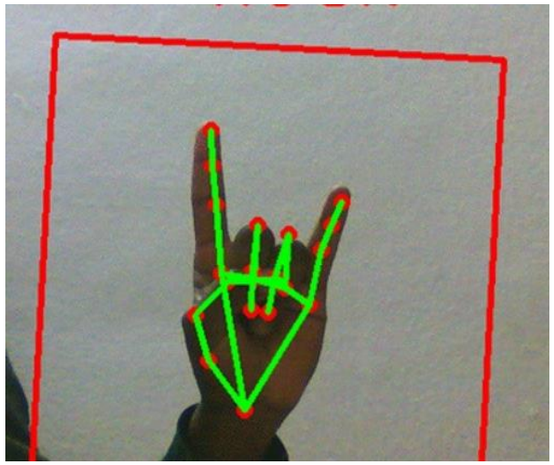


Static vs. Dynamic Gestures

Static: a hand posture

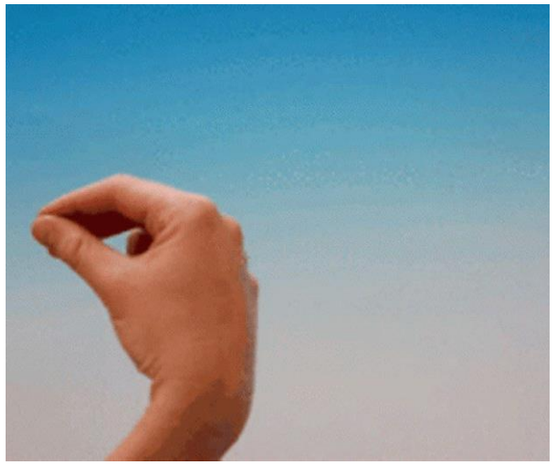
The shape of the hand carries the meaning

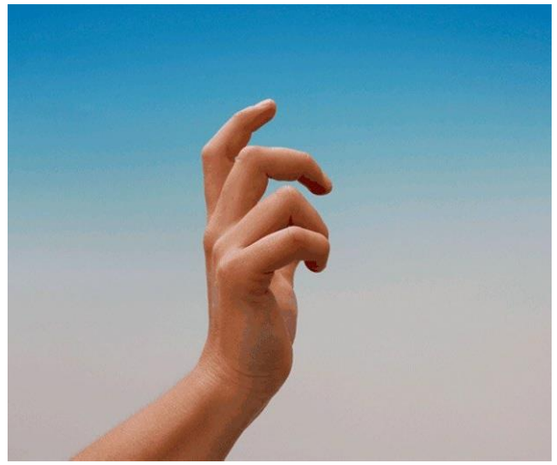




Dynamic: a movement over time

Trajectory and changing hand shape carry the meaning





Gesture phases → **Preparation** → **Stroke** → **Hold** → **Retraction**

The stroke carries the meaning; preparation and retraction frame it (Kendon, 1980; McNeill, 1992). For a designer: where does a command start and end?

Application Domains





Extended reality (XR)

Hand pose and motion captured by camera-based trackers within head-mounted displays

Multimodality: combined with speech (“put that there”), haptic feedback, visual feedback and guidance

Automotive & public displays

Depth or motion sensors facing the user

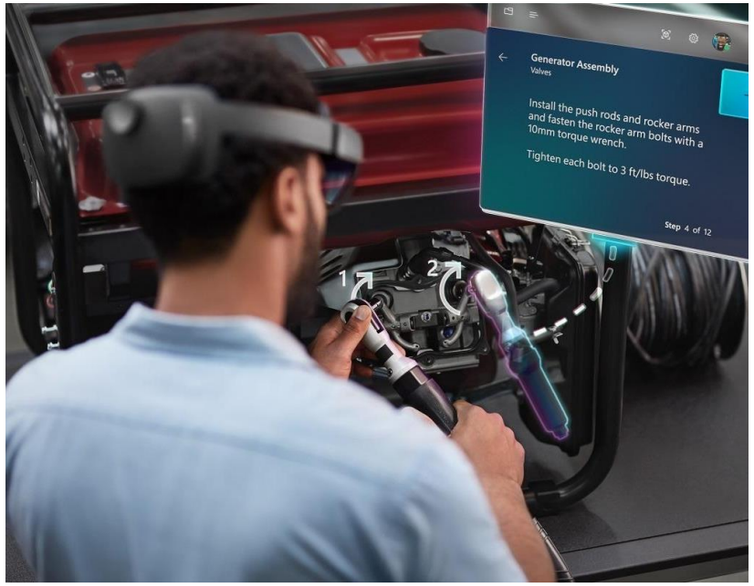
Close-range interaction: hand position itself may be used to activate the system

Other Applications



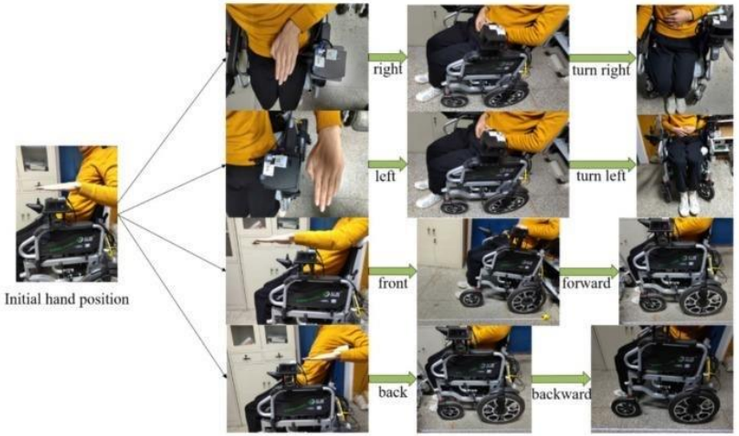
Human-robot interaction

Commanding and cooperating with robots through natural signals



Industrial XR

Hands-free guidance for assembly and maintenance tasks



Assistive technologies

Alternative control, e.g., driving a wheelchair with hand gestures

Real challenges

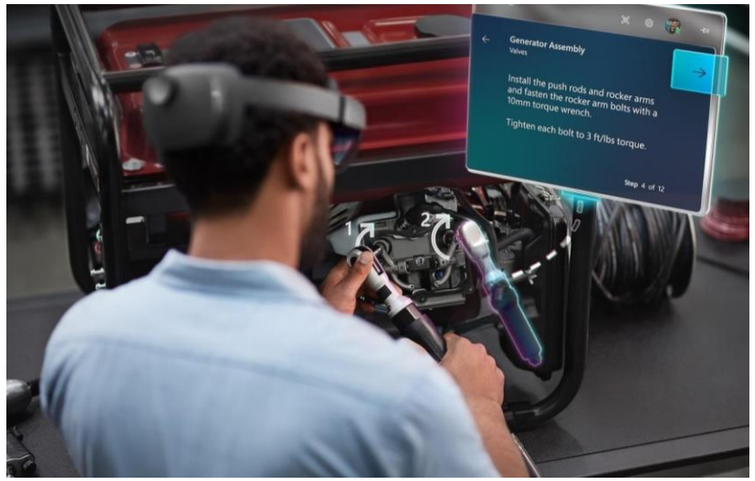
- Challenging? Yes: few applications and limited dictionaries
  - Why?
- Limited accuracy of gesture recognition
  - Example: occlusion issues for systems based on HMD cameras
- Non-trivial design of dictionary and semantic mapping
  - Ambiguity, ergonomics, fatigue, cultural variability
- Gesture detection and false positive issues
  - Can we accept false positives and missed detections? A very small rate can destroy usability in many contexts

Ambiguity

- Same form, different meanings
  - A raised palm can mean “stop”, “hello”, or “five”
- Limited Intentional or not?
  - Scratching, adjusting glasses, talking with the hands
- Variable execution
  - Speed, amplitude, and hand shape differ within and between people

- <u>Resolving ambiguity</u>
  - Context: application state, object in focus
  - Other modalities: speech and gaze narrow the interpretation
  - Interaction design: explicit activation, confirmation of critical actions, easy undo

Capturing Hands for Interaction







Head-mounted cameras

Built into XR headsets; no extra hardware, but hands can leave the field of view or be occluded

Depth and infrared sensors

Precise at close range; fixed interaction volume (desk, dashboard, public display)

Wearables

Data gloves, IMUs, EMG bands: robust to occlusion, but must be worn

Enabling Technologies

- Vision-based hand tracking
  - Marker-based and markerless tracking
  - RGB / RGB-D cameras
- Wearable sensors and data gloves

- Data quality issues and challenges
  - Analysis focused on mid-air gestures captured with sensors mounted on XR headsets
  - Practical implications for real-time gesture-based interaction (can be extended to different modalities)

Wearables

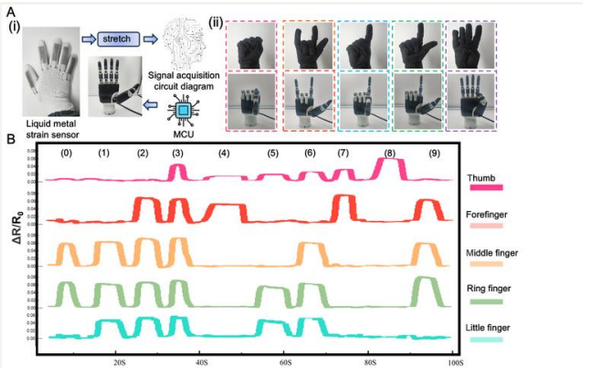
- Gloves
  - Accurate finger tracking. Enabling potentially accurate gesture recognition
  - Also providing haptic feedback
- Too cumbersome and expensive to implement general-purpose, practical interfaces
  - Used only on simulators and specific applications (e.g., remote control)







https://www.manus-meta.com/



Li, Dongsheng, et al. "A soft closed-loop glove with gesture sensing and haptic feedback for real-time human-machine interfaces." *Soft Science* 6.1 (2026): N-A.

Wearables

- **Wristbands:** IMU sensors detect wrist rotation, acceleration, taps, and arm movements; EMG sensors can detect muscle contractions for subtle finger gestures.
- **Rings:** IMUs, capacitive sensing, or optical sensors detect finger bends, taps, pinches, and hand poses.
- Potentially suitable for the development of gestural interfaces.
- Big hype since 2010s
  - But still currently not applied in the practice



Myo armband (discontinued in 2018)



https://www.tapwithus.com/



https://genki.is/wave

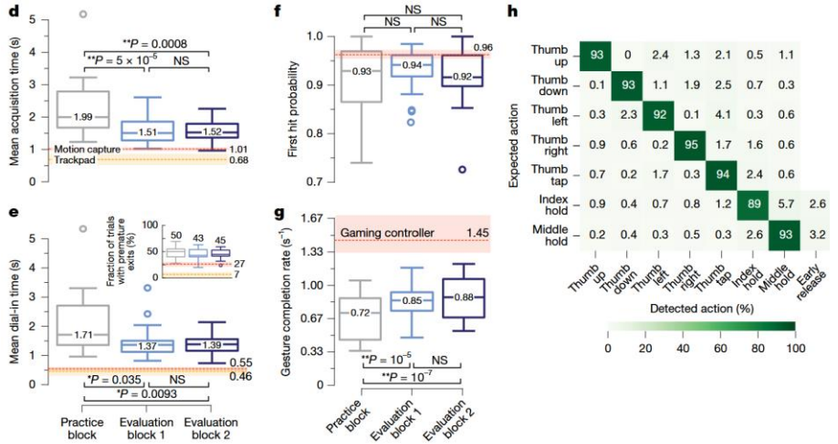
Wearables

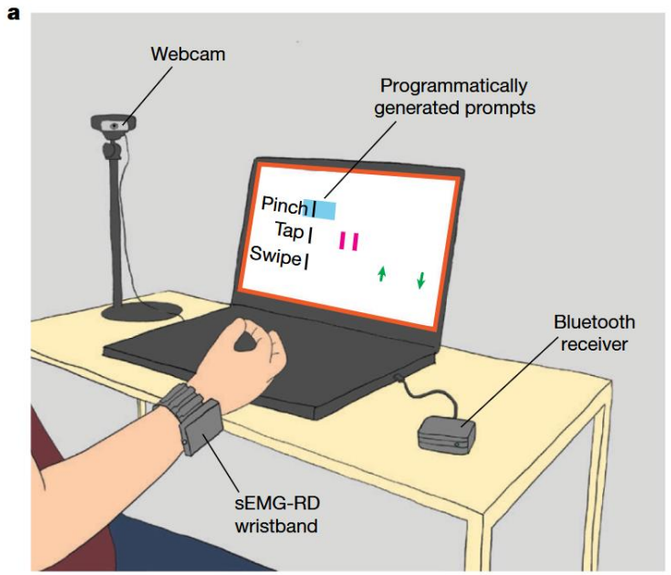
- **Key challenges:** latency, accidental gestures, calibration across users, battery life, comfort, and distinguishing intentional gestures from normal movement.
- Poor or totally missing research on validating interface design, gesture dictionaries
- Lots of announcements
  - Still poorly used

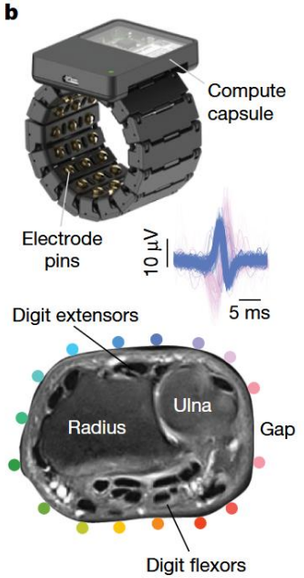


Current research results

- Meta Research (2025)
- Still focus on improvement of machine learning tools
- Good improvements with more training data, but suboptimal results







Kaifosh, P., Reardon, T.R. & CTRL-labs at Reality Labs. A generic non-invasive neuromotor interface for human-computer interaction. *Nature* **645**, 702–711 (2025). https://doi.org/10.1038/s41586-025-09255-w

Microwave/millimeter-wave sensing

- Based on a radar transmitting electromagnetic wave
- Hand motion changes the reflected signal. Signal processing extracts features such as Doppler frequency, Range, Direction of arrival, Micro-Doppler patterns, Time–frequency spectrograms
- A classifier recognizes gestures such as swipes, taps, rotations, or finger movements



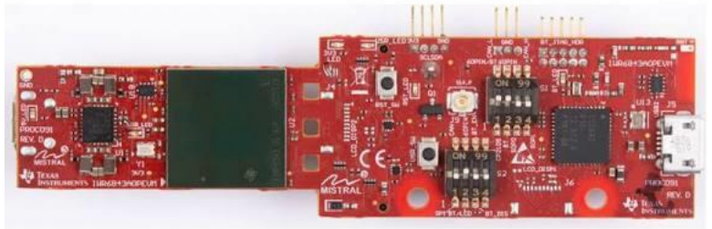
Microwave/millimeter-wave sensing

+ Works in darkness and bright light, preserves privacy better than cameras, can operate through smoke, dust, or some materials, detects motion without physical contact

- Limited range/FOV, multipath reflections from walls and furniture, sensitivity to hand distance and orientation, interference from other people or radar devices
- Classification can be hard for similar motion patterns

Sensors and practical applications

- Texas Instruments, Infineon sensors available at limited cost
- Google Soli available on some devices
- Practically used to recognize few gestures close to the sensor
  - Automotive, screen devices, appliance
- More challenging tasks proposed in the literature, but typically they are not testing continuous classification in a data stream making the methods not usable in the practice

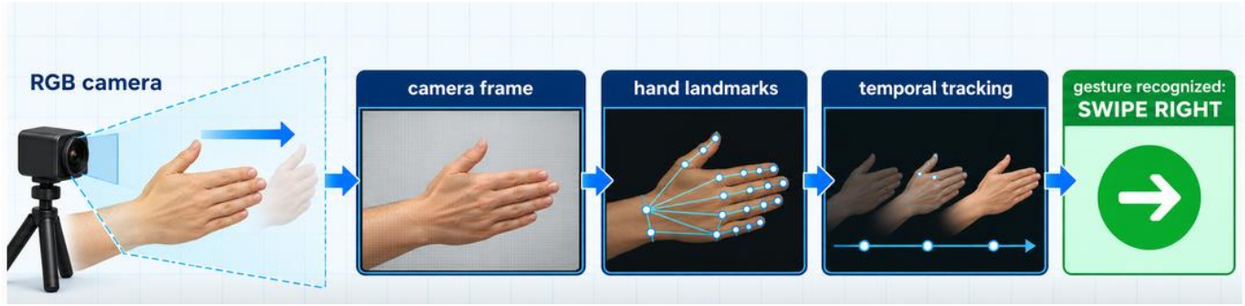




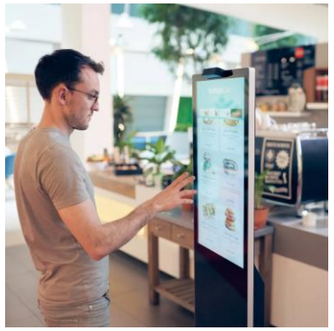


Vision-based hand tracking

- Computer vision provides advanced tools to capture arms' and hands' motions
- Marker-based or markerless
- Different camera configurations
  - Egocentric, exocentric
- Different camera types
  - RGB, RGBD
- Different gesture encodings
  - segmented images, skeletons,







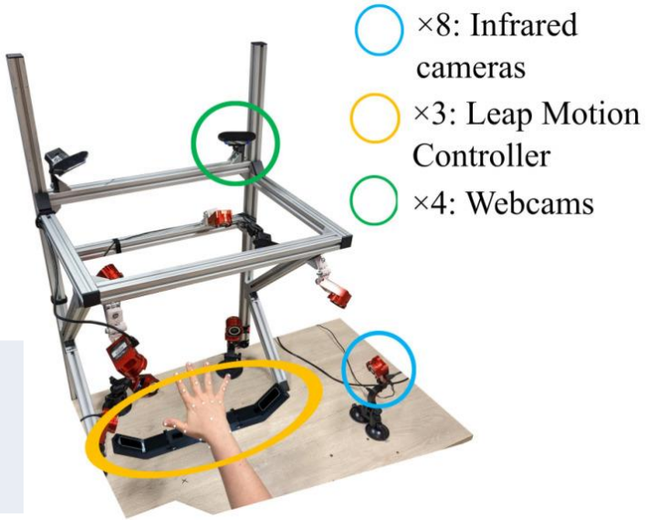


Marker-based

- Expensive
- Not practical for interactive applications
- Used to build dataset with accurate ground truth
  - Usable to train markerless methods based on different sensors

Veldhuijzen, Ben, et al. "SHREC 2024: Recognition of dynamic hand motions molding clay." *Computers & Graphics* 123 (2024): 104012.





Maggioni, Valentin, et al. "Optimisation and comparison of markerless and marker-based motion capture methods for hand and finger movement analysis." Sensors 25.4 (2025): 1079.

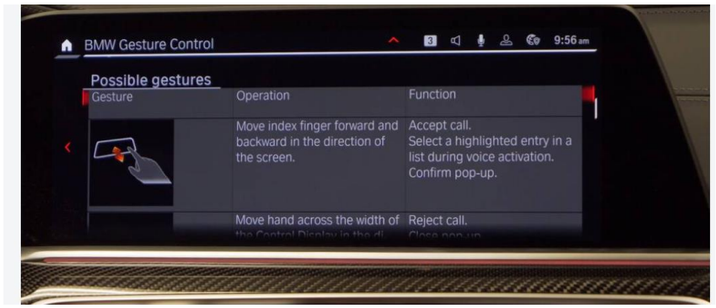
Markerless hand tracking

- Apparently a solved problem
  - Complete finger tracking with >20 joints tracked at more than 20fps
    - Using RGB-D sensors (e.g. UltraLeap or headset trackers)
    - But also on low end webcams (e.g., Google Mediapipe)
    - Limited computational cost
- Widely used in VR apps and games
- Used in remote control and automotive

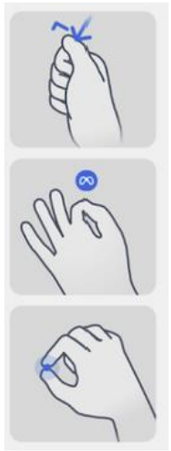


Vision-based hand tracking

- Reliable for simple gestures/interaction, limited vocabularies
  - Point, pinch, grab, now some microgestures





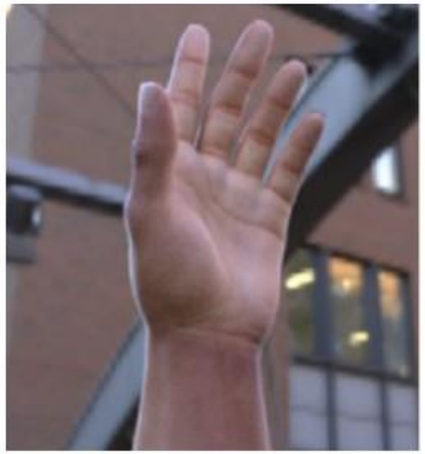






Hand Gesture Data Representation

The same gesture can be captured and stored in very different forms:



RGB

Appearance: colour, texture, context



RGB-D

Adds the distance of each pixel



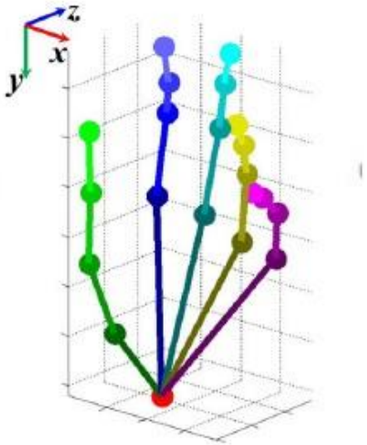
Optical Flow

Apparent motion between frames



Infrared (IR)

Active illumination, works in the dark



Skeleton

3D positions of the hand joints

Gestures as Image Sequences

A dynamic gesture is a sequence of T frames:

T × H × W × C

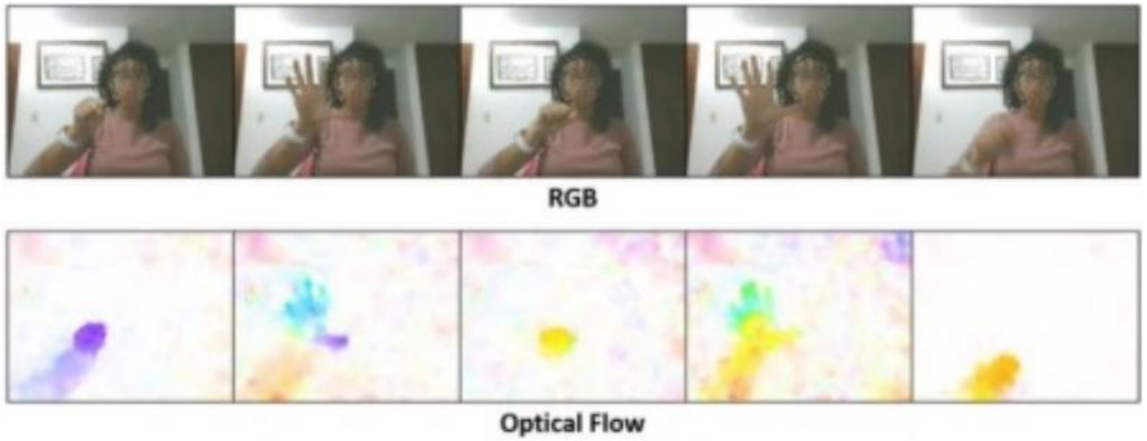
frames × height × width × channels (3 for RGB, 1 for depth)

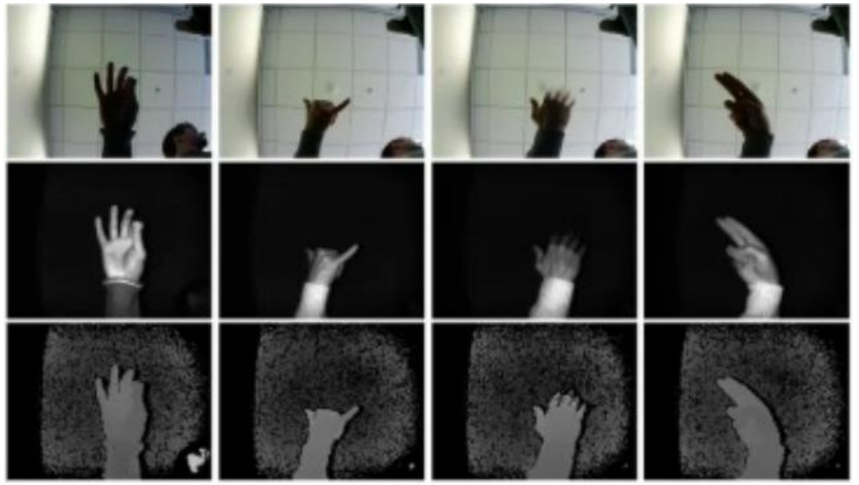
- **RGB:** rich appearance, but sensitive to lighting, background, skin tone, and clothing
- **Depth:** geometry, robust to illumination, easier hand segmentation; noisier and lower resolution
- **High-dimensional:** a 640 × 480 RGB frame holds about 920,000 values



Image, depth, and skeleton views of the same gesture: SHREC'17 (De Smedt et al., 2017)

Motion and Infrared Streams





Optical flow

Encodes the apparent motion of pixels between consecutive frames

Highlights the moving hand and removes the static background, but is costly to compute

Example: IPN Hand (Benitez-Garcia et al., 2020)

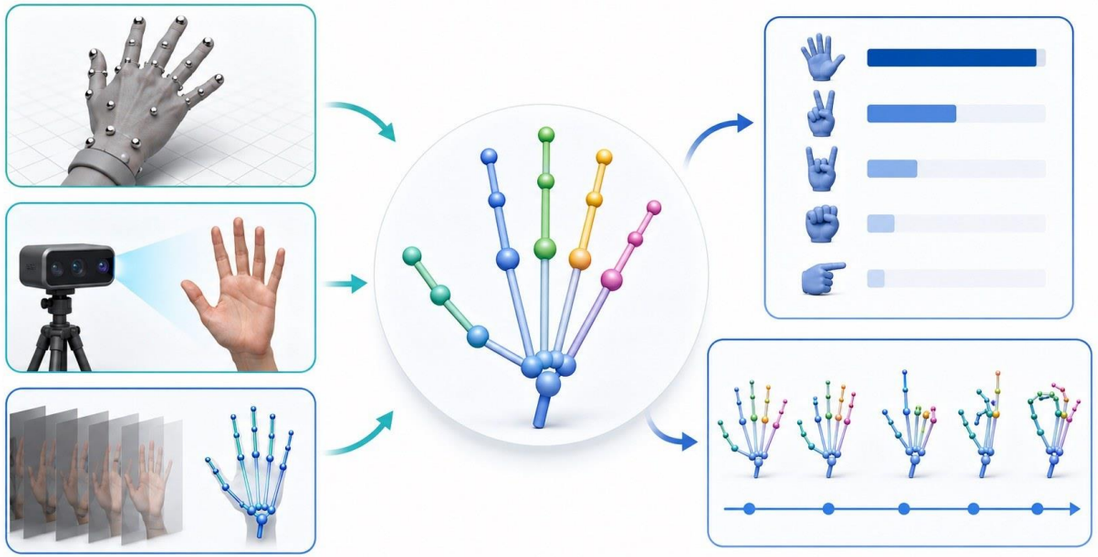
Infrared and multi-stream

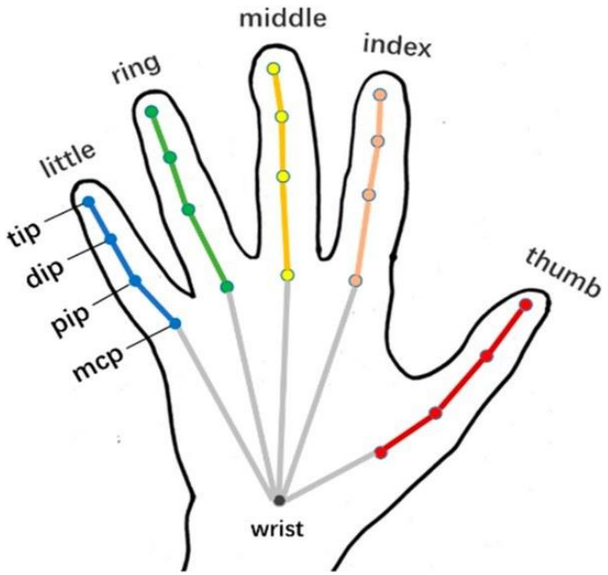
Active illumination makes infrared independent of ambient light, e.g., inside a car cabin

Combining synchronized streams (RGB, depth, IR) increases robustness

Example: Briareo (Manganaro et al., 2019)

Skeleton-Based Representation





N keypoints per hand

21 in MediaPipe, 22 in SHREC'17 (with the palm center), with 2D or 3D coordinates

Source-agnostic

Optical MoCap, pose estimation from RGB/RGB-D (MediaPipe, WiLoR), depth and IR sensors

Compact

Tens of values per frame

Appearance-independent

No background, lighting, or skin

Anatomically grounded

Fixed bones, limited degrees of freedom

Graph-ready

Joints and bones form a natural graph

Skeleton Sequences

A gesture becomes a small tensor:

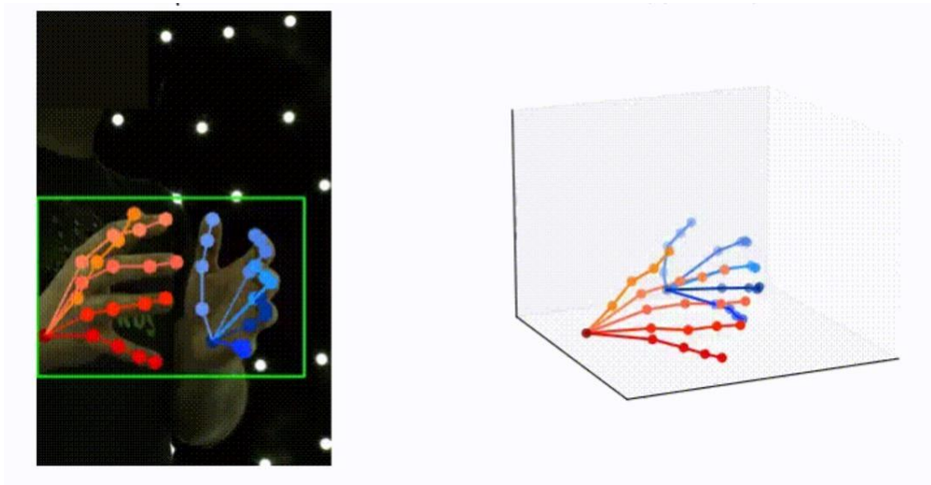
T × N × C

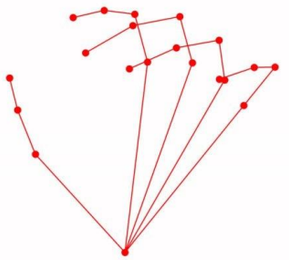
frames × joints × coordinates

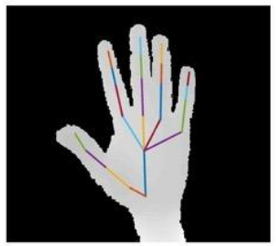
22 joints × 3 coordinates = 66 values per frame

versus about 920,000 values for one RGB frame

Temporal information is explicit: each joint draws a 3D trajectory over time.









Preparing Skeleton Data

Raw joint coordinates depend on where the user stands and how big the hand is. Typical normalization:

Center
Subtract the wrist or palm position

→

Scale
Divide by a reference bone length

→

Align
Remove the global hand rotation

→

Resample
Bring every sequence to a fixed length T

Features derived from the skeleton

Joints
Normalized 3D positions

Bones
Vectors between connected joints

Motion
Velocities: differences between frames

Angles
Finger flexion and inter-finger angles

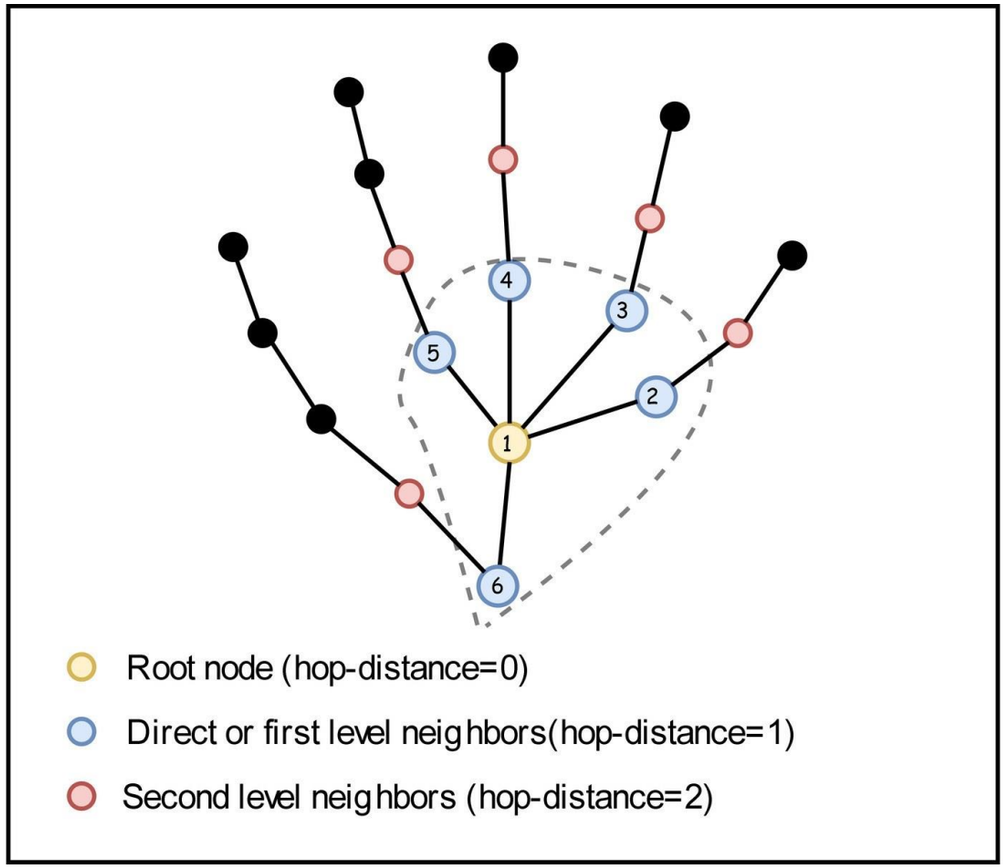
Joint, bone, and motion streams are often used together, as complementary views of the same gesture.

From Skeleton to Graph

G = (V, E)

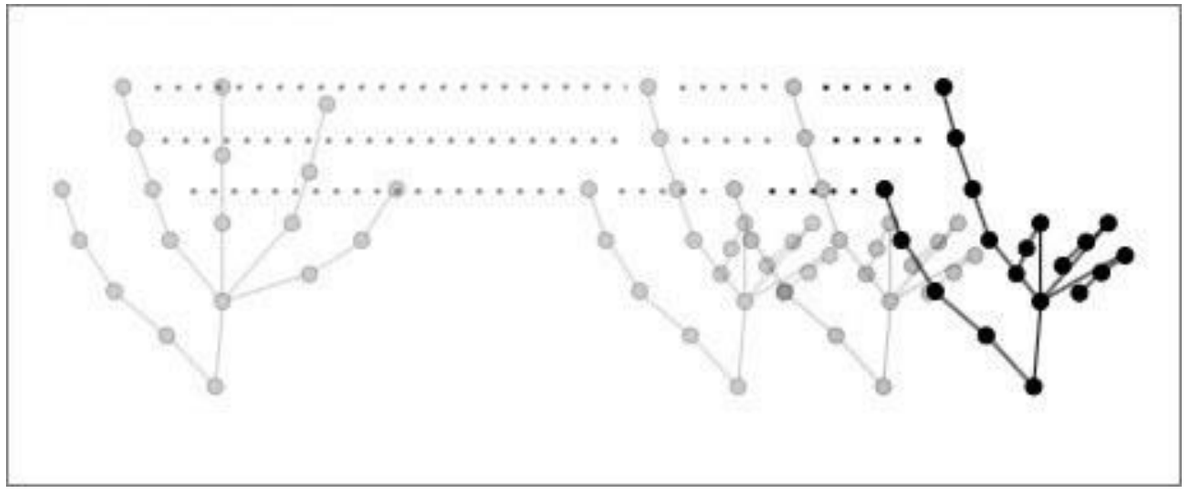
- **Nodes:** hand joints
- **Edges:** bones (physical connections)
- **Adjacency matrix A:** A(i, j) = 1 if joints i and j are connected

Nodes	N1	N2	N3	N4	N5	N6
N1	1	1	1	1	1	1
N2	1	1	0	0	0	0
N3	1	0	1	0	0	0
N4	1	0	0	1	0	0
N5	1	0	0	0	1	0
N6	1	0	0	0	0	1



Neighbors grouped by hop distance from a root joint (0, 1, 2 hops)

Spatio-Temporal Graphs



Spatial edges

Bones connect joints within the same frame

Temporal edges

Each joint is linked to itself in the next frame

Graph convolution

Each joint aggregates features from its neighbors:

$X' = \sigma(\hat{A} X W)$, with $\hat{A}$ the normalized adjacency

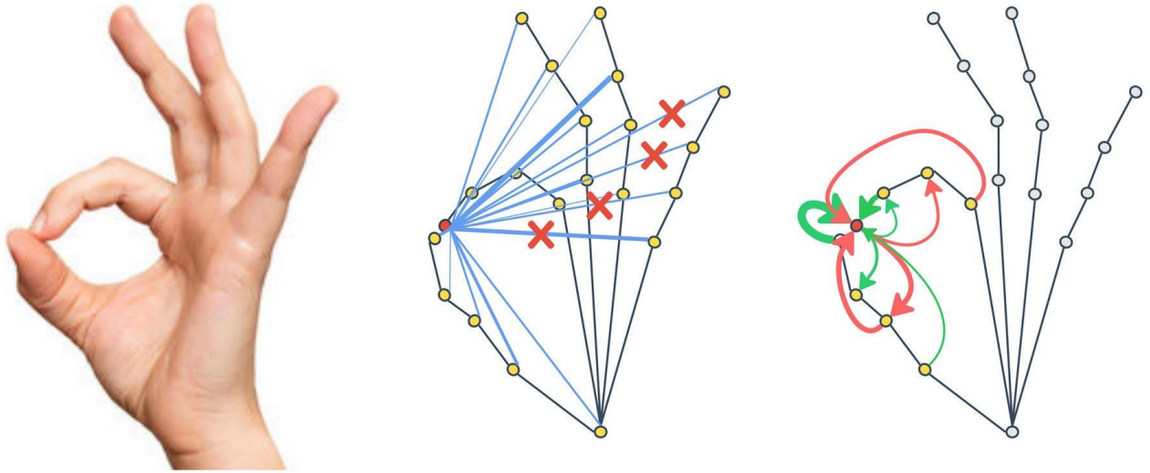
The graph encodes the hand structure directly in the representation (e.g., ST-GCN, Yan et al., 2018).

Rethinking the Hand Graph

Graph-based models improved spatio-temporal learning, but they:

- mostly rely on dense, undirected skeletal graphs
- assume all joints contribute equally
- ignore the directional flow of articulation (wrist → fingertips)

→ redundant connections, more computation, weaker results on fine-grained gestures



(A) (B) (C)

Two key observations

(i) Sparsity

Only a subset of joint-to-joint dependencies is functionally relevant for a given gesture

(ii) Direction

The flow of information between joints conveys cues about gesture meaning and dynamics

Comparing Representations

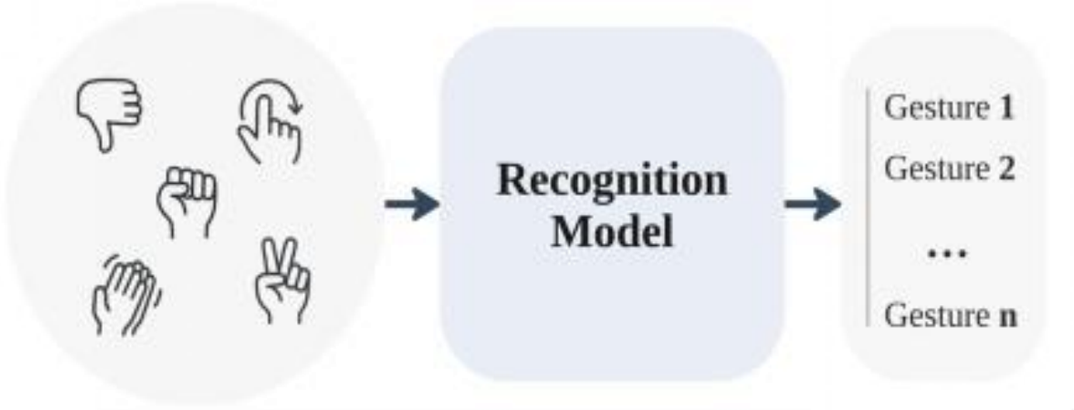
Representation	Captures	Strengths	Limitations
RGB video	Appearance, context	Cheap cameras, rich cues	Lighting, background, high dimension
Depth	Geometry	Robust to light, easy segmentation	Noise, range, resolution
Optical flow	Pixel motion	Focus on movement	Costly, camera motion
Infrared	Actively lit image	Works in the dark	Sunlight interference
Skeleton	Joint positions	Compact, appearance-free	Depends on pose estimator
Graph	Joints + relations	Encodes hand structure	Graph design choices

Skeletons offer the best trade-off between abstraction, computation, and biomechanical fidelity.

Recognition & Generation – Two Complementary Problems

Recognition

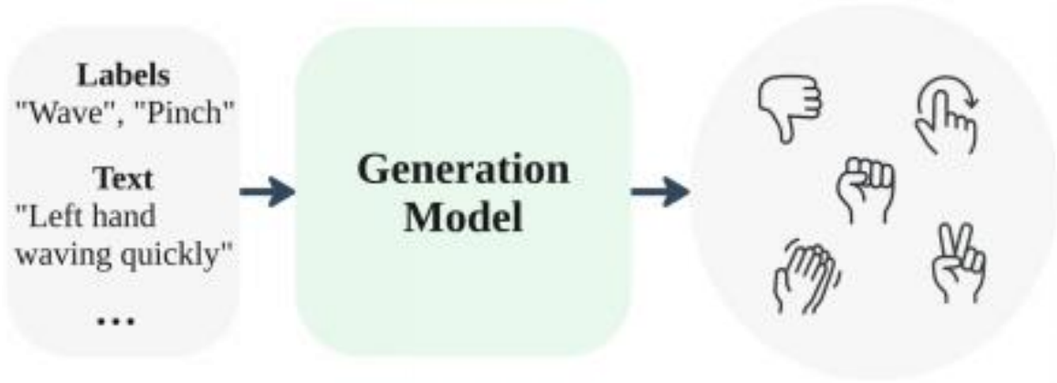
"What is the hand doing?"



- **Task:** Classify a sequence of hand motion into a predefined category (e.g., "wave," "pinch").
- **Challenge:** Learn from limited data; recognize gestures in real-time.
- **A discriminant model:** In probabilistic language: find for each x, the value y that maximizes P(y|x)

Generation

"How to make the hand do it?"



- **Task:** Synthesize a realistic, continuous sequence of hand motion from a high-level cues (e.g., "wave hello").
- **Challenge:** Ensure motions are anatomically plausible and semantically correct.
- **A Generative model:** In probabilistic language: find for each y the values of x which contribute to the joint distribution P(x, y)

Question

- Given that
  - hand tracking seems to work well
  - We have effective static/dynamic representations of hand structures
  - Machine learning is a mature technology



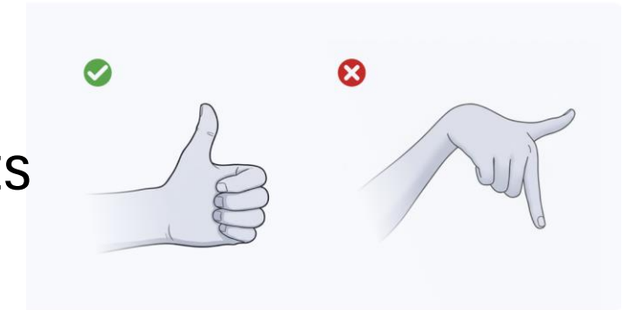
- Why it is not easy to find applications with more compled hand-based interaction and interfaces are limited to simple grab, pinch gestures, and user experience not always optimal?
- What makes the design of mid-air gesture-based interfaces hard?

Interaction design issues

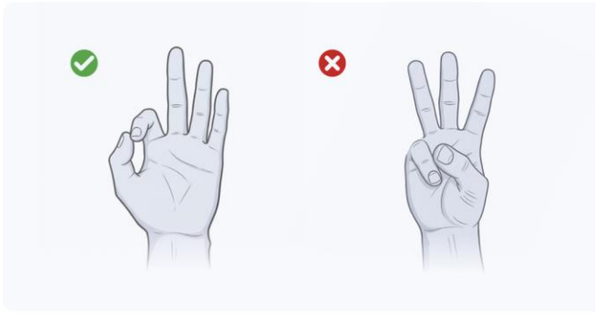
- Example: XR applications
- User experience often frustrating
- Issues: line of sight, latency. Why?
  - Constrained sensor placement
  - Missed or false detection
  - Few errors or unwanted activation

  can destroy the user experience
- How to improve?
  - Understanding the system constraints
  - Considering human factors



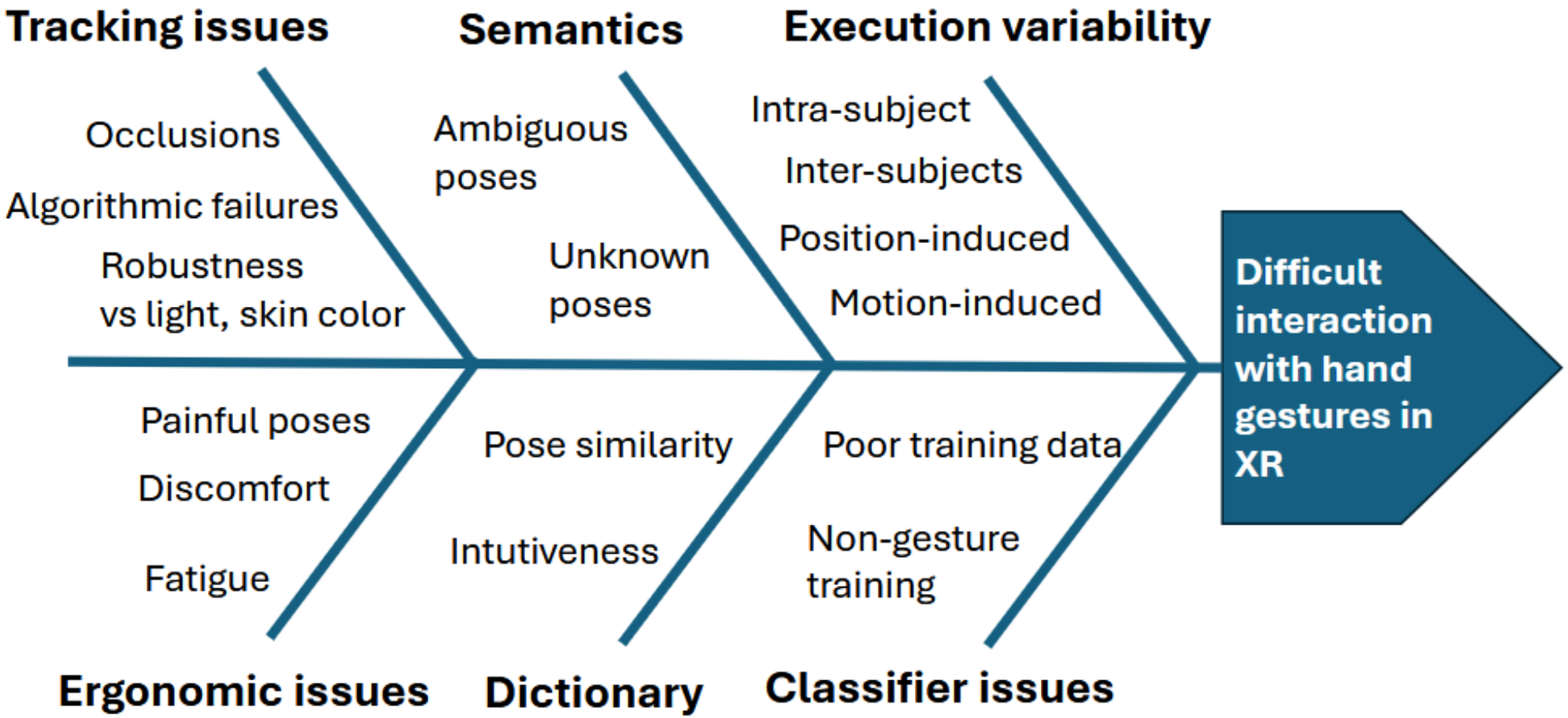


Gestures should face the headset cameras to avoid self-occlusion



Accuracy diminishes when using the middle, ring, or pinkie fingers.

Example challenges in XR gesture interaction



Emporio, M., Fattahi, H., Caputo, A., & Giachetti, A. (2026). Meaningful and ergonomic hand poses for extended reality interaction. *Virtual Reality*.

Acquisition issues

- Simple hand poses not always well captured (Emporio et al. 2026)
  - We captured static, meaningful poses performed by different subjects in different conditions with a HMD tracker and analyzed the data





Emporio, M., Fattahi, H., Caputo, A., & Giachetti, A. (2026). Meaningful and ergonomic hand poses for extended reality interaction. *Virtual Reality*.

Semantic of gestures?

- Hand poses and movements are relatively complex
- The same gestures are not always executed in the same way
  - «Because gestures are unconstrained, they are apt to be performed in an ambiguous or uninterruptable manner, in which case constructive feedback is required to allow the person to learn the appropriate manner of performance and to understand what was wrong with their action» (Norman, 2010)
- Semantics is not associated through conscious rules (apart a few simple cases)
  - For gestures that we commonly use in communication it is learned by examples (Kendon 2004)

Norman, D. A. (2010). "Natural User Interfaces Are Not Natural." *Interactions, 17*(3).

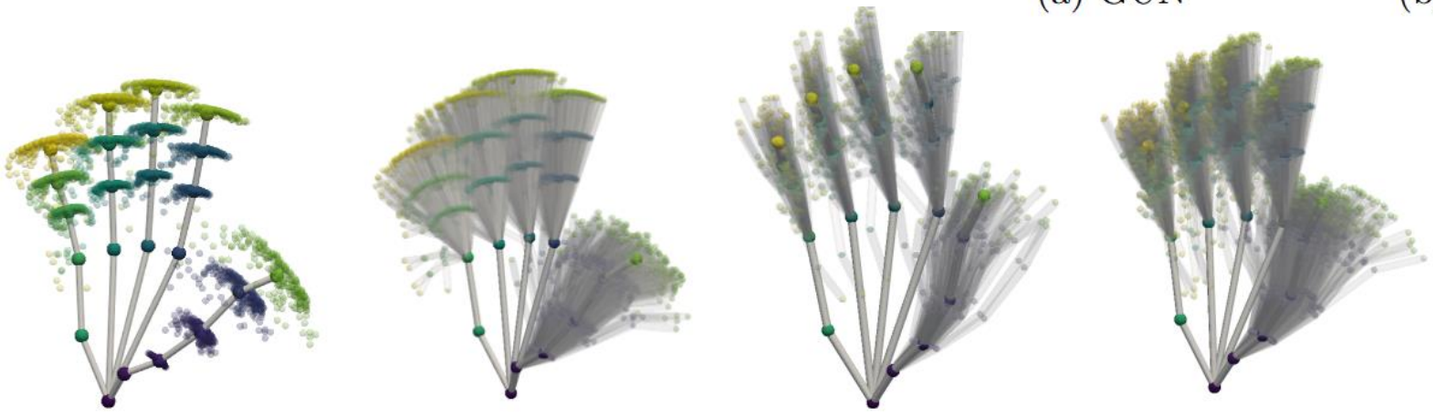
Kendon A (2004) Gesture: Visible action as utterance. Cambridge University Press

Gesture variability and tracking failures

- Large variability in the poses performed
- When the hand is rotated the pose is also inadvertently changed



(a) GUN (b) THREE



(a) Five, optimal (b) Five, rotated (c) Relaxed hand, optimal (d) Relaxed hand, rotated

How to verify?

- We performed a visual verification by rendering captured skeletons and performing human labeling, still based on the examples provided

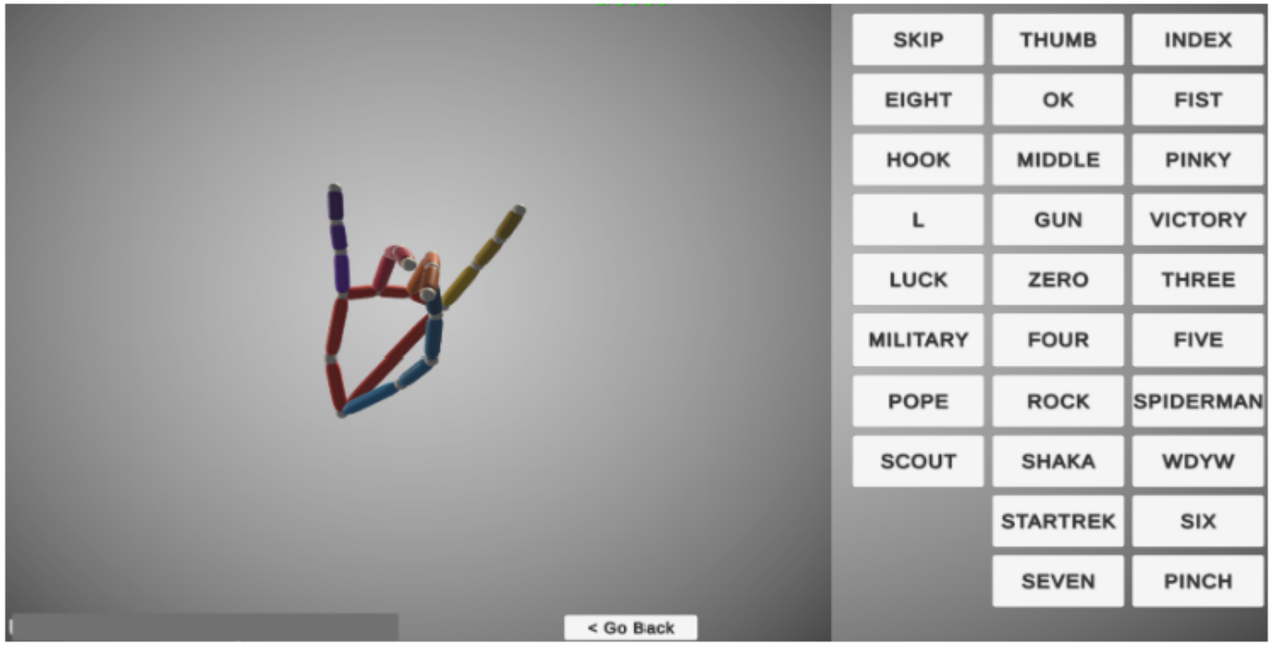
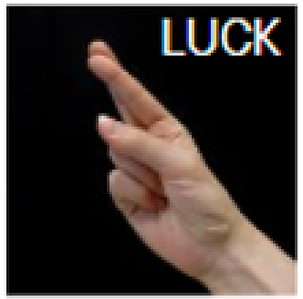


Fig. 10: Screenshot of the visual labeling application.

Gesture variability and tracking failures

- Visual verification of reconstructed skeletons reveal a non negligible amount of badly tracked poses, due to occlusions and model failure
- Many errors for poses acquired with rotating hands
- In some cases it is clearly due to the failure of the tracker (e.g., in case of crossed fingers)







(c) SPOCK



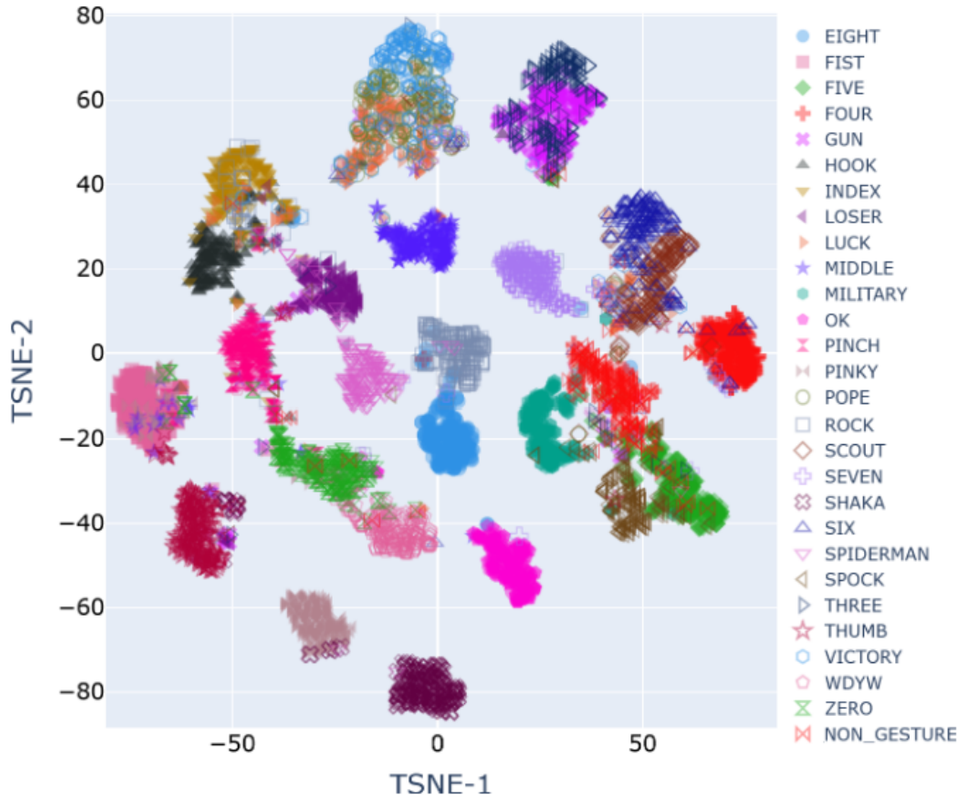
(d) LUCK

	Prec.	Rec.	F-Sc.
OK	0,99	0,92	0,95
SHAKA	0,99	0,89	0,94
WDYW	0,97	0,90	0,94
FOUR	0,92	0,93	0,93
MILITARY	0,95	0,88	0,92
SEVEN	1,00	0,84	0,91
THUMB	0,86	0,94	0,90
EIGHT	1,00	0,81	0,89
PINCH	0,88	0,91	0,89
LOSER	0,87	0,88	0,88
MIDDLE	0,97	0,80	0,88
FIST	0,78	0,98	0,87
PINKY	0,99	0,78	0,87
SIX	0,92	0,82	0,87
SPIDERMAN	0,99	0,78	0,87
VICTORY	0,88	0,84	0,86
FIVE	0,85	0,83	0,84
HOOK	0,98	0,73	0,84
ZERO	0,86	0,81	0,83
ROCK	0,90	0,75	0,82
SCOUT	0,99	0,70	0,82
THREE	0,93	0,74	0,82
GUN	0,87	0,75	0,81
INDEX	0,70	0,91	0,79
POPE	0,63	0,77	0,69
SPOCK	1,00	0,52	0,68
NON_GESTURE	0,23	0,96	0,38
LUCK	1,00	0,01	0,02
Average	0,89	0,80	0,81

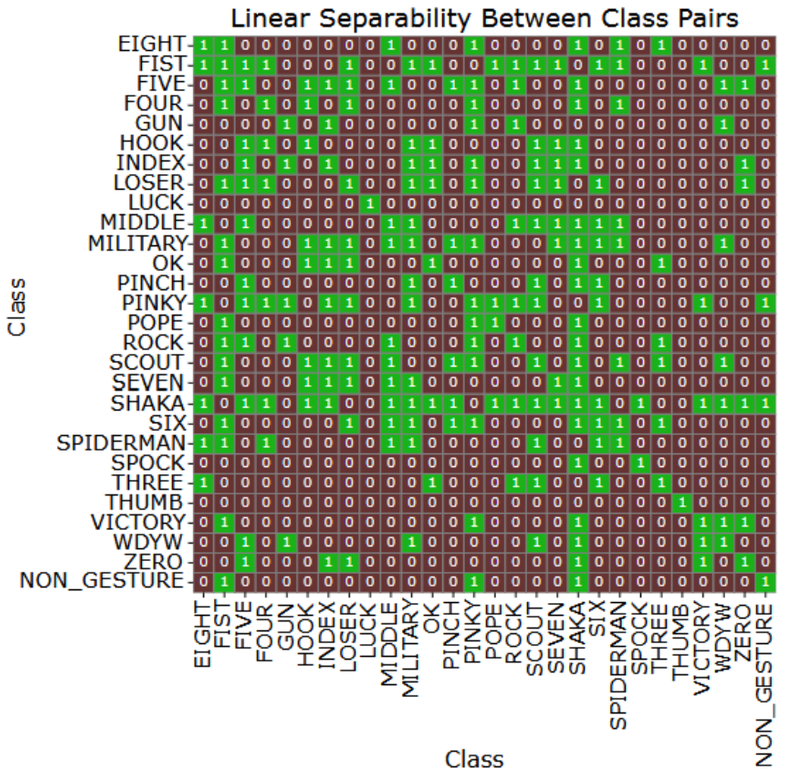
Table 4: Classification scores in visual labeling of skeletons captured generic orientations. Gestures are sorted by F-Scores, taking both missed and false detections. Top rated gestures are the most robust against orientation changes during the executions.

Gesture class overlap

- Critical for selected groups of poses



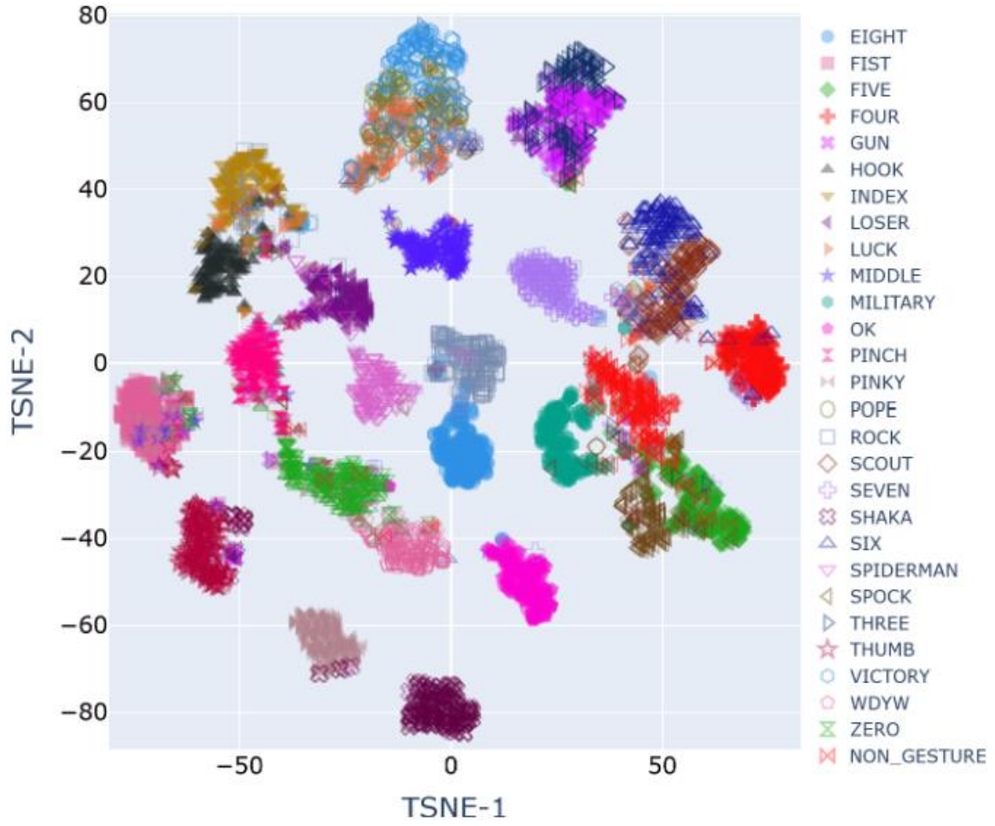
(a)



(b)

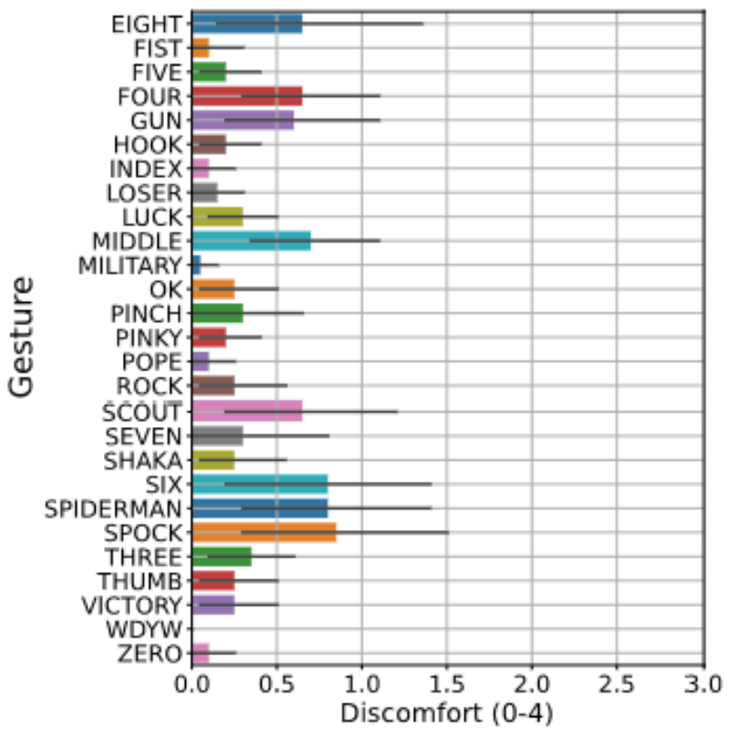
Other challenge: non-gesture signals

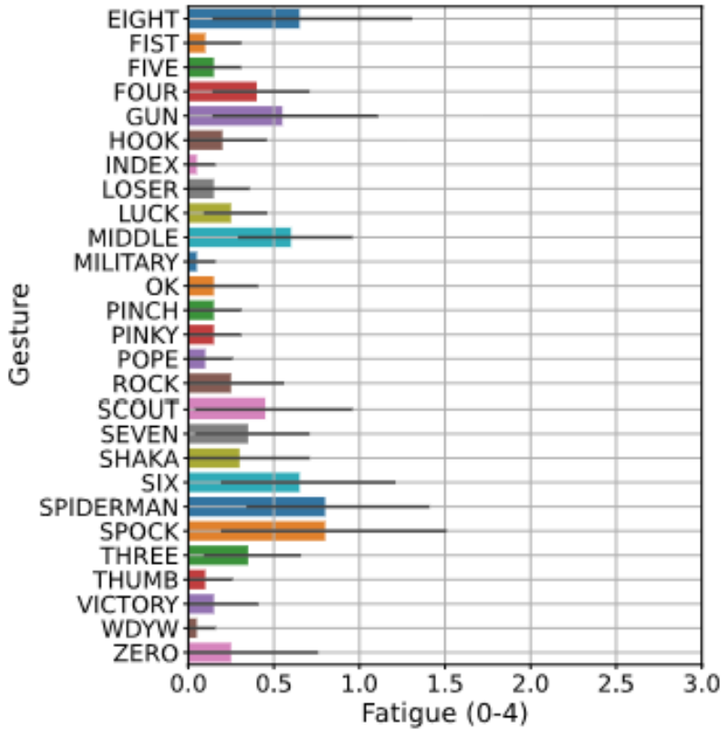
- Problem: feature space of captured gestures can be complex
  - E.g., joints distance vectores for tracked hand poses
- We have examples of labeled gestures, but they cover a small part of the feature space
- How to classify avoiding false positives for meaningless gesticulation?



Fatigue and discomfort

- Can be non negligible even in case of limited duration of the interaction





Gestures and natural interaction

- «Natural interaction» has been (unfortunately) seen as a goal for interaction design, especially in the case of gesture-based interfaces
- Typical idea:
  - Think about gestures we do in real life and reuse in interfaces
  - Ask user to choose which gesture they associate to an interface action
- Does it work?
  - Typically not. Naturalness is learned. And the success of interaction design rely on reliable input channels and easily learnable mappings

Norman, D. A. (2010). "Natural User Interfaces Are Not Natural." *Interactions, 17*(3).

Natural? Effective?

- Tentatives to build «Minority Report»-like interfaces, often motivated by «natural» interaction fail.
- Mouse control was extremely successful. Is it «natural»?

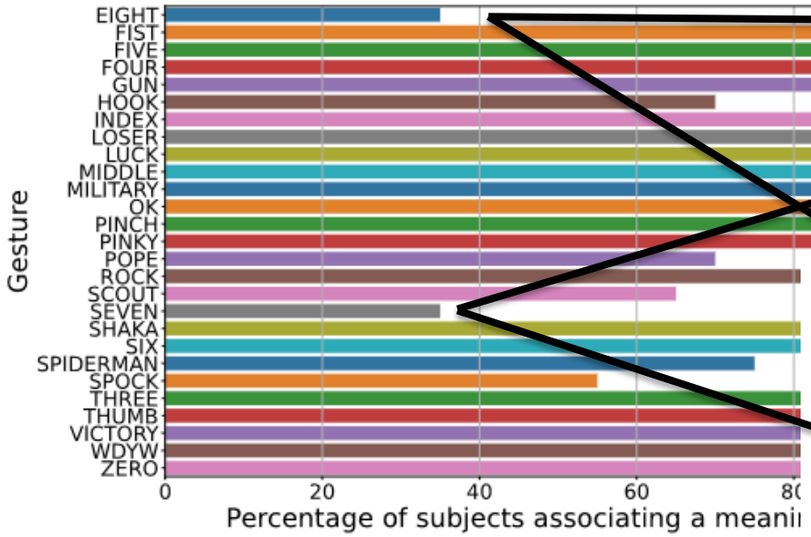


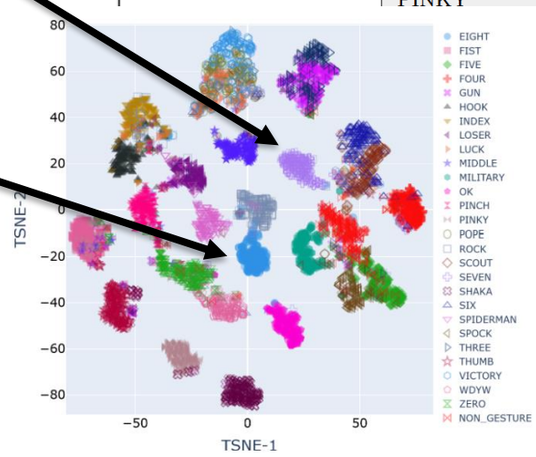


Semantic check

- Are poses more often associated with meanings easier to perform consistently and be recognized? We asked for associations and checked.
  - Indicative answer seems no…







	Prec.	Rec.	F-Sc.
OK	0,99	0,92	0,95
SHAKA	0,99	0,89	0,94
WDYW	0,97	0,90	0,94
FOUR	0,92	0,93	0,93
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PINKY	0,99	0,78	0,87
	0,92	0,82	0,87
	0,99	0,78	0,87
	0,88	0,84	0,86
	0,85	0,83	0,84
	0,98	0,73	0,84
	0,86	0,81	0,83
	0,90	0,75	0,82
	0,99	0,70	0,82
	0,93	0,74	0,82
	0,87	0,75	0,81
	0,70	0,91	0,79
	0,63	0,77	0,69
	1,00	0,52	0,68
RE	0,23	0,96	0,38
	1,00	0,01	0,02
	0,89	0,80	0,81

Lessons learned

- We need probably to redesign the approach used to think about gesture mapping
- Don’t start from elicitation studies or science-fiction
- Start from the constraints of the acquisition system and ergonomics
  - Design hand pose/actions that can be consistently executed by users without issues and are accurately captured by the sensor used
  - Create mappings that are easy to learn (not necessarily similar to existing tasks)

Emporio, M., Fattahi, H., Caputo, A., & Giachetti, A. (2026). Meaningful and ergonomic hand poses for extended reality interaction. *Virtual Reality*.

Next: how to detect and classify?

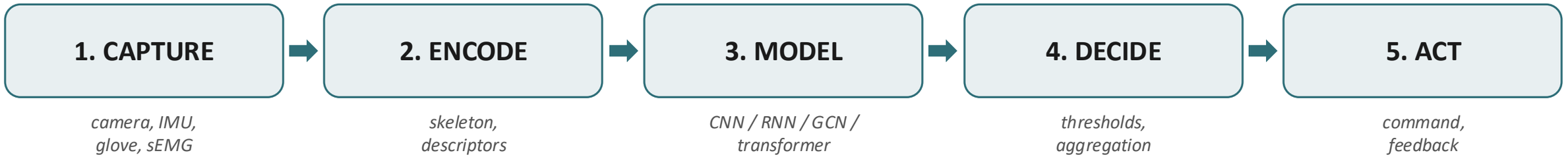
- In the next presentation we will see an analysis of the related issue based on our experience on vision-based methods and interaction design for extended reality
- The same concepts are valid for all the other types of sensors
  - We need to classify temporal sequences
  - For real mid-air generic interaction, we need to separate meaningful gestures from gesticulation (gesture segmentation)
  - We need dictionaries datasets and benchmarks (difficult to find…)

3 - HAND GESTURE RECOGNITION: METHODS, BENCHMARKS, AND APPLICATIONS

Hand Gesture Recognition: Methods, Benchmarks, and Applications

- Gesture representations and encoding strategies
- Static, dynamic, segmented, and continuous gesture recognition
- Temporal modeling and sequence-based recognition
- Learning-based methods for gesture classification and detection
- Continuous recognition challenges: segmentation, false positives, temporal ambiguity, and online inference
- Benchmark datasets and evaluation protocols
- Practical tools and implementation considerations
- Demonstrations of gesture-based applications

From sensor to command: the whole pipeline to create a good gestural inteface



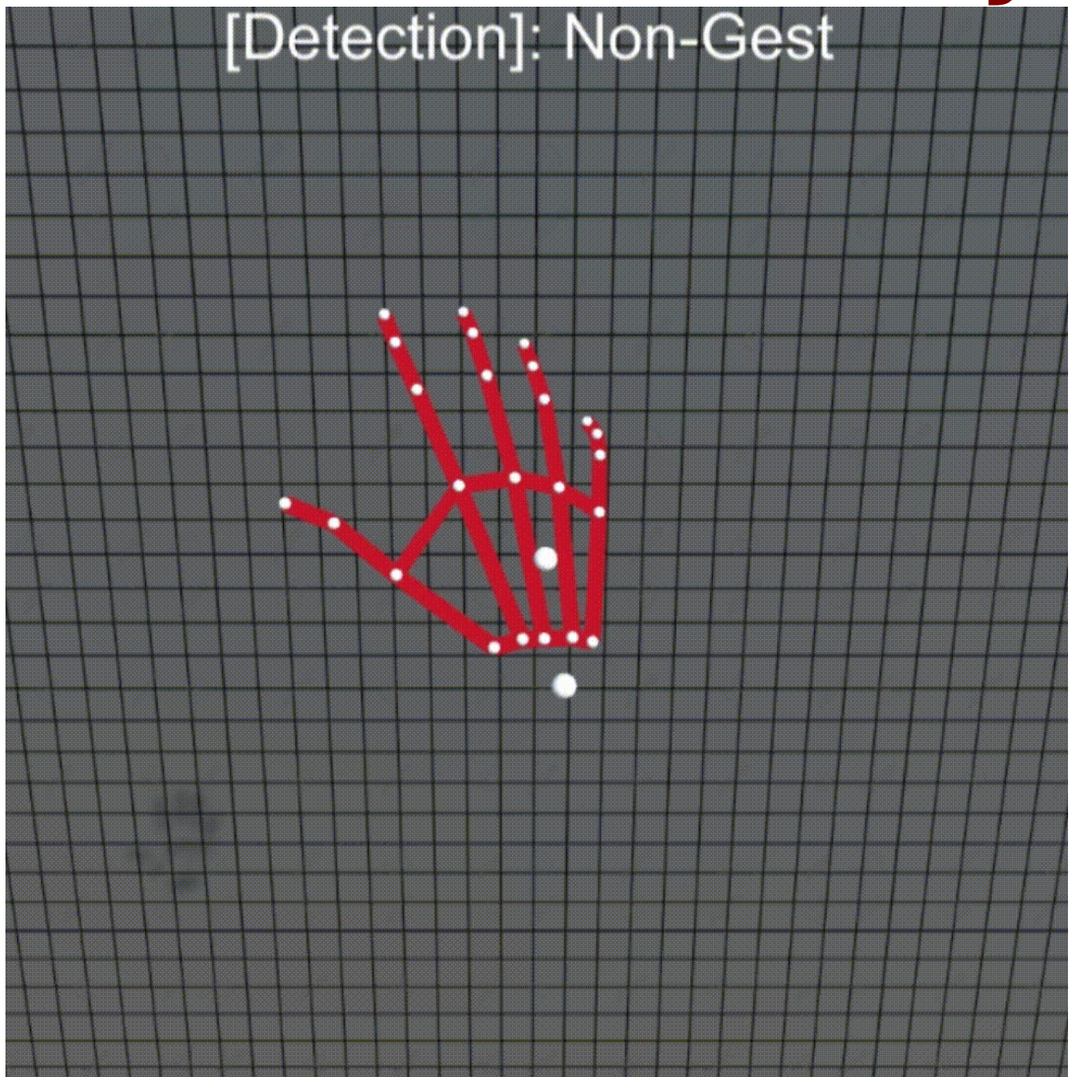
Challenges in Gesture Recognition

- Hand movements tracking
- Variability in gestures:
  - Different users, styles, speeds and variations due to fatigue or environment.
- Gestures don’t happen in isolation
- Fast recognition time
- Minimizing false positives and misclassifications.



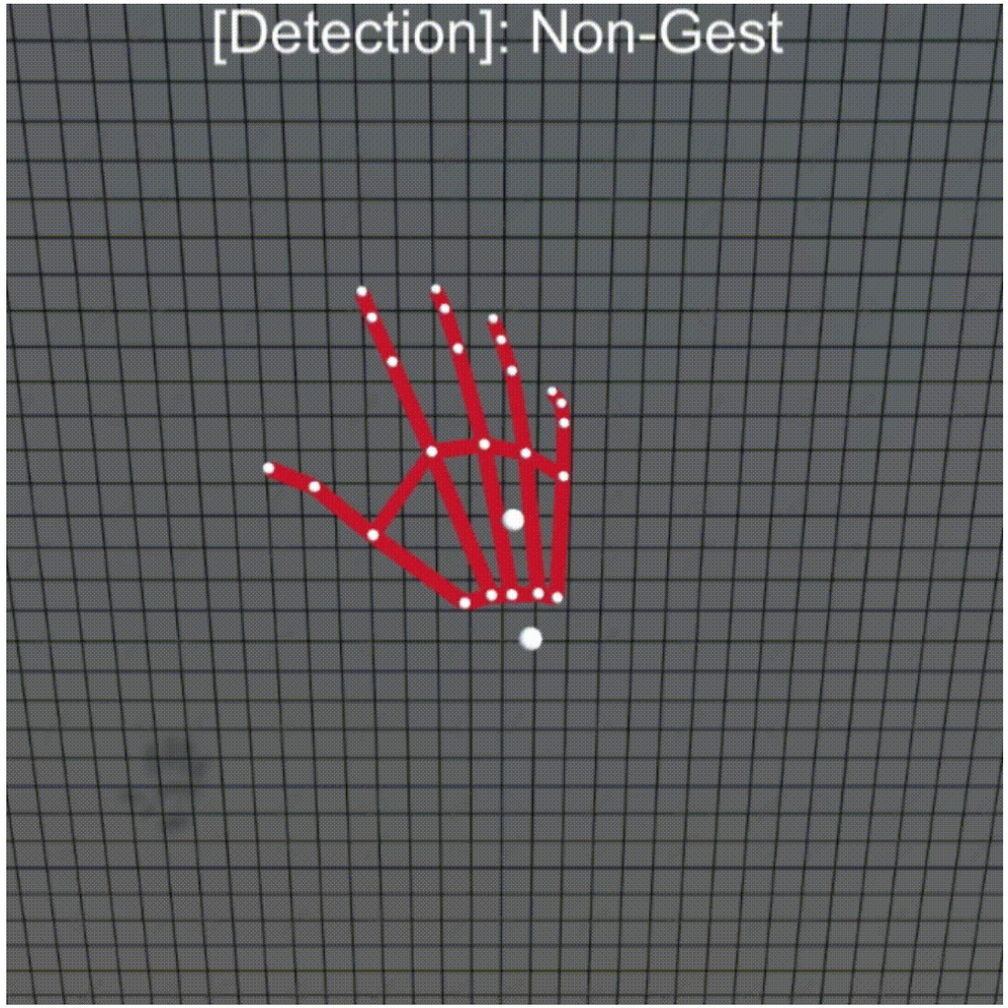
Which gestures can be used in Mixed Reality?

Gestures that we expect that could be used in interfaces must be significative, intuitive, easy to esecute, ergonomic and enogh different from any other gesture in the dictionary



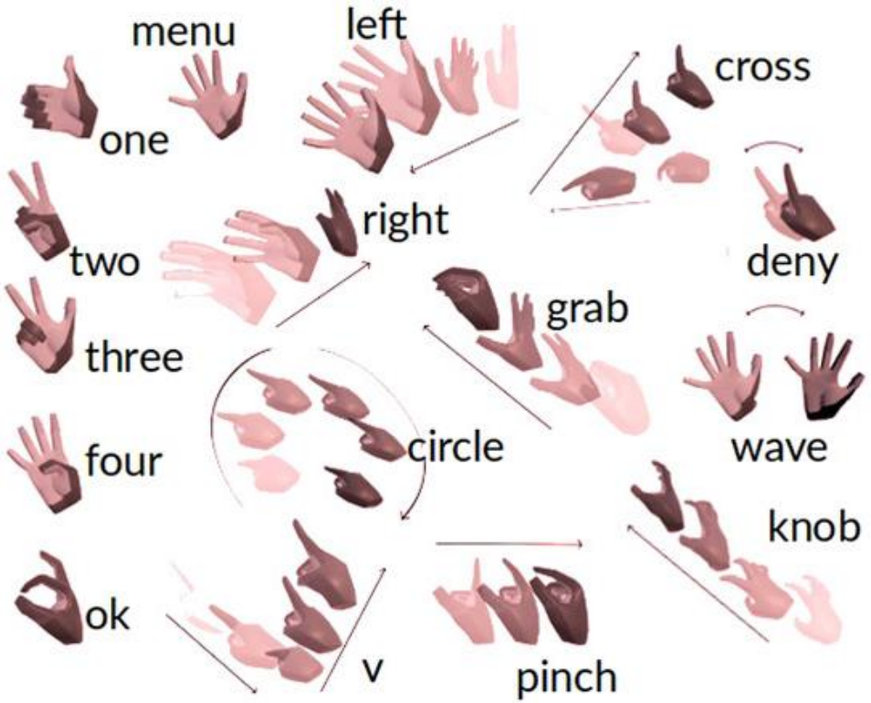
Heterogeneous dictionaries

- Functional Classification:
  - Semaphoric / Language gestures
  - Gesticulation / Non-gesture movements
- Temporal Classification:
  - Static (poses kept fixed for a certain time)
  - Dynamic (characterized by variations)
    - Coarse (gestures characterized by the hand trajectory)
    - Fine (gestures characterized by variations of hand articulation)
    - Periodic (Gestures characterized by periodic movements)



Heterogeneous dictionaries

- These gestures can have an huge difference in execution time
- How can we recognise all the categories in a single system?
- Current XR systems libraries are not supporting recognition of dictionaries composed by many gestures of different types



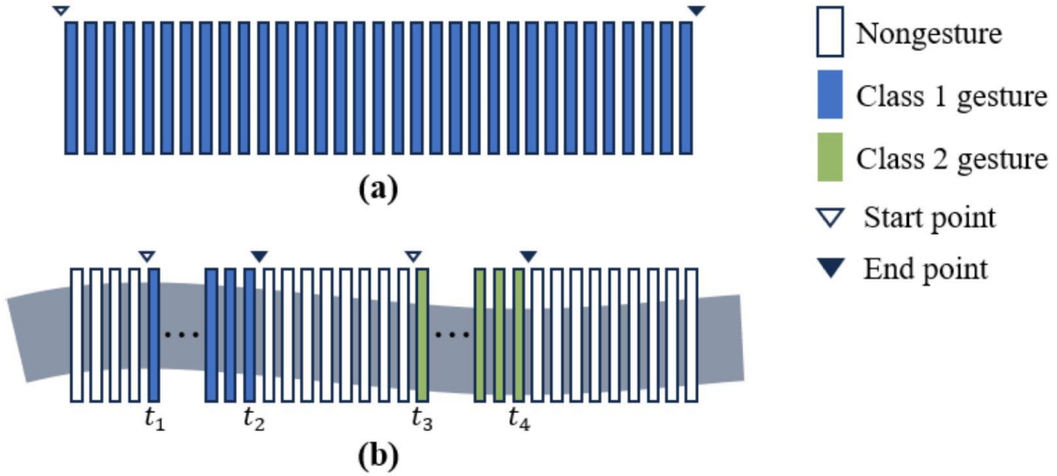
Gesture Invariance

- Position-Invariant:
  - Gesture meaning remains the same regardless of translation.
- Orientation-Invariant:
  - Gesture meaning is unchanged despite hand rotation.
- Direction-Invariant:
  - Gesture meaning is unaffected by movement direction.



Segmented VS Continous

- Segmented recognition
  - input: a trimmed clip containing exactly one gesture
  - output: a label - evaluated with classification accuracy
- Continuous / online recognition
  - input: an unbounded stream of poses, causal access only
  - output: a list of events (start, end, label) - and everything else rejected
- In an interface, gestures trigger events: only the second task is real
- Offline accuracy tells you almost nothing about online behaviour



Shrec'17 (DHG) benchmark

- Heterogeneous gesture dictionary
  - Coarse and fine dynamic gestures
- Segmented sequences (skeletons and RGBD)
- Many methods tested in research
  - 152+295 citations
- Several architectures tested
  - RNN, CNN, GCN, attention,
  - Use of handcrafted features as input
  - Very good results (accuracy >97%)
    - Is this really useful?

Gesture	Label	Tag name
Grab	Fine	G
Expand	Fine	E
Pinch	Fine	P
Rotation CW	Fine	R-CW
Rotation CCW	Fine	R-CCW
Tap	Coarse	T
Swipe Right	Coarse	S-R
Swipe Left	Coarse	S-L
Swipe Up	Coarse	S-U
Swipe Down	Coarse	S-D
Swipe X	Coarse	S-X
Swipe V	Coarse	S-V
Swipe +	Coarse	S-+
Shake	Coarse	Sh



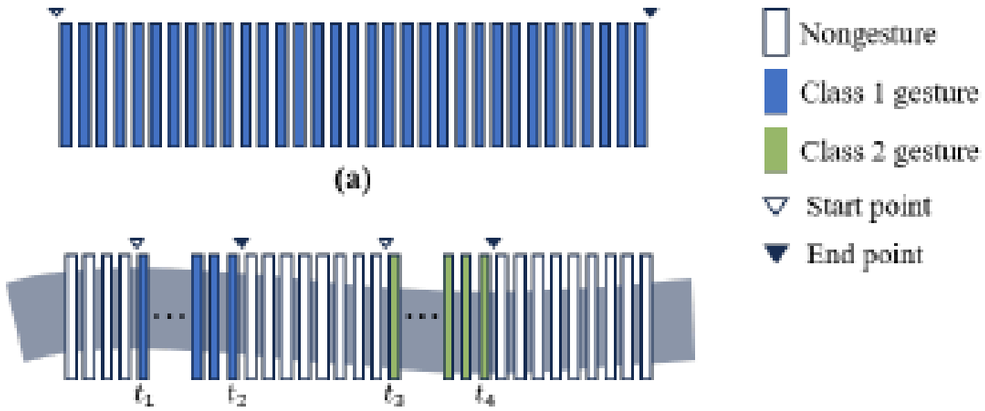
Segmented benchmarks are saturated

- Seven years of architectural work for roughly three and four points
  - SHREC'17 tells you nothing about your interface (latency, false positive, …)
- On 2,800 clips from 20 subjects:
  - no non-gesture class
  - one gesture at a time

Method on SHREC'17	Year	14 cls	28 cls
DDNet	2019	94.6%	91.9%
CTR-GCN	2021	96.1%	94.4%
STDA-GCN	2024	97.1%	95.8%
MA-GCN	**2026**	97.5%	**96.0%**
DSTSA-GCN	**2025**	**97.7%**	95.4%

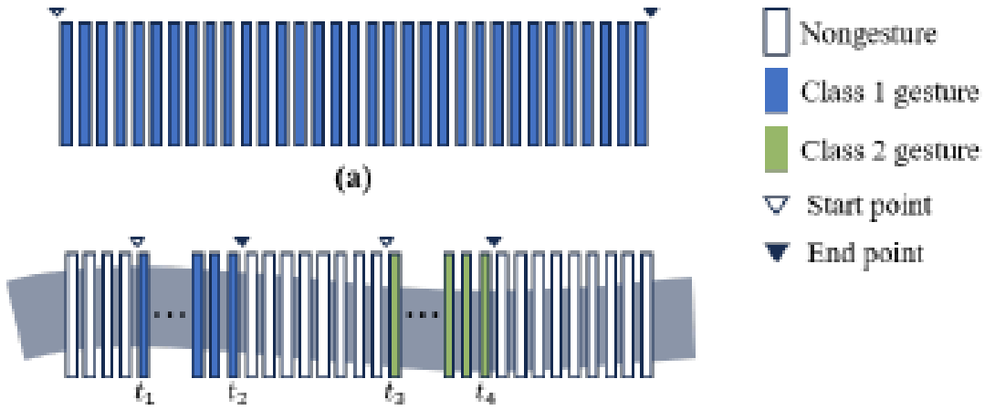
Practically useful?

- Maybe to publish a paper, but not so much for practical applications
  - Why trying to improve nearly optimal accuracy on a single benchmark, especially if the benchmark is not designed to solve the real-world problem?
- In a realistic setting gestures are not automatically segmented
  - We need to work on continuous (online) gesture recognition



Continuous recognition

- Segmented gesture recognition does not test how the recognizer could work in a practical setting:
  - Gestures are done to trigger events and the evaluation should be event-based
  - We need to identify gesture start/stop and discriminate the gesture sequences from background (non-gesture)
  - Avoid classification of background as gesture is critical
  - Difficult if gestures have varying duration



Problems in online benchmarks

- Segmentation: where exactly does the gesture start?
  - Annotations themselves disagree by several frames
- High False positives
  - the non-gesture space is huge (class imbalance) and badly sampled
- Temporal ambiguity: sub-parts are shared between classes
  - the first 10 frames of GRAB and PINCH are the same
- Online inference
  - causal, bounded latency, bounded compute

Benchmarking continuous recognition

- We ran a systematic review
  - Not too many datasets (we found 12, including ours up to 2024)
  - Most of them not using skeleton data or egocentric view
- Non standardized metrics
- Practical problems to be solved
  - Trigger actions with gestures / we must evaluate accuracy in **event** detection
    - Avoiding missing event but also duplicating events
  - Need to detect gestures with low latency / we must evaluate end-to-end delay
  - Need to correctly classify gestures
  - Need to avoid false positives / we must introduce a specific metric

Emporio et al., Continuous Hand Gesture Recognition: Benchmarks and Methods (2025)

Existing benchmarks

- Not too many

Table 2
Characteristics of the gesture dictionaries used in the reviewed benchmarks: types of gestures (S = static, DC = dynamic coarse, DF = dynamic fine, DP = dynamic periodic) and invariance (P = position, O = hand orientation).

Name	Year	Type of gestures	Invariance	Application domain
ChAirGest	2013	DC, DP	P	General interaction
Montalbano T3	2014	DF, DP	P	Verbal communication
ConGD	2016	S, DF, DC, DP	P	Many domains
NVGesture	2016	S, DF, DC, DP	P	Automotive
EgoGesture	2018	S, DF, DC, DP	P	Mixed reality
SHREC'19	2019	DC	P	Mixed reality
IPN Hand	2020	S, DF, DC	P, O	Touchless screens
MlGesture	2020	DC,DP	P, O	Automotive
LD-ConGR	2022	S, DC	P, O	Long distance recognition
SHREC'22	2022	S, DF, DC, DP	P	Mixed reality
ODHG 14/28	2022	DF, DC	P, O	General interaction
ZJUGesture	2023	DF, DC	P, O	Mobile

Emporio et al., Continuous Hand Gesture Recognition: Benchmarks and Methods (2025)

Existing benchmarks

- Not too many

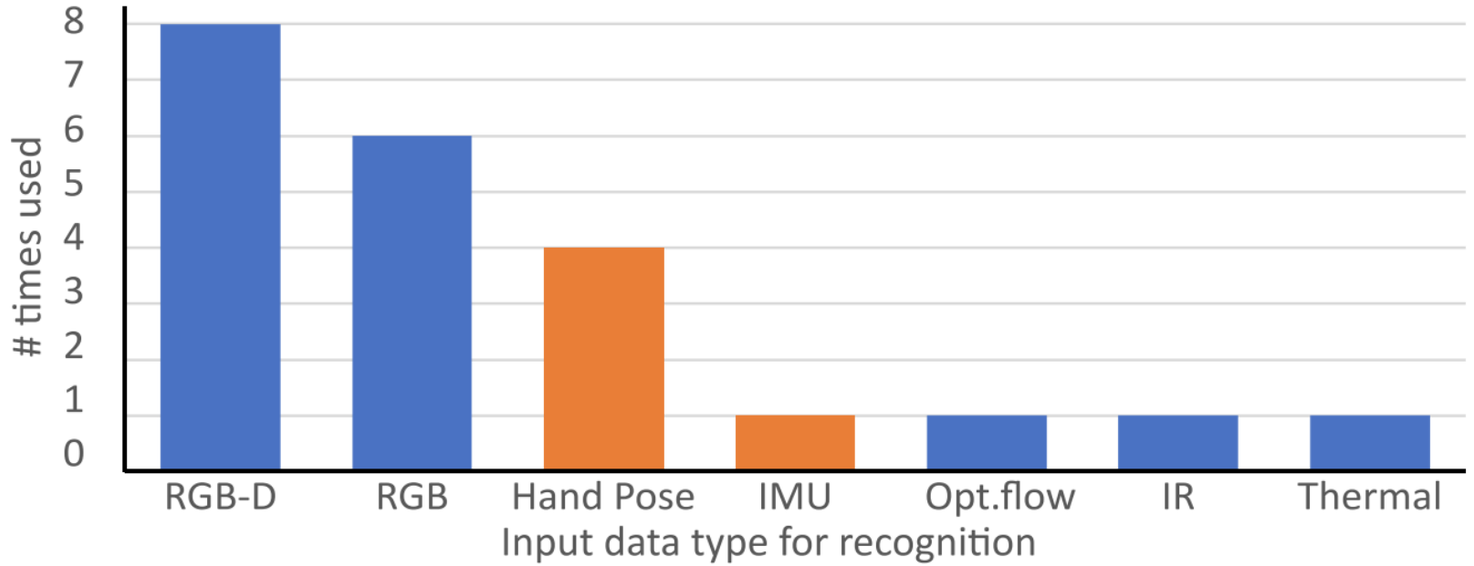
Table 3
Characteristics of the acquisition setups for the datasets included in the reviewed benchmarks.

Name	Data types	View	Input devices	FPS
ChAirGest	RGB, RGB-D, Hand pose, IMU	EGO, EXO	Kinect/Xsens MTw IMU	30/50
Montalbano T3	RGBD	EXO	Kinect	20
ConGD	RGB-D	EXO	RGB-D Camera Setup	NA
NVGesture	RGB-D, Optical flow, IR	EXO	SoftKinetic DS325/DUO 3D	30
EgoGesture	RGB-D	EGO	Intel RealSense SR300	30
SHREC'19	Hand pose	EGO	Leap Motion	NA
IPN Hand	RGB	EXO	Webcam	30
MlGesture	RGB-D, RGB, Thermal	EXO	Industrial cameras	8/12/16
LD-ConGR	RGB-D, RGB	EXO	Kinect v4	30
SHREC'22	Hand pose	EGO	Hololens 2	20
ODHG 14/28	RGB-D, Hand pose	EXO	Intel Real-Sense	30
ZJUGesture	RGB	EXO	Mobile phone	30

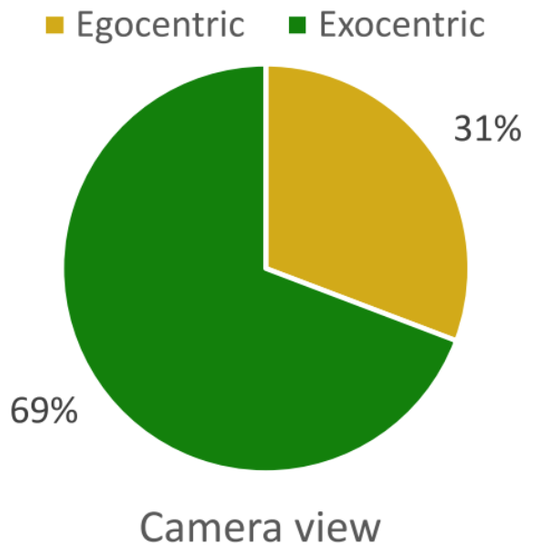
Emporio et al., Continuous Hand Gesture Recognition: Benchmarks and Methods (2025)

Existing benchmarks

- And even less using hand pose data, XR setup



(a)

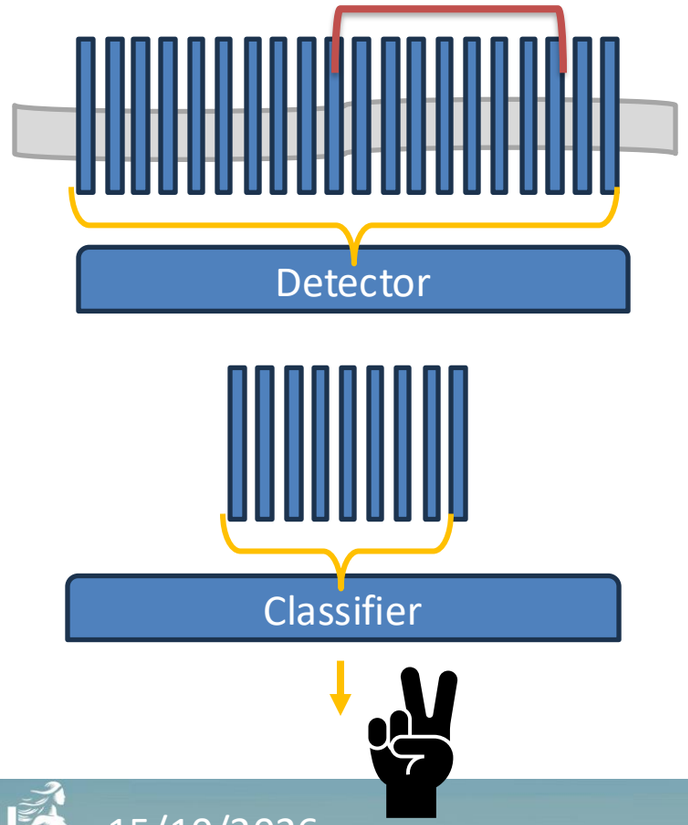


(b)

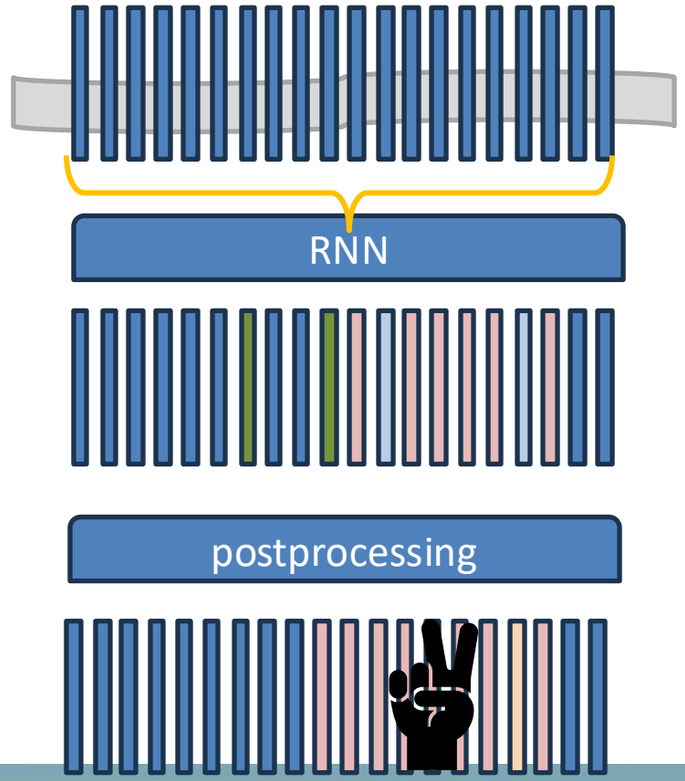
Detection methods in Continuous

- Three strategies

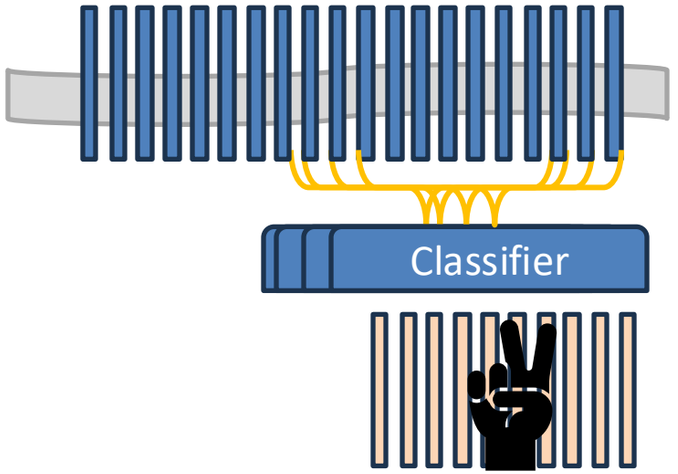
Detection/classification modules



Recurrent network + postprocessing



Sliding windows



Detection+classification

- Pros
  - Segmented gesture recognition based on recent networks works well
  - Gestures can be resampled, making recognizer not affected by varying duration and speed
- Cons
  - Difficult to have a good detector for «heterogeneous» gestures
    - Usually based on «energy»
    - Supposing that gesture starts/end can be characterized by high speed change
    - Not working well for many gesture types
  - A large support for detection may cause a relevant latency in the response

Recurrent networks

- Continuous output labels/probabilities
- Per frame classification
- Converted in events with simple aggregation of postprocessing (filtering)
- Example: uDeepGRU, applied successfully to the SHREC'19 benchmark

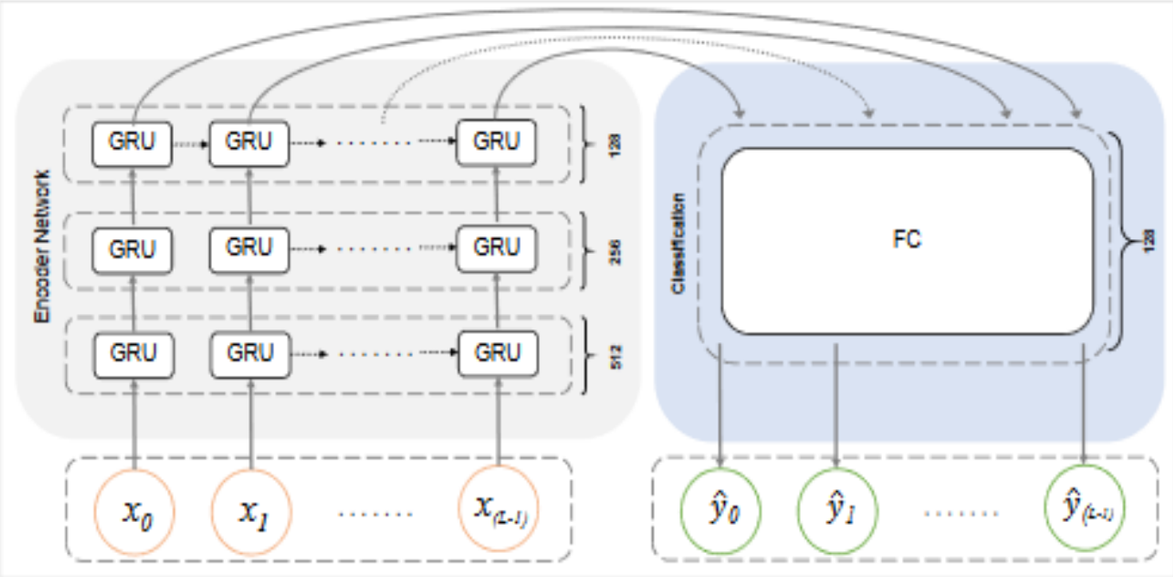


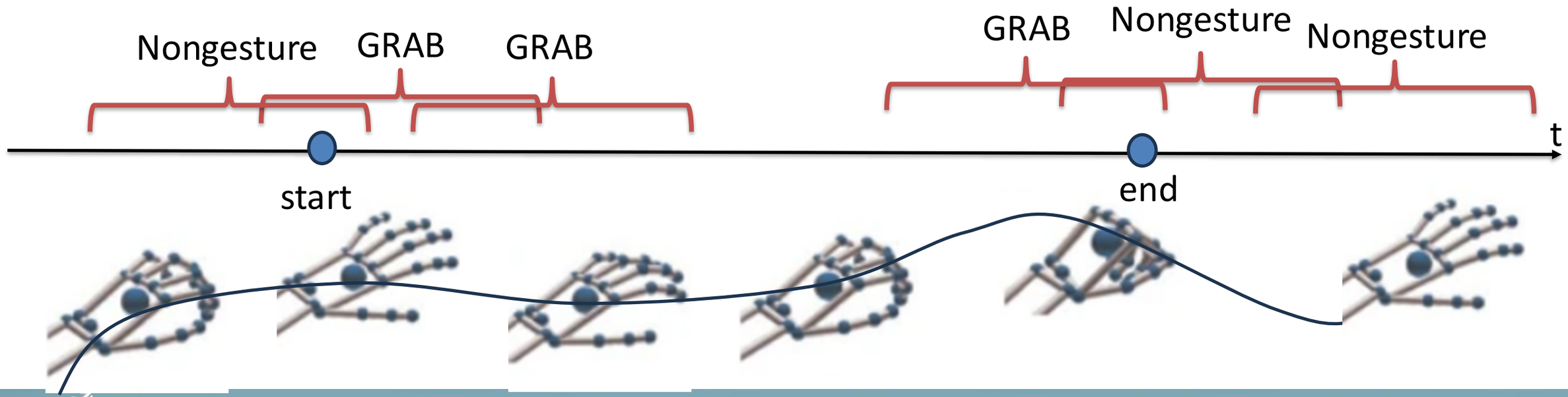
Figure 4: *uDeepGRU architecture which consists of an encoder network and a classification subnetwork.*

Sliding windows

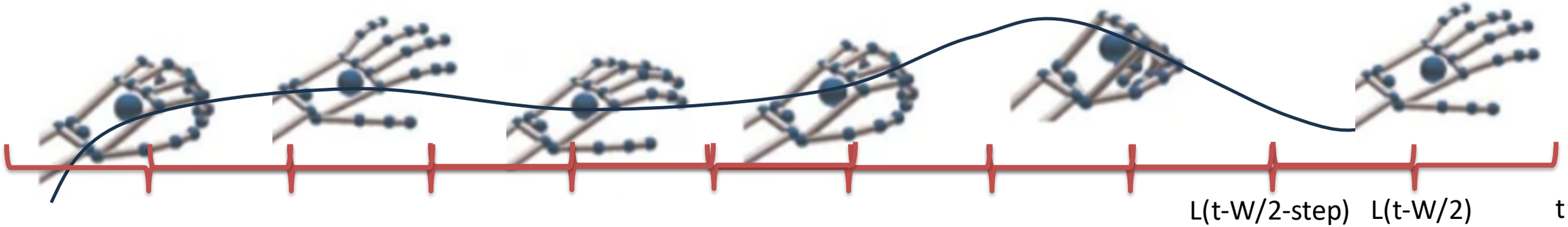
- Idea: use just one module trained to recognize gesture «windows» of fixed size
- We can use different classifiers, all those applied in the segmented setting are usable
  - Just training them on windows
- Training set creation:
  - Extract windows from train sequence
  - assign label (e.g., according to the majority of frame labels)
    - Including a non-gesture class!
- Train windows classifier
- At runtime, sample windows and assign

Sliding windows - training

- Assign label to partial gesture executions, and learn to recognize them
  - Issue 1: window size? May depend on gesture length/type
  - Issue 2: window characterization? Gesture parts can be included in other gestures/nongestures
    - Some issues can be addressed with postprocessing or with dictionary design.



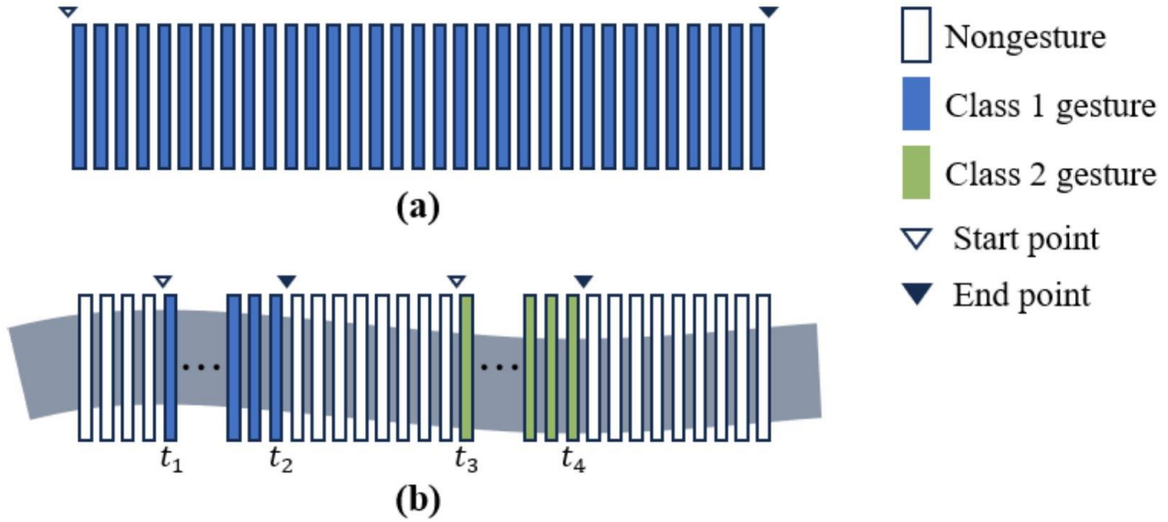
Sliding windows-runtime



- With a delay of half window size we have a window label
  - We can detect gesture start at the first «gesture» label and end at the first different (nongesture) label
  - Postprocessing may be applied to reduce false detection (e.g., gesture length check)
    - Adding delay

Evaluation metrics

- **Frame-Based** : treat gesture as labels **sequences** for comparison
- **Event-Based**: treat gestures as **events** (start time, end time, label)
- **Efficiency-Based**: memory requirements, computational complexity, real-time performance



Frame-based vs event based evaluation

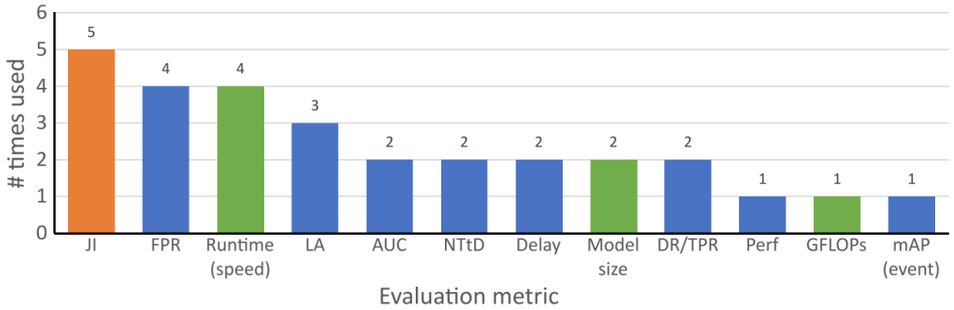
- Frame based: assign a gesture (non gesture) label to each time sample of the sequence
  - Usual metric for time sequence: Jaccard Index
    - Used in many continuous time gesture detection challenges/benchmarks
    - But does not capture all the problems (e.g., repeated detections, short gestures missing)

- Event-based: detect event, assigning starting (end) time and label
  - Critical aspect: they need to define criteria for «gesture detection», typically parametric
    - For example a gesture is considered correctly detected if the system detects it within a certain distance in time from its execution

Important event-based metrics

- **Detection rate**
  - number of correctly detected gestures of the class and the corresponding ground truth number
- **False positives' score**
  - ratio between the number of gesture predictions not corresponding to real ones and the total number of gestures of the same class included in the test set
- **Delay, normalized time to detect**
  - Absolute or relative (respect to the duration) delay in detection

- Note: many continuous benchmarks use **Jaccard Index** (JI), e.g. the average relative overlap between the ground truth and the predicted annotations of frames belonging to each gesture class in the sequences, that is a frame-based metric



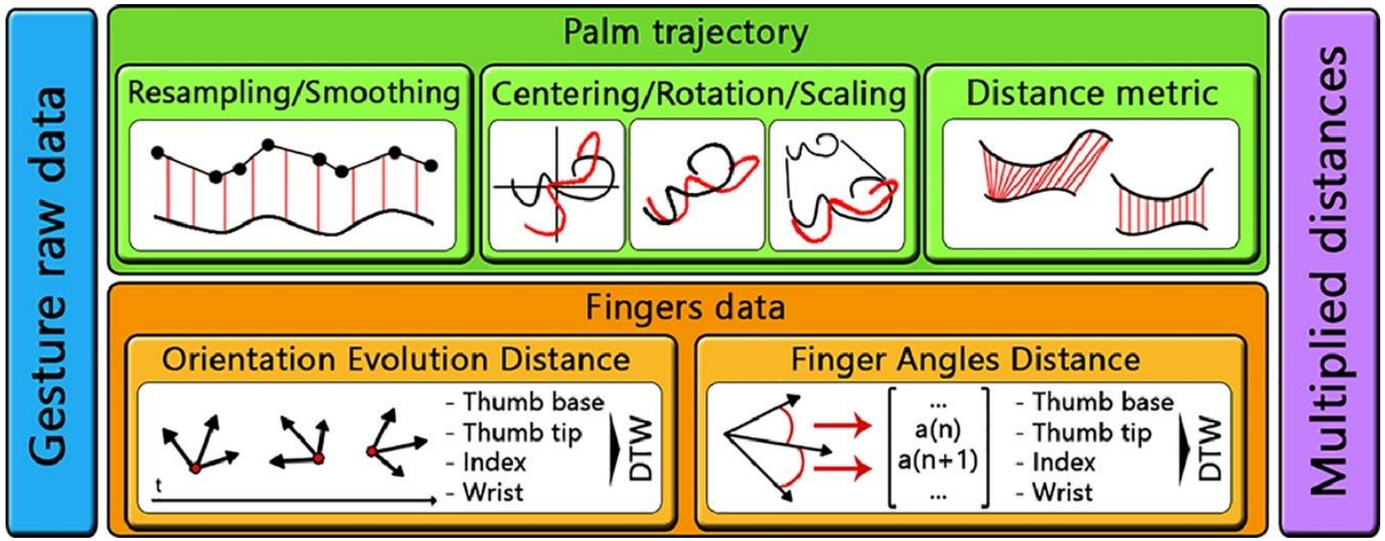
What are the best models to use?

TEMPLATE MATCHING	RECURRENT LSTM / GRU	TEMPORAL CONVOLUTION	GRAPH ST-GCN	ATTENTION + SSM
3-Cent *no training*	*DeepGRU, DLSTM* *native streaming*	*DDNet, causal TCN* *fast, small*	*STA-GCN, TD-GCN* *anatomy as prior*	*DSTA, transformers* *Mamba (2025–26)*

Non-neural approach – Template matching

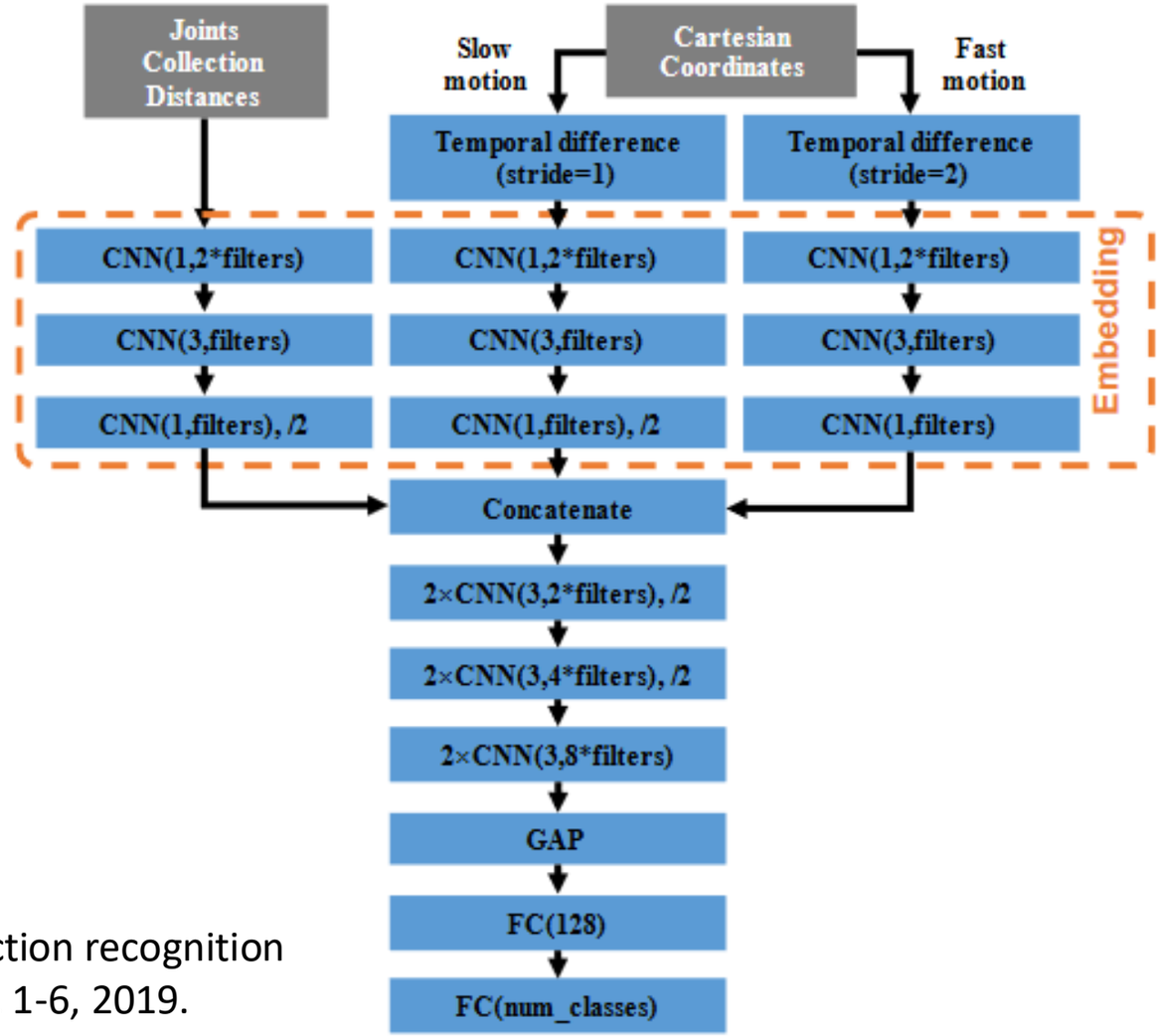
- 3-Cent classifier (Caputo et al. 2018):
  - Resample segmented gestures
  - Compare with themplates
    - Trajectory distances
    - Orientation/articulation distances
      - With dynamic time warping

90% accuracy on Shrec 17



Convolutional network

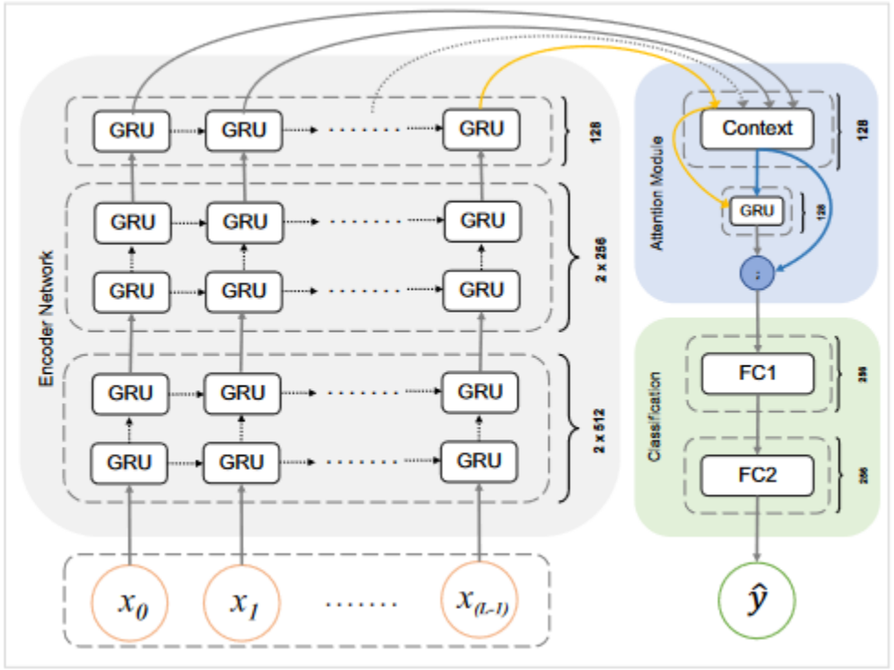
- Double-Motion/Double Feature (DDNet), Yang et al. 2019
- Input are not raw coordinates, but handcrafted features
  - Motion Fast/slow (temporal differences of coords)
  - Joint collection Distances (JCD)
- 94.6% accuracy



F. Yang, Y. Wu, S. Sakti and S. Nakamura, "Make skeleton-based action recognition model smaller faster and better", *Proc. ACM Multimedia Asia*, pp. 1-6, 2019.

Recurrent networks

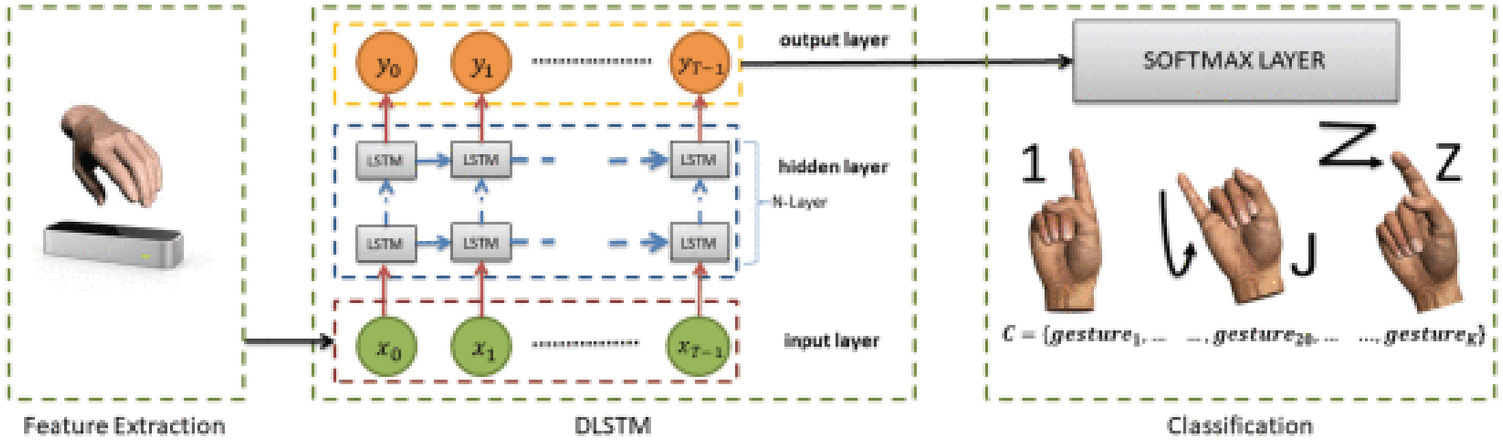
- DeepGRU (Maghoumi et al., 2018)
- Raw input data (skeleton coordinates)
- Deep GRU architecture+attention module
  - 94.5% accuracy on SHREC'17

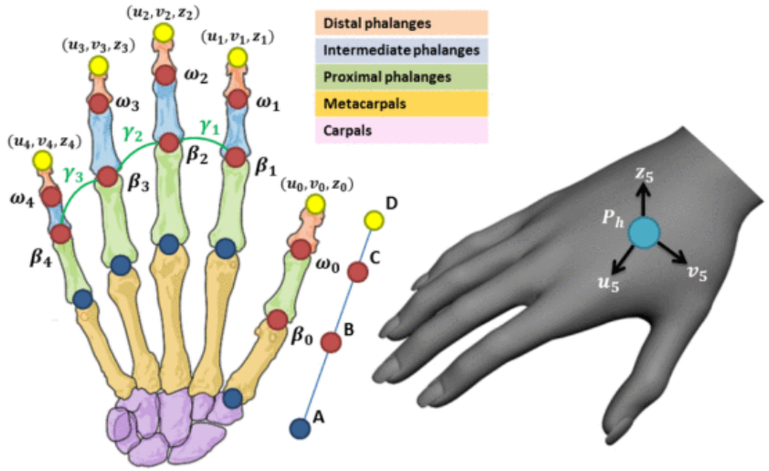


Maghoumi, M., & LaViola, J. J. (2019). DeepGRU: Deep gesture recognition utility. In *Advances in Visual Computing: 14th International Symposium on Visual Computing, ISVC 2019*

Recurrent networks

- DLSTM (Avola et al. 2019)
  - Huge set of features (handcrafted)
    - Internal angles
    - Fingertip, palm positions
    - Intra-finger angles
  - Temporal filter (Savitzky-Golay)
  - Deep LSTM architecture
    - 97.6% accuracy

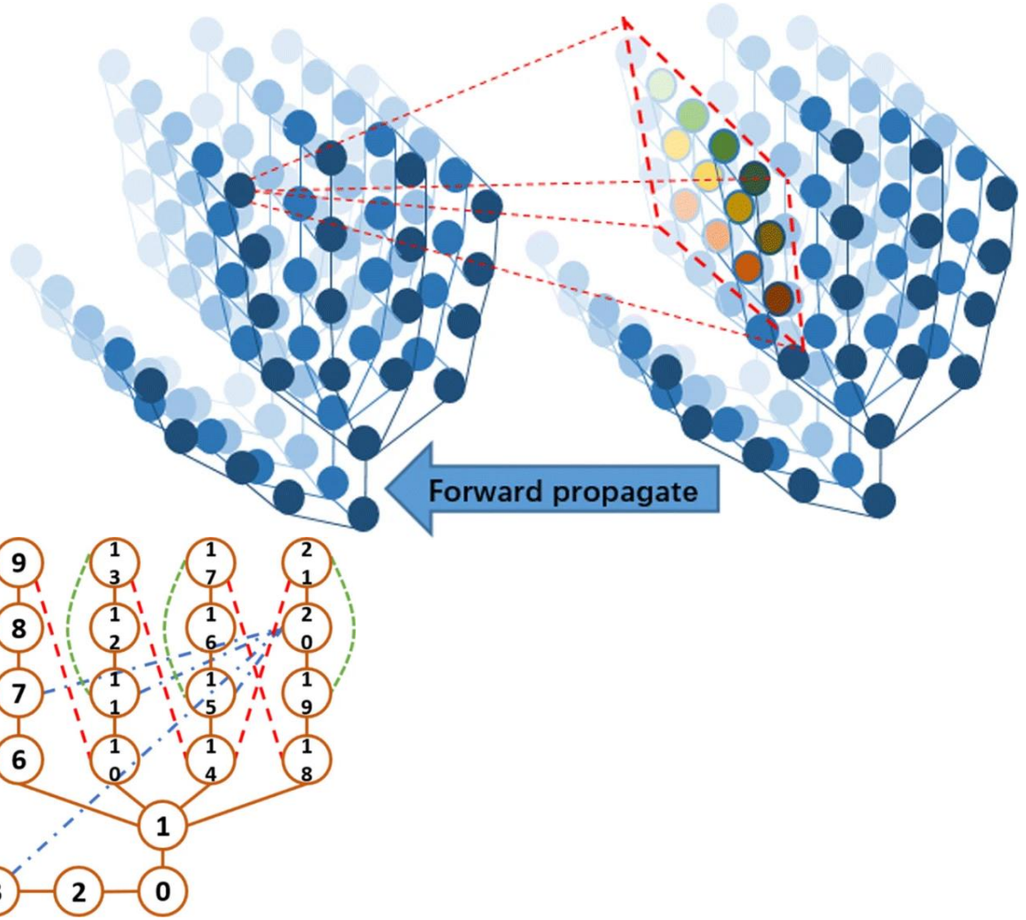




D. Avola, M. Bernardi, L. Cinque, G. L. Foresti, and C. Massaroni. Exploiting recurrent neural networks and leap motion controller for the recognition of sign language and semaphoric hand gestures. IEEE Transactions on Multimedia

Graph networks

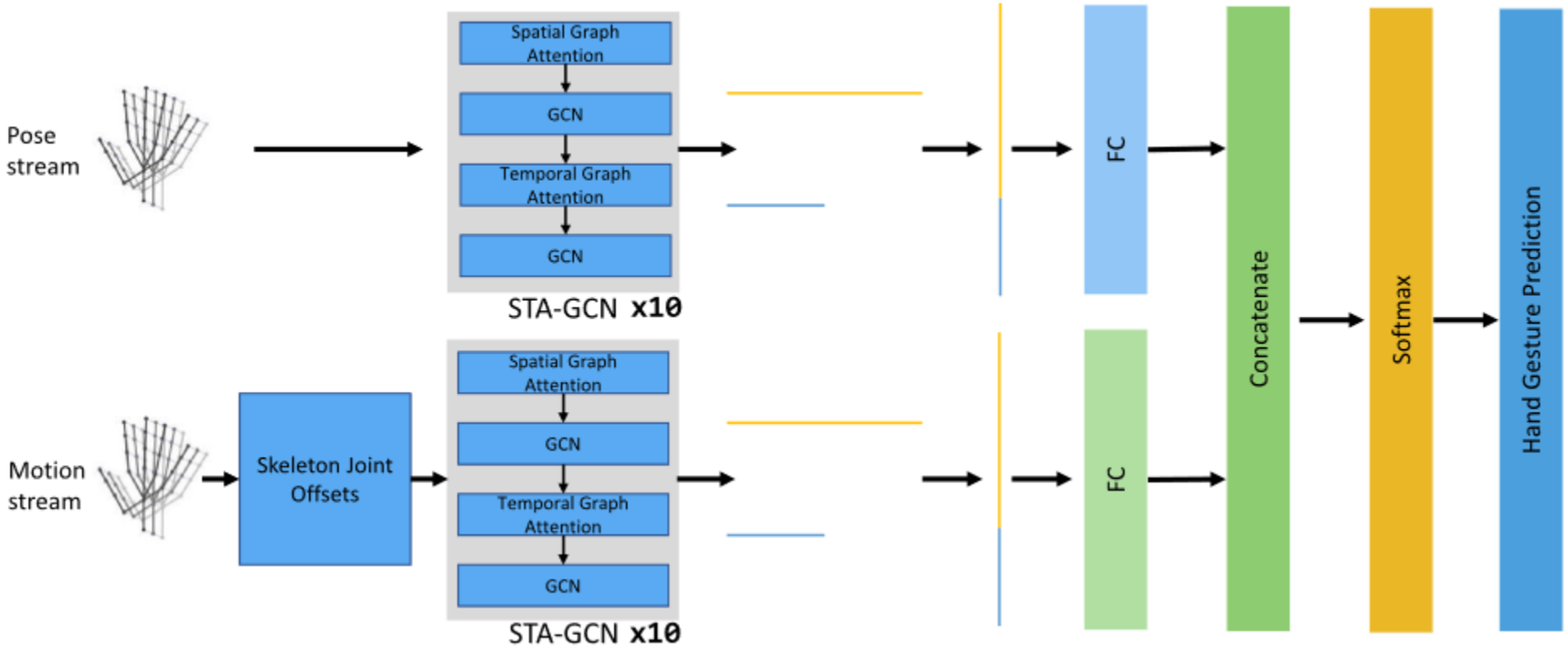
- Adaption of techniques used for body action recognition (ST-GCN Yan et al. 2018)
- Spatio-temporal network structure
- Raw data input
  - But tested 3D+2D, spherical coordinates
  - 92,6 accuracy (efficient)



LI, Yong, et al. Spatial temporal graph convolutional networks for skeleton-based dynamic hand gesture recognition. *EURASIP Journal on Image and Video Processing*, 2019, 2019: 1-7.

Improved graph networks

- Multiple streams (pose, velocity)
- Attention module
- Accuracy 95.4



Zhang, Wei, et al. "Sta-gcn: two-stream graph convolutional network with spatial–temporal attention for hand gesture recognition." *The Visual Computer* 36 (2020): 2433-2444.

Features extraction

- Raw coordinates are rarely what goes into the network
- JCD -Joint Collection of Distances
  - all pairwise Euclidean distances between joints
  - invariant to global position and orientation
- M_slow / M_fast – joint velocities at one- and two-frame strides

$$S^k = \{J_1^k, J_2^k, \ldots, J_N^k\}.$$

$$JCD^k = \begin{bmatrix} \left\|\overrightarrow{J_2^k J_1^k}\right\|_2 & & & \\ \vdots & \ddots & & \\ \vdots & \cdots & \ddots & \\ \left\|\overrightarrow{J_N^k J_1^k}\right\|_2 & \cdots & \cdots & \left\|\overrightarrow{J_N^k J_{N-1}^k}\right\|_2 \end{bmatrix}; \quad (1)$$

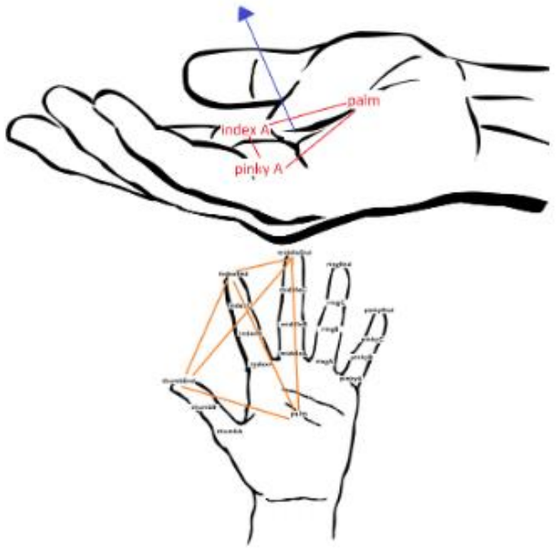
where $\left\|\overrightarrow{J_i^k J_j^k}\right\|_2 (i \neq j)$ denotes the Euclidean distance between J_i^k and J_j^k.

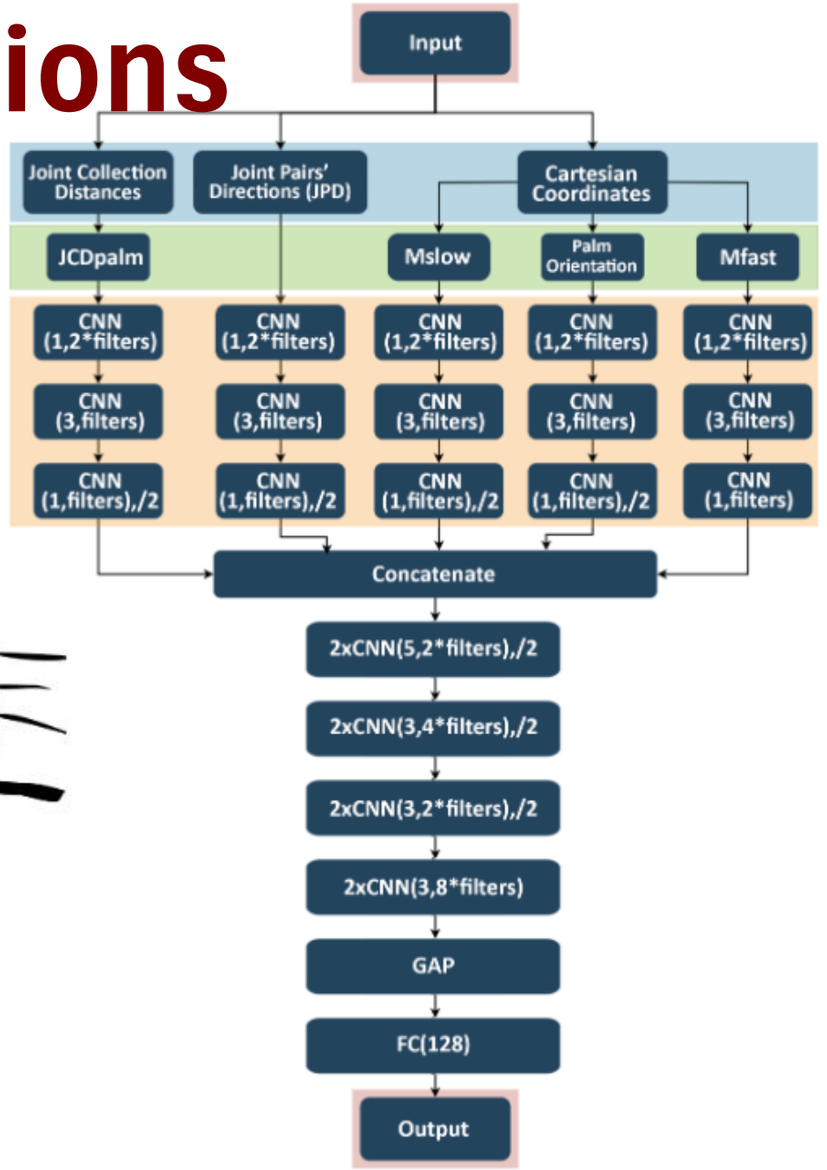
$$M_{slow}^k = S^{k+1} - S^k \text{ for } k \in \{1, 2, 3, \ldots, K-1\};$$

$$M_{fast}^k = S^{k+2} - S^k \text{ for } k \in \{1, 3, \ldots, K-2\};$$

DDNet modifications

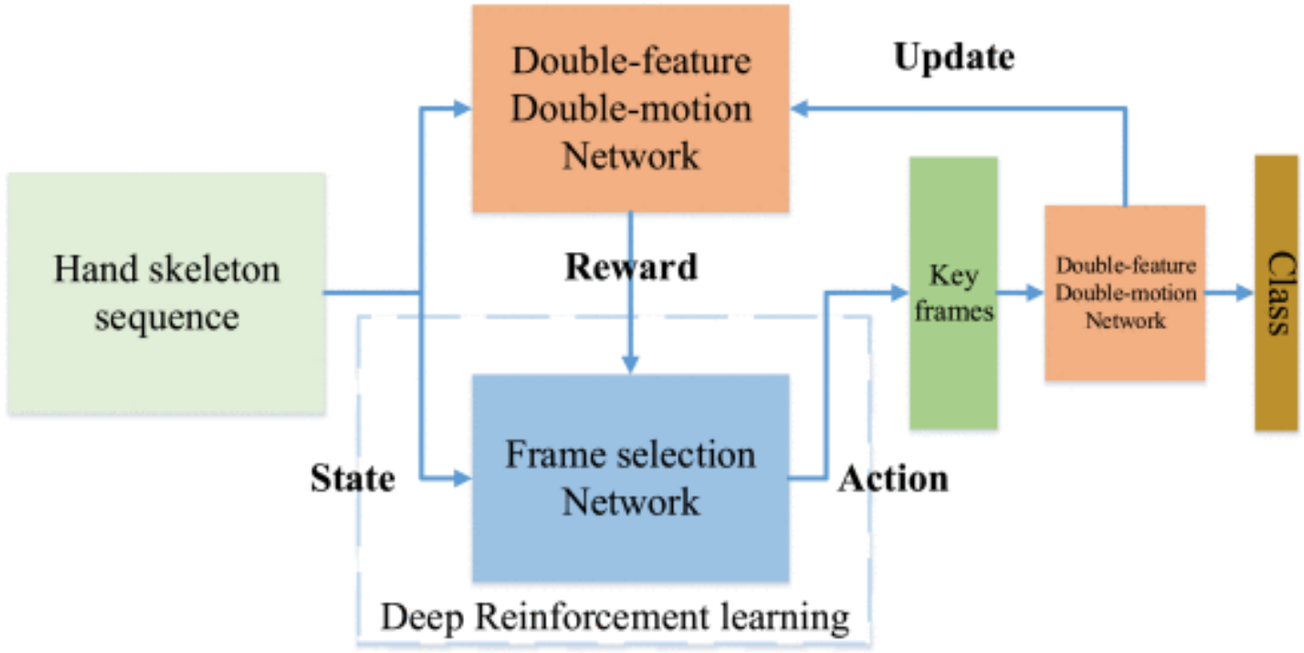
- STRONGER (Emporio et al. 2021)
  - Added features
    - Joint pairs direction
    - Palm orientation evolution
    - 95% accuracy on shrec'17
    - 91% on shrec'21





DDNet+RL

- DDNet coupled with reinforcement learning
  - Used for keyframe selection
  - 96.5% accuracy



Gan, Minggang, et al. "Keyframe Selection Via Deep Reinforcement Learning for Skeleton-Based Gesture Recognition." *IEEE Robotics and Automation Letters* (2023).

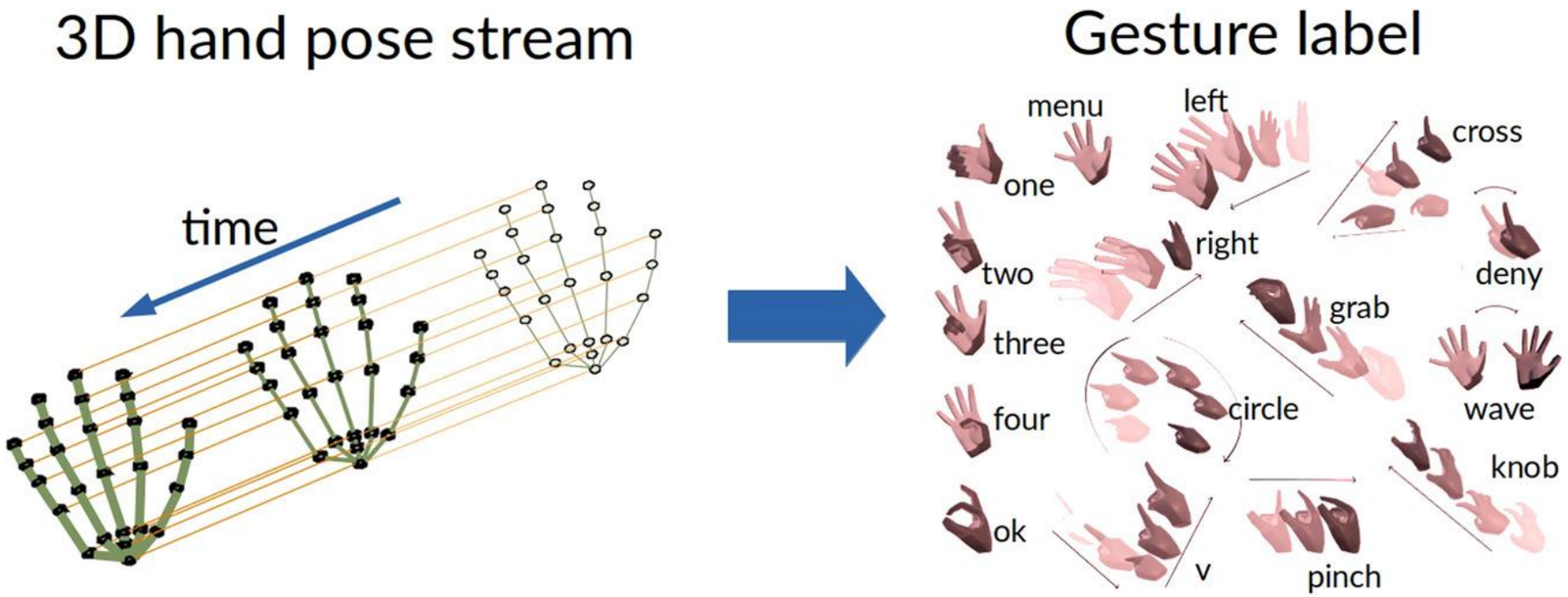
Benchmarks proposed at UniVR

- SHREC'19 Track: Online Gesture Recognition
  - Captured with Leapmotion
  - Only dynamic coarse gestures
- SHREC'21 Track: Skeleton-based Hand Gesture Recognition in the Wild
  - Captured with Hololens 2
  - Heterogeneous gestures/egocentric view
  - Some issues in the dictionary/labeling
- SHREC'22 Track: online detection of heterogeneous gestures
  - Captured with Hololens 2
  - Heterogeneous gestures/egocentric view

All related to contests organized within the Eurographics SHREC initiative, presented at 3DOR conferences

SHREC'22

- Benchmark to evaluate online skeleton based detection of heterogeneous gestures



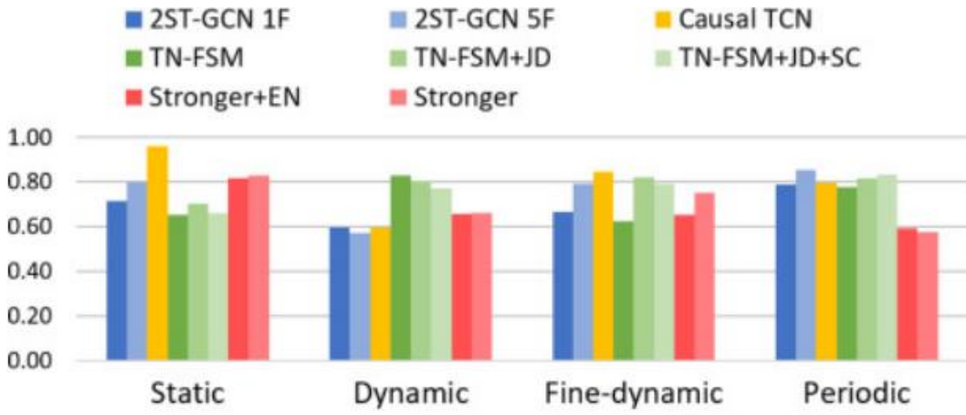
	ONE	TWO	THREE	FOUR	OK	MENU	LEFT	RIGHT	CIRCLE	V	CROSS	GRAB	PINCH	DENY	WAVE	KNOB
Min length	21	24	15	23	22	20	11	11	25	17	24	19	22	31	27	37
Max length	72	59	71	69	52	59	37	32	64	37	50	70	61	81	73	79
Avg. length	**34.1**	**37.4**	**38.6**	**37.3**	**34.0**	**33.0**	**19.6**	**18.6**	**45.4**	**27.2**	**37.5**	**39.9**	**36.2**	**47.8**	**45.4**	**54.7**

Table 1: Minimum, maximum and average lengths of the gestures (background colors indicate gestures' types.

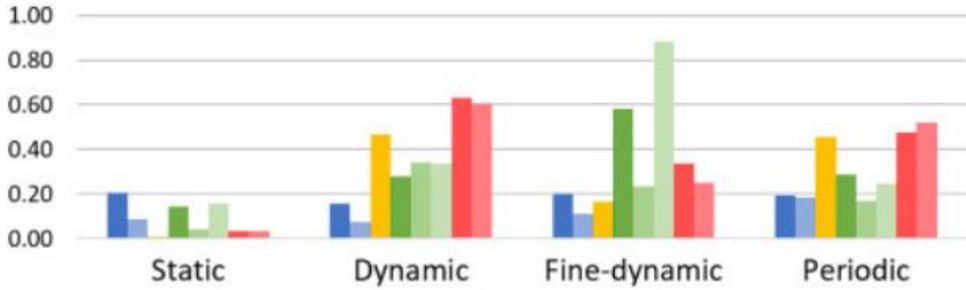
SHREC'22 contest results

- Methods are fast
- Methods are not so accurate
- False positives are many
  - Not acceptable: issue of many unwanted commands
- Sliding windows work well

Group	Method	DR	FP	JI	Delay (fr.)	Time (ms)
B.Line	Stronger + EN	0.70	0.34	0.56	16.40	100
	Stronger	0.72	0.33	0.59	14.79	100
G1	2ST-GCN 1F	0.68	0.19	0.52	12.55	**2.1**
	2ST-GCN 5F	0.74	**0.10**	0.61	13.28	**2.1**
G2	Causal TCN	**0.80**	0.26	**0.68**	19.00	28
G3	TN-FSM	0.73	0.34	0.27	**10.00**	4.65
	TN-FSM + JD	0.77	0.18	0.63	**10.00**	4.65
	TN-FSM + JD + SC	0.74	0.32	0.56	**10.00**	4.65



(a) Detection rate

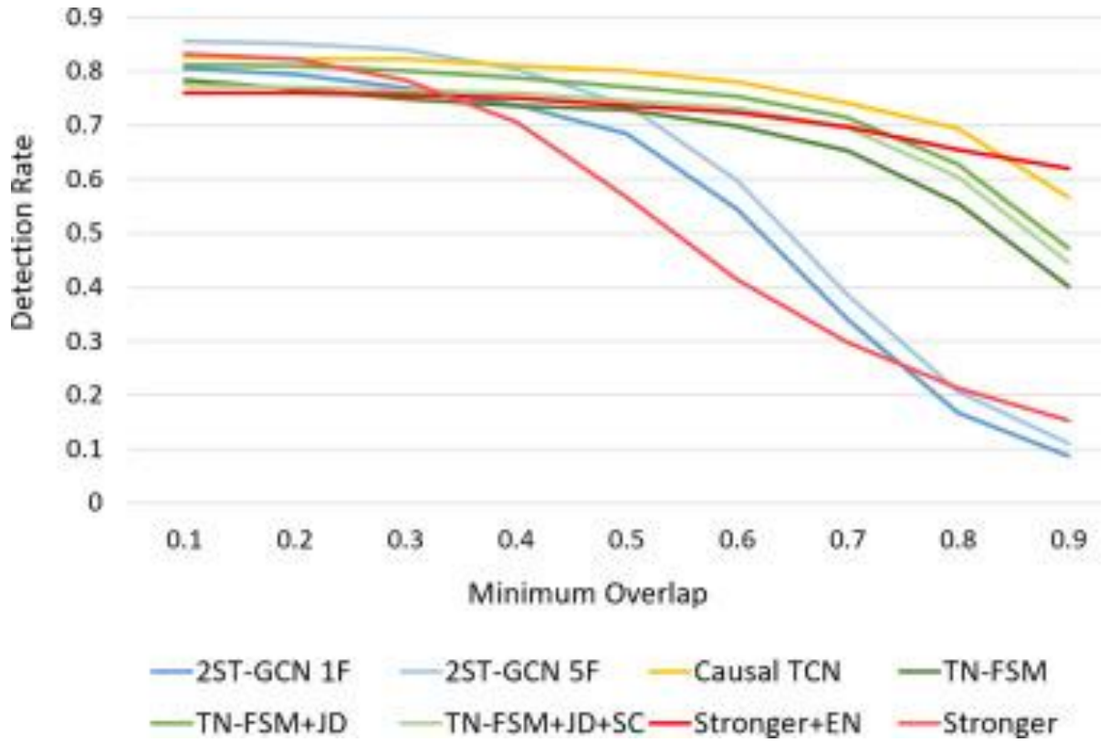


(b) False positives

Issue in continuous evaluation

- DR depends on a parameter
- Methods ranking can change if the threshold is changed

- The family matters far less than the literature suggests. What changes the result is the training protocol and the rejection rules.



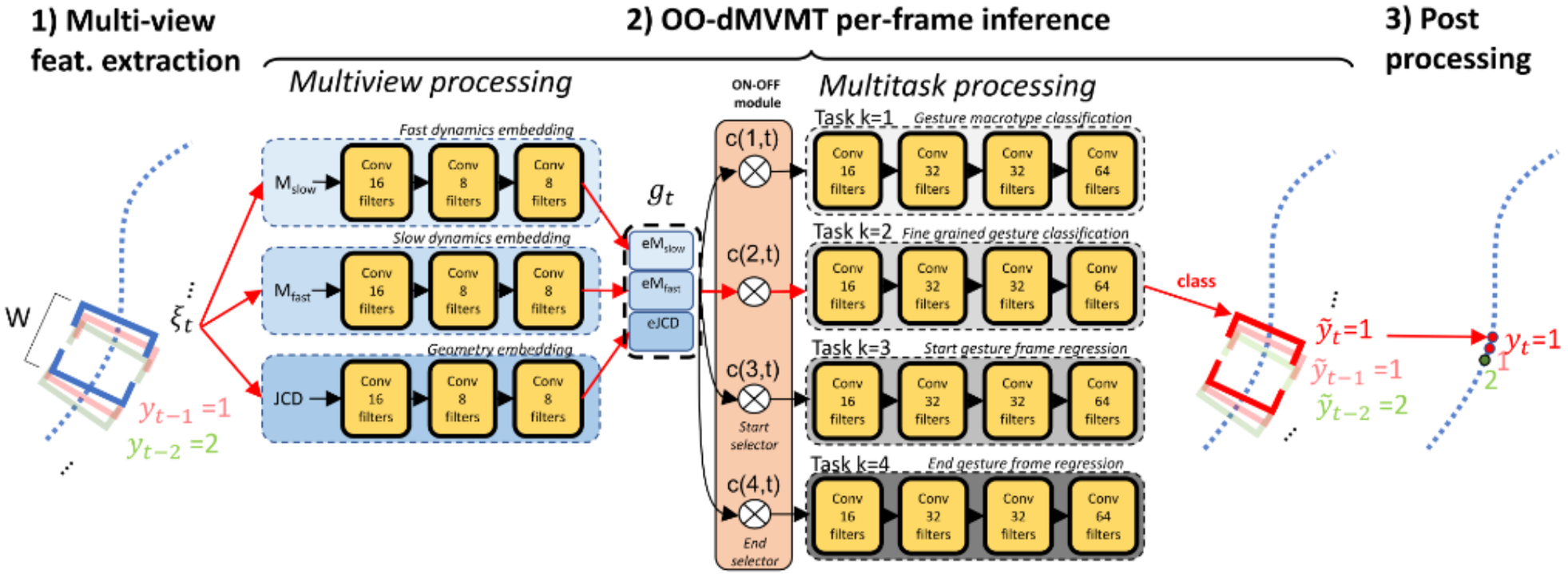
Testing different networks

- We trained several networks used in the segmented task for the continuous one using the sliding window approach. DDNet was the best and we improved it

Method	DR ↑	FP ↓	JI ↑	Delay(fr.) ↓	time(ms) ↓	SV	MV	ST	MT
DeepGRU [24] (2018)	0.26 *(.14)*	0.25 *(.23)*	0.21 *(.09)*	8.00	3.1	✓		✓	
DG-STA [8] (2019)	0.51 *(.10)*	0.32 *(.20)*	0.40 *(.20)*	8.00	4.2	✓		✓	
SeS-GCN [25] (2022)	0.60 *(.13)*	0.16 *(.09)*	0.53 *(.13)*	8.00	1.8	✓		✓	
PSUMNET [29] (2022)	0.62 *(.14)*	0.24 *(.15)*	0.52 *(.15)*	8.00	24.4		✓	✓	
MS-G3D [22] (2020)	0.68 *(.11)*	0.21 *(.15)*	0.57 *(.14)*	8.00	29.3	✓		✓	
Stronger [13] (2022)	0.72 *(.11)*	0.34 *(.26)*	0.59 *(.18)*	14.79	100.0		✓	✓	
DSTA [27] (2020)	0.73 *(.07)*	0.24 *(.13)*	0.61 *(.12)*	8.00	9.2	✓		✓	
2ST-GCN 5F [13] (2022)	0.74 *(.12)*	0.23 *(.05)*	0.61 *(.11)*	13.28	2.1	✓			✓
TN-FSM+JD [13] (2022)	0.77 *(.06)*	0.23 *(.12)*	0.63 *(.03)*	10.00	4.6		✓	✓	
Causal TCN [13] (2022)	0.80 *(.15)*	0.29 *(.22)*	0.68 *(.24)*	19.00	28.0	✓		✓	
DDNet [31] (2019)	0.88 *(.06)*	0.16 *(.18)*	0.78 *(.14)*	8.00	2.2		✓	✓	
FG + SDN	0.86 *(.06)*	0.17 *(.06)*	0.75 *(.11)*	8.00	4.1		✓		✓
FG + SDN + GC	0.90 *(.11)*	0.10 *(.11)*	0.83 *(.13)*	8.00	4.0		✓		✓
OO-dMVMT	**0.92** *(.06)*	**0.09** *(.09)*	**0.85** *(.11)*	8.00	4.1		✓		✓

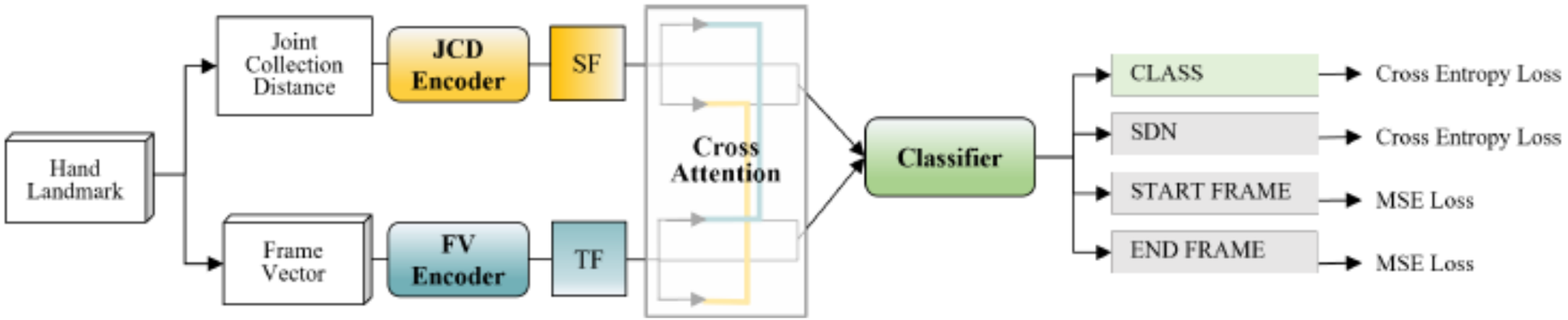
Improving the method

- Multi-task learning can improve the network accuracy
  - Idea: windows annotation also include start/end annotations (and also gesture categorization). We can use multi-task learning with specific losses



Improved results

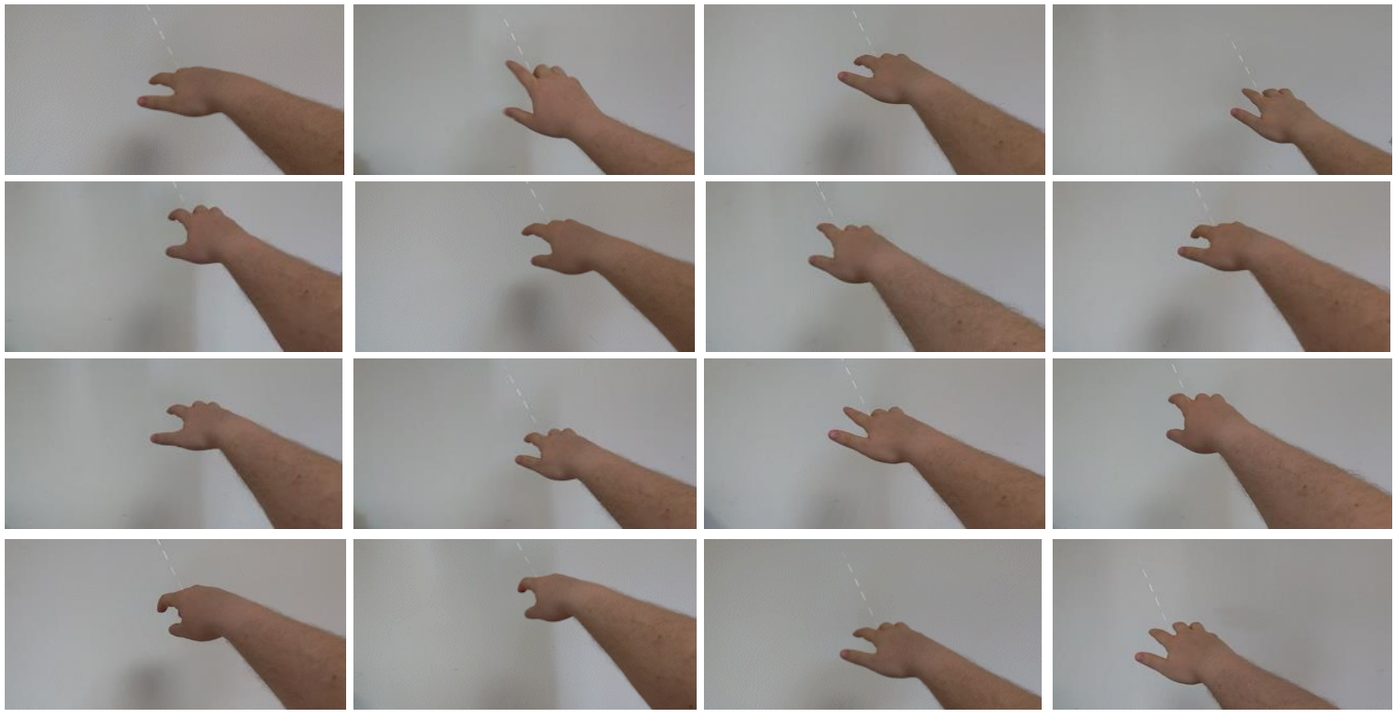
- Attention seems to help



Method	DR↑	FP↓	JI↑	Delay (fr.)	Time (ms)
DeepGRU [12] (2018)	0.26 (0.14)	0.25 (0.23)	0.21 (0.09)	8.00	3.1
DG-STA [15] (2019)	0.51 (0.10)	0.32 (0.20)	0.40 (0.20)	8.00	4.2
SeS-GCN [23] (2022)	0.60 (0.13)	0.16 (0.09)	0.53 (0.13)	8.00	1.8
PSUMNET [24] (2022)	0.62 (0.14)	0.24 (0.15)	0.52 (0.15)	8.00	24.4
MS-G3D [25] (2020)	0.68 (0.11)	0.21 (0.15)	0.57 (0.14)	8.00	29.3
Stronger [13] (2022)	0.72 (0.11)	0.34 (0.26)	0.59 (0.18)	14.79	100.0
DSTA [16] (2020)	0.73 (0.07)	0.24 (0.13)	0.61 (0.12)	8.00	9.2
2ST-GCN 5F [13] (2022)	0.74 (0.12)	0.23 (0.05)	0.61 (0.11)	13.28	2.1
TN-FSM+JD [13] (2022)	0.77 (0.06)	0.23 (0.12)	0.63 (0.03)	10.00	4.6
Causal TCN [13] (2022)	0.80 (0.15)	0.29 (0.22)	0.68 (0.24)	19.00	28.0
DDNet [5] (2019)	0.88 (0.06)	0.16 (0.18)	0.78 (0.14)	8.00	2.2
OO-dMVMT [14] (2023)	0.92 (0.06)	0.09 (0.09)	0.85 (0.11)	8.00	4.1
Proposed method	0.95 (0.01)	0.03 (0.01)	0.92 (0.02)	8.00	3.4

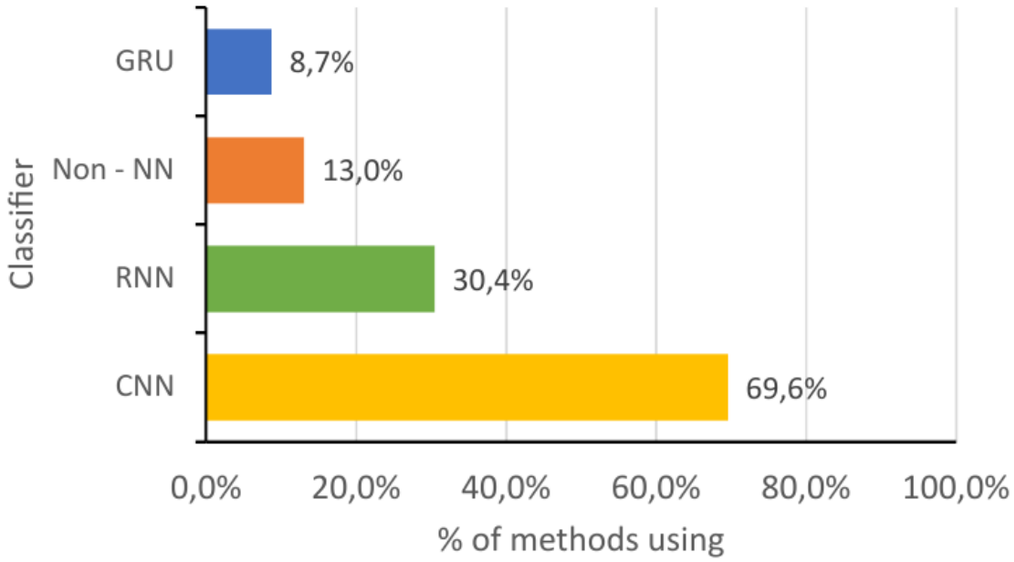
What “good” means in practice

- A false positive score of 0.1 sounds excellent, BUT:
  - SHREC’22 sequences: about 30 seconds, 4 gestures each
  - 0.1 → roughly 4 spurious commands every 5 minutes of hand movement
  - The best contest method scored 0.23. The baseline, 0.34
- In an industrial cell or an operating room that is not a metric, it is an incident report



What the literature actually uses

- Across the 23 best-performing methods in our survey:
  - CNN in ~70%, RNN in ~30%, GRU ~9%, non-neural ~13%
- CNNs win on latency, which is what the task rewards
- And in most of them the input is not raw data
  - features, distances, speeds, angles
- Emporio et al., CVIU 259 (2025)



Comments / research directions

- Not too bad results, but not optimal, improvement possible with known methods
  - E.g., adding attention modules
- Limited number and quality of benchmarks
  - Results can heavily depend on the dictionary, and on the data capture settings
  - Network-based methods may struggle for the limited number of training examples
    - Data augmentation may be useful
  - Analysis and optimization of dictionaries is mandatory for practical applications
    - Design them possibly to avoid
      - Long ambiguous sequences (shared by different gestures in the dictionary)
      - Poses/movements difficult to be tracked
      - Gestures prone to false detection

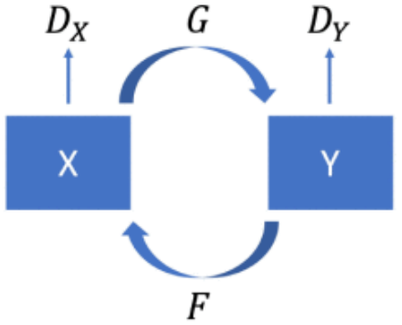
Data augmentation

- These works are based on limited datasets
  - Gesture examples are scarce
  - non-gesture examples are unrepresentative (the training non-gesture covers a vanishing fraction of what a hand can do)
- Augmentation can be effective. Are there related works?
  - Many on single trajectory augmentations, for 2D recognition
  - Very few on 3D hand pose sequences
- Dictionary design

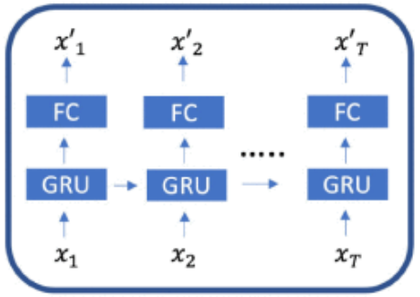
Augmentation techniques

- Applied in segmented gesture classification (Shrec'17)
- Classical augmentation
  - Apply noise to joint position
  - Resample trajectory
  - Scale, Shift, rotate (?)
- GAN-based augmentation
  - Imaginative GAN, based on a Cycle GAN
  - Trained with 4 gestures, can generalize
    - On unseen classes
  - Need a good number of training data
    - Transfers style across different examples

Shen, Junxiao, John Dudley, and Per Ola Kristensson. "The imaginative generative adversarial network: Automatic data augmentation for dynamic skeleton-based hand gesture and human action recognition." *2021 16th IEEE international conference on automatic face and gesture recognition (FG 2021)*. IEEE, 2021.



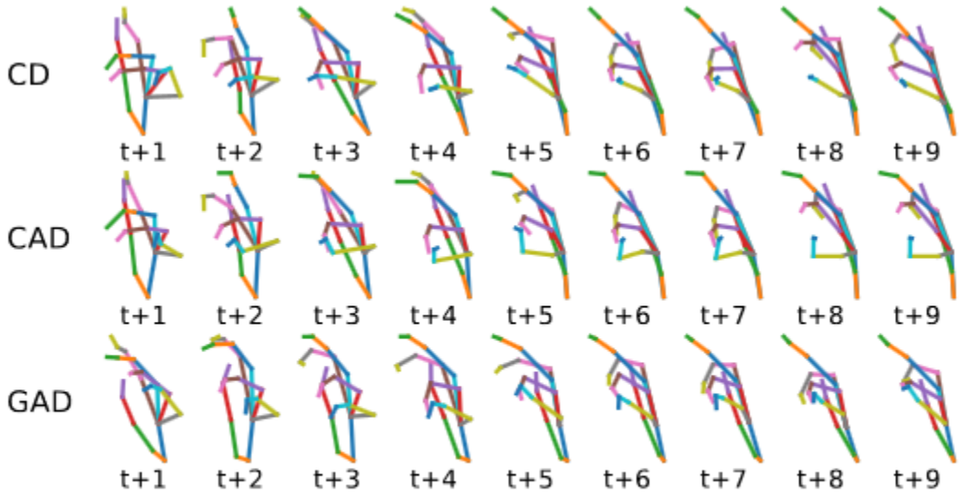
(a) The Imaginative GAN is trained using a CycleGAN structure with a cycle consistency loss. The latent attributes are transferred within the input datasets X and Y. G and F are the two generators, D_X and D_Y are the two discriminators.



(b) The generator in the Imaginative GAN has one GRU layer and one fully connected layer. It has a teacher forcing-like mechanism so the input is a ground truth sample and the output is an altered sample. T is the length of the sample.

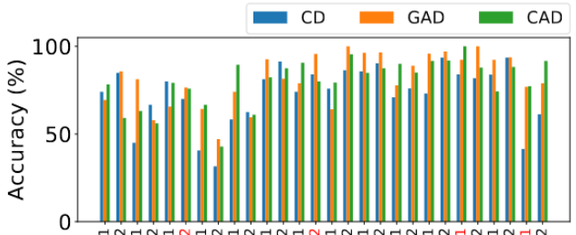
Augmentation results

- Better than «classical», but with limited «classical» methods
  - Not on all gestures
- Slight improvements in the classificati accuracy with best methods, but not clear how it was trained (all data? Cleaned?)
- Not necessarily available in practice



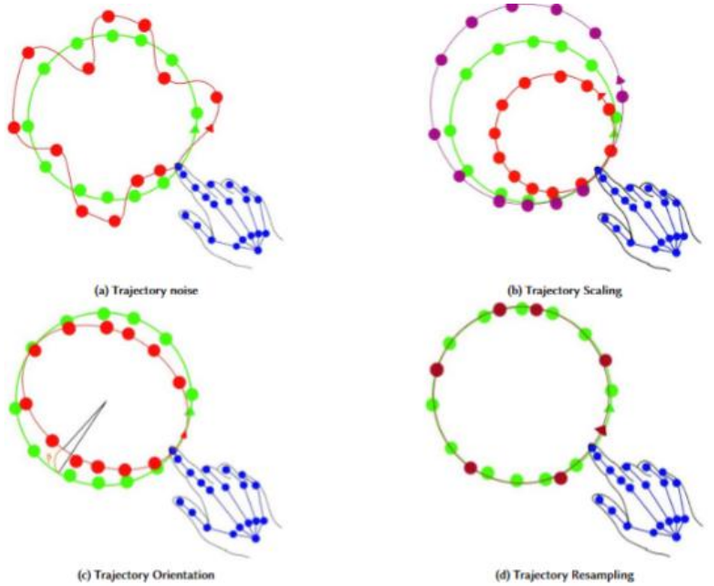
(a) SHREC'17 Track: Swipe Up with one finger

Model	SHREC'17 Track		DHG-14/28	
	14G	28G	14G	28G
DG-STA [1]	+0.7%	+0.8%	+0.6%	+0.4%
DD-Net [36]	+0.7%	0.7%	+0.6%	+0.6%
DSTA-Net [30]	+1.0%	0.7%	+0.8%	+0.5%



Augmentation on continuous detection

- Tests done with classical augmentation
  - Gesture augmentation effective with limited data but only on detection rate
  - Non-gesture augmentation effective to reduce false positives
  - Difficult tuning



Technique	DR ↑	FPS ↓	JI ↑
Baseline (no augmentation)	0,72	0,33	0,58
AAD	0,83	0,32	0,66
NGAD	0,82	0,25	0,69
AAD + NGAD	0,84	0,25	0,70

rts the online evaluation scores (DR – Detection Rate, FPS – False Positive Score, and J
ll augmented gestures (AAD), the union of all augmented non-gestures (NGAD), and

Augmentation challenges

- Both GAN and classical augmentation create gestures similar to the existing examples.
- But the goal would be to make the system robust to different executions not included, **especially for non-gestures**
- Non-gesture class is quite different
  - It may be completely different from the examples seen in the training set

Dictionary design

- Recognition is only a single aspect of a dictionary design
- Recent analysis and literature survey
- Joint optimization of different factors is another interesting research direction

Table 3. A Comparison of the Degree to which Expert-Led, User-Led, and Computationally-Based Gesture Design Methodologies Focus on the 13 Factors Identified by the Literature Review

Factor	Expert-Led	User-Led	Computationally-Based
Context	◐	●	○
Modality	●	○	◐
Social Acceptability	◐	●	○
Discoverability	○	◐	○
Intuitiveness	○	◐	○
Learnability	○	◐	○
Transferability	●	○	○
Complexity	●	◐	●
Efficiency	●	◐	◐
Ergonomics	●	○	◐
Occlusion	●	○	◐
Feedback	●	○	◐
Recognition	●	○	●

▮ – Situational factors; ▮ – Cognitive factors; ▮ – Physical factors; ▮ – System factors; ● – Yes; ◐ – Maybe; ○ – No

Xia, Haijun, et al. "Iteratively designing gesture vocabularies: A survey and analysis of best practices in the HCI literature." *ACM Transactions on Computer-Human Interaction (TOCHI)* 29.4 (2022): 1-54.

- Authors reviewed how the different aspects have been addressed in the literature

Ergonomics	The human anatomical, anthropometric, physiological and biomechanical characteristics as they relate to physical activity (i.e., performing a gesture) [239]	Look for: – Use of outer positions [229] – Use of repetition [229] – Use of static positions [229] – How relaxed the muscles are [229] – How relaxed the "middle" neutral position is between outer positions [229]	– Change size of the gesture – Change where the gesture is performed – Change the geometry, pressure of the gesture to avoid issues described on the left
			(Continued)
Occlusion	How much of a system's output is covered or blocked when a gesture is performed [336]	– Use Hand Occlusion Model [336] – Visually inspect for occlusion [51]	– Change size of the gesture – Perform at dedicated or offset area [50, 381] – Use callout of the occluded content [336, 337]
Recognition	Process of tracking a gesture from its initial representation through to its later conversion into a semantically meaningful command [262]	– Compute accuracy/error	– Reduce similarity of gestures – Utilize alternative recognition algorithms [150, 222, 254, 307]

Xia, Haijun, et al. "Iteratively designing gesture vocabularies: A survey and analysis of best practices in the HCI literature." *ACM Transactions on Computer-Human Interaction (TOCHI)* 29.4 (2022): 1-54.

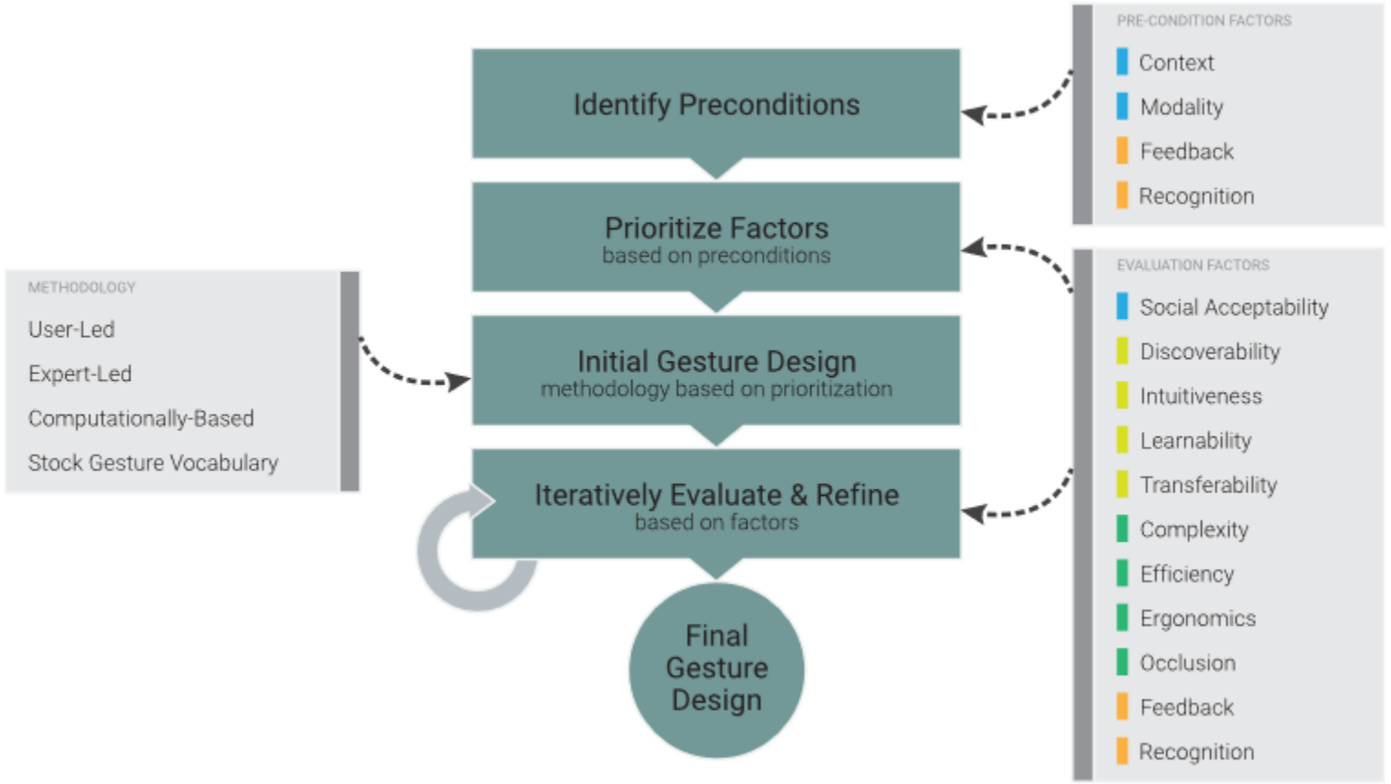
Design factors in the literature

- Authors reviewed how the different aspects have been addressed in the literature

Discoverability	The quality of gestures that enables a user to access intended referents of those gestures despite a lack of knowledge about the gesture [361]	– Compute agreement score [362] – In the wild deployment and observation [289]	– Utilize adopted gestures – Mimic common actions [121] – Reveal gestures when appropriate [18]
Intuitiveness	The degree to which a gesture makes use of transferable and existing knowledge [260]	– Lab study with matching gestures to commands [228, 362] and matching commands to gestures [228] – Lab study to measure differences between expert and novice duration [3]	– Leverage existing and transferable knowledge
Complexity	The state or quality of a gesture being intricate or complicated [243]	– Inspect number of steps or movements required by gestures [285] – Inspect movements that require substantial effort [381] – Indicated by the measurement or predication of production time	– Reduce the number of segments in gesture movement – Reduce the number of figures and hands required – Match the complexity of the gesture to that of the task
Efficiency	The accuracy and completeness with which users perform an action or gesture [239]	– Compute production time [278] – Apply computational and estimation models for pen input[59, 129, 321], and touch input [35, 268]	– Reduce the size of gesture – Reduce the duration of gestures (i.e., shorter dwell time) – Reduce the number of repetitions

Xia, Haijun, et al. "Iteratively designing gesture vocabularies: A survey and analysis of best practices in the HCI literature." *ACM Transactions on Computer-Human Interaction (TOCHI)* 29.4 (2022): 1-54.

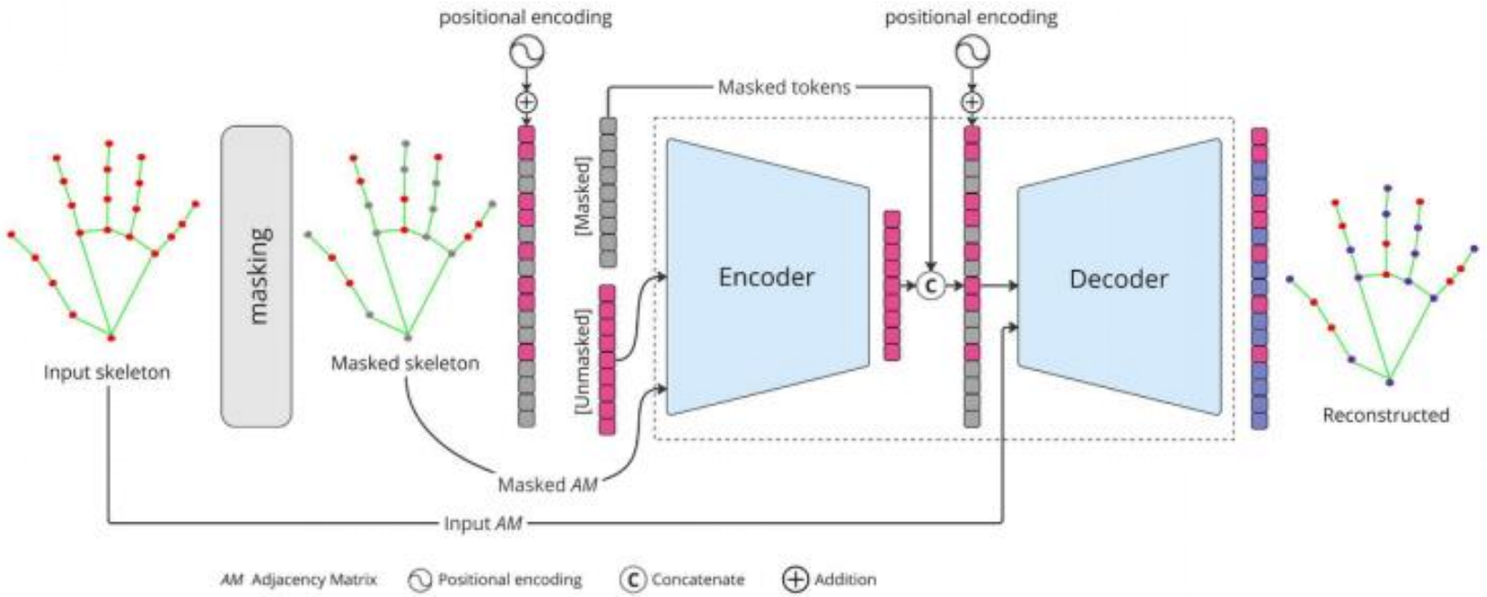
Proposed approach



Xia, Haijun, et al. "Iteratively designing gesture vocabularies: A survey and analysis of best practices in the HCI literature." *ACM Transactions on Computer-Human Interaction (TOCHI)* 29.4 (2022): 1-54.

Graph-based Methods for Dynamic HGR

Self-Supervised Feature Extraction - A Topology-aware Masked Auto-Encoder

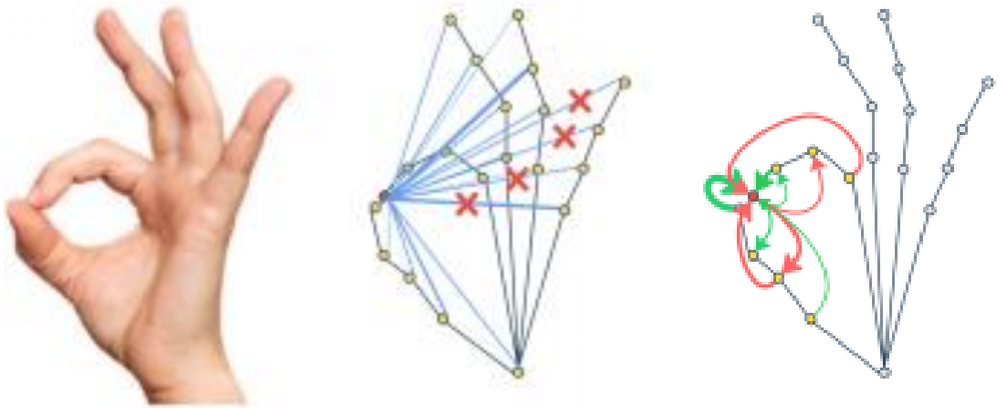


Adapting Vision Transformer (ViT) architecture for skeletal data processing.
A modified attention mechanism that incorporates adjacency matrix information.
Integration of Fourier feature mapping, showcasing its superiority over linear mapping.

O. Ikne, B. Allaert, and H. Wannous. Skeleton-based self-supervised feature extraction for improved dynamic hand gesture recognition. In International Conference on Automatic Face and Gesture Recognition IEEE FG 2024.

Graph-based Methods for Dynamic HGR

Spatio-Temporal Sparse Graph Convolution Network (1/2)



Graph-based advancements improved spatial–temporal learning, but…

✗ mostly rely on dense, undirected skeletal graphs

✗ assume all joints contribute equally

✗ ignore directional flow of articulation (wrist → fingertips)

→ leads to redundant connections, higher computation, and weaker performance for fine-grained gestures

The model is motivated by two key observations:

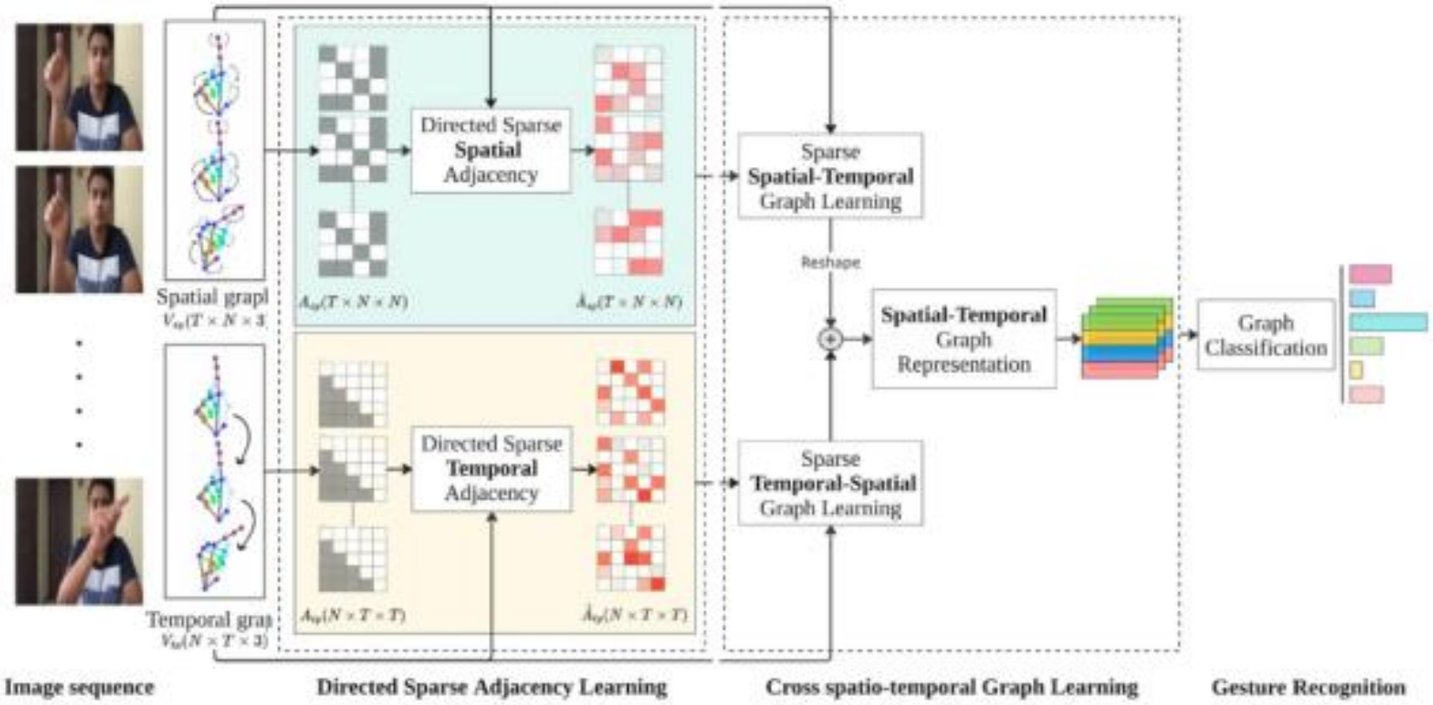
(i) Only a subset of joint-to-joint dependencies is functionally relevant for a given gesture

(ii) The flow of information between joints conveys distinctive cues about gesture meaning and movement dynamics

Graph-based Methods for Dynamic HGR

Spatio-Temporal Sparse Graph Convolution Network (2/2)

- Decoupled learning of directed and sparse spatial and temporal interactions
- Capturing cross-spatial-temporal interactions for graph representation
- Gesture classification



[Ikne et al. FG 2024]

Graph-based Methods for Continuous Dynamic HGR

Self-Supervised Learning with Anatomical Regularizer for Improved Continuous HGR (1/2)

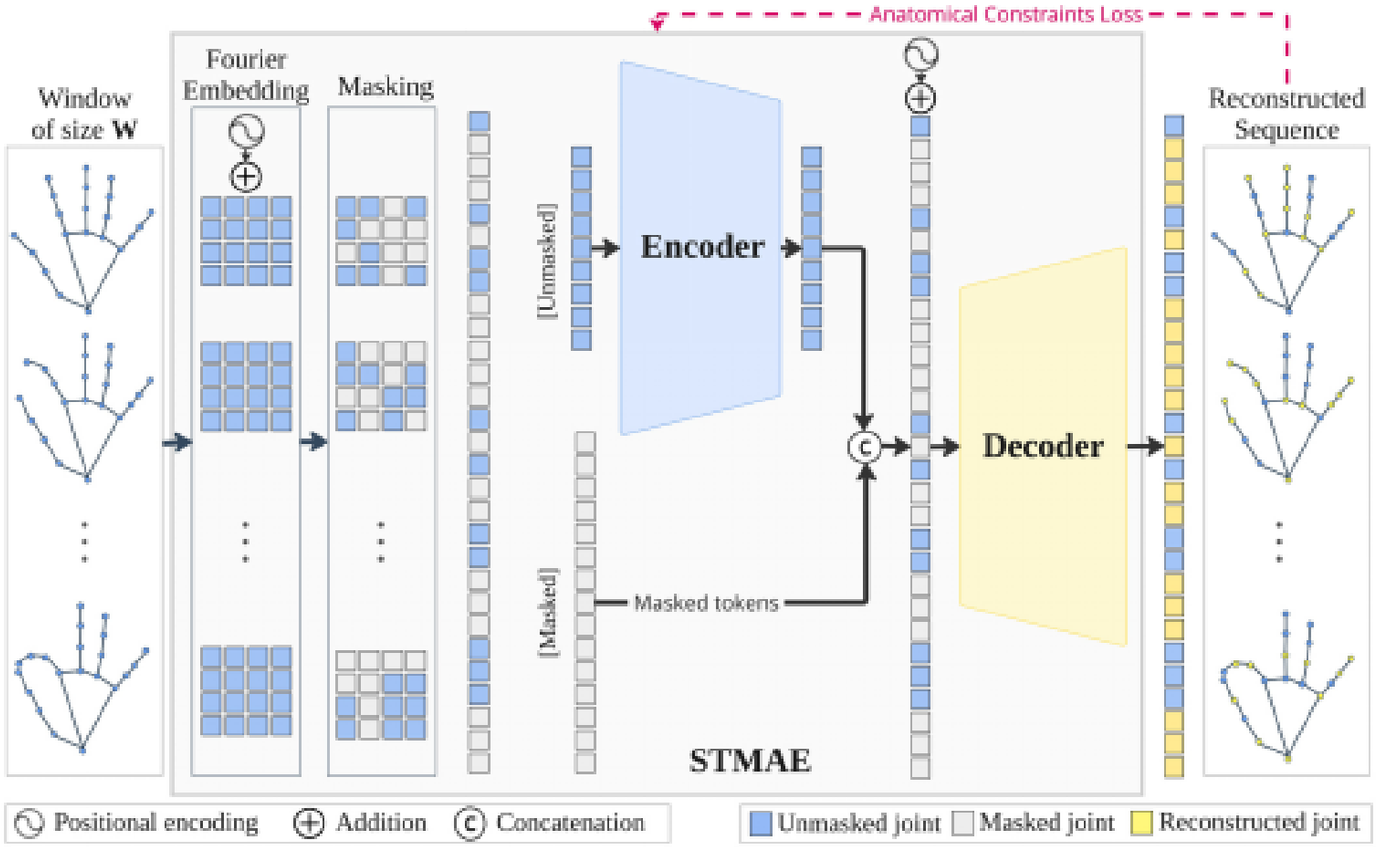
- Joints-based Masker Auto-Encoder
- Spation-temporal Transformer backbone
- Spatio-temporal masking strategies

Bones length loss:

$$\mathcal{L}_{BL}(P) = \frac{1}{N_b}\sum_{i=1}^{N_b}\mathcal{I}(\|b_i\|_2; b_i^{\min}, b_i^{\max}),$$

Joints angle loss:

$$\mathcal{L}_{JA}(P) = \frac{2}{N(N-1)}\sum_{i=2}^{N}\sum_{j=i+1}^{N}\mathcal{I}(\angle(\overrightarrow{J_wJ_i}, \overrightarrow{J_wJ_j}); a_{ij}^{\min}, a_{ij}^{\max}),$$



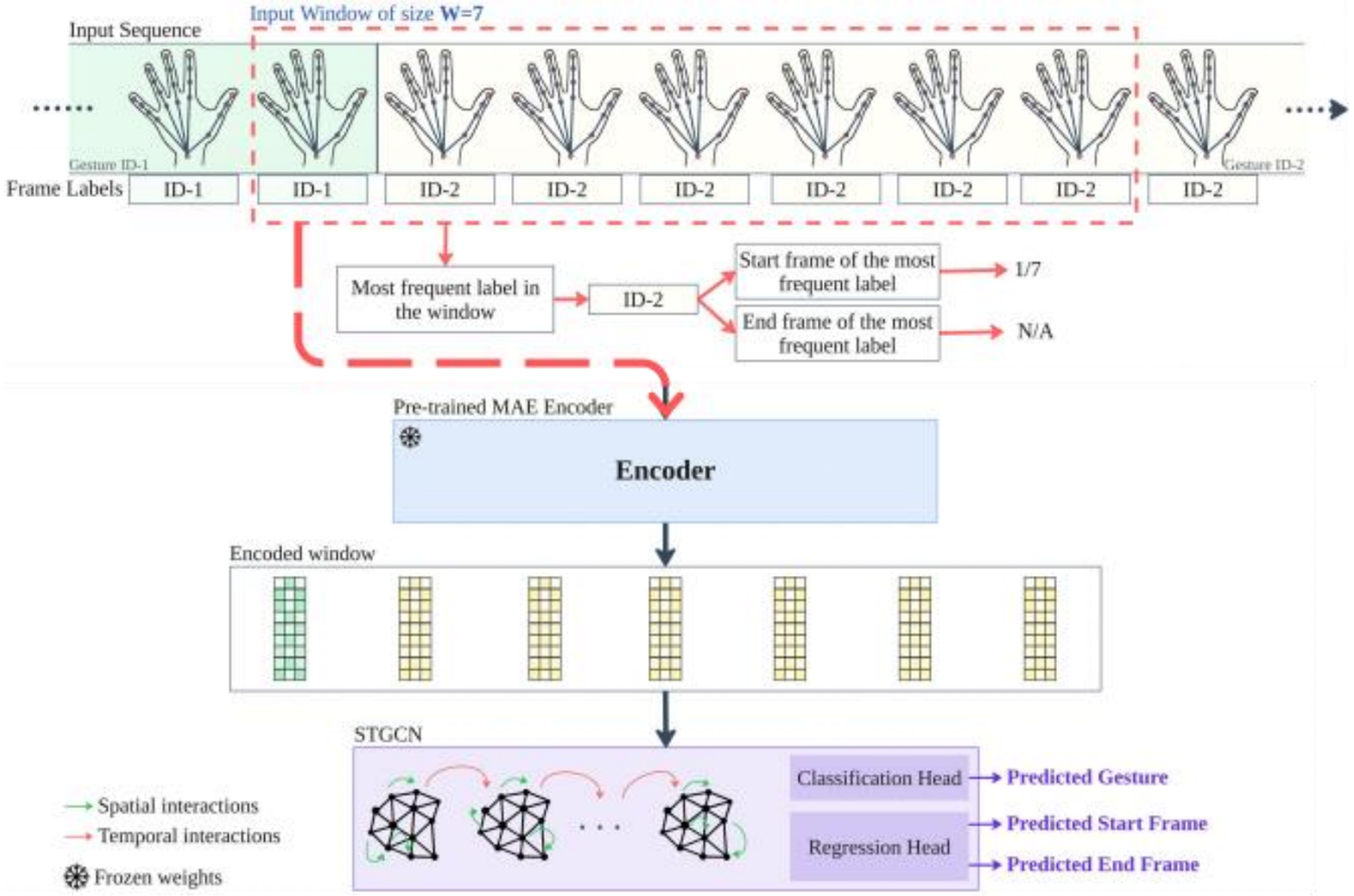
[Ikne et al. 3DV 2025]

Graph-based Methods for Continuous Dynamic HGR

Self-Supervised Learning with Anatomical Regularizer for Improved Continuous HGR (2/2)

1. Data preprocessing

2. Online Training



Integration of Extended Reality with a Cyber-Physical Factory Environment

- **XR Integration in Industry 4.0/5.0**
  - Real-World Use Cases & Gesture-Based Interaction
  - Synchronizing Data Between Physical and Digital Twins
- **Cyber-Physical Factory & Digital Twin**
  - Enhancing Worker Productivity & Safety
  - Real-Time Monitoring and Control

Marco Emporio, Ariel Caputo, Deborah Pintani, Dong Seon Cheng, Thomas De Marchi, Gianmaria Forte, Franco Fummi, and Andrea Giachetti. *"Integration of Extended Reality with a Cyber-Physical Factory Environment and its Digital Twins". Proc. ACM Hum.-Comput. Interact. 8, EICS,2024*

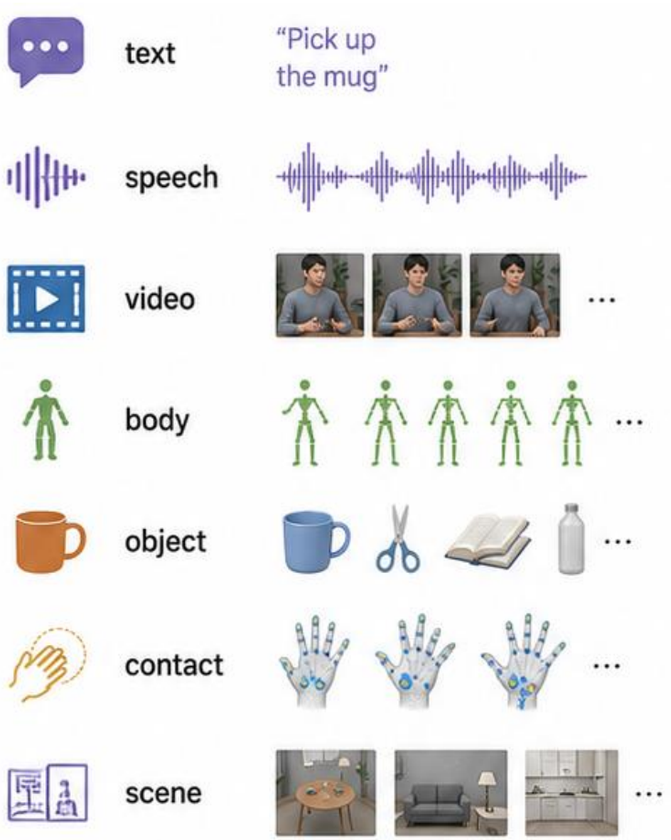
4 - Hand Motion Generation: Recent Advances and Challenges

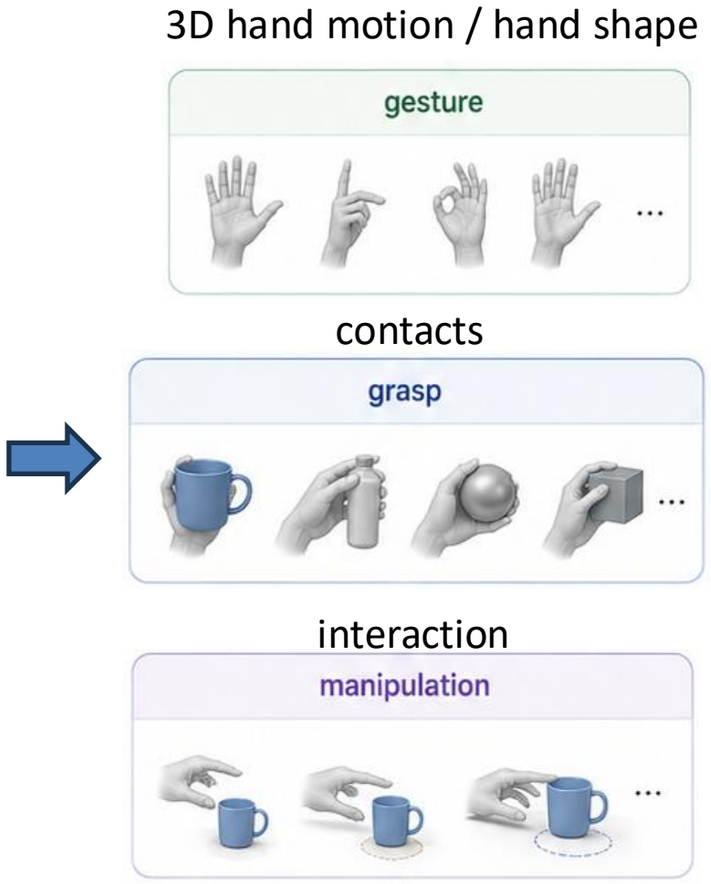
Outline

- Introduction to Hand Motion Generation
- Multimodal conditioning
- Ensuring Physical Plausibility
- Evaluation and Common Pitfalls
- Open research challenges

Introduction to Hand Motion Generation

What exactly are we generating?





- **Pose generation**: one frame
- **Motion generation:** temporal sequence
- **Gesture generation:** communicative motion
- **Interaction generation:** motion + object/contact
- **Embodied generation:** motion that can actually execute a task

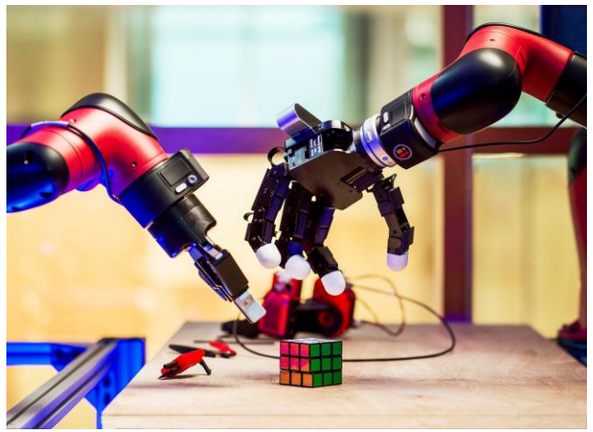
Introduction to Hand Motion Generation

Why generate hand motion?

- Applications in virtual agents, sign language, XR, robotics, and embodied AI





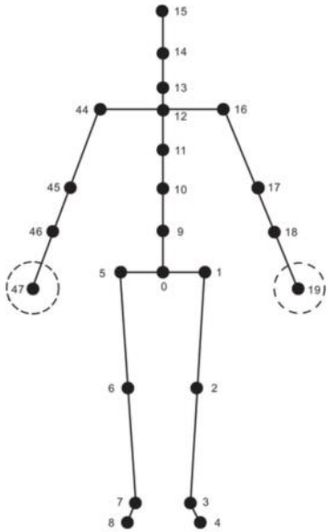




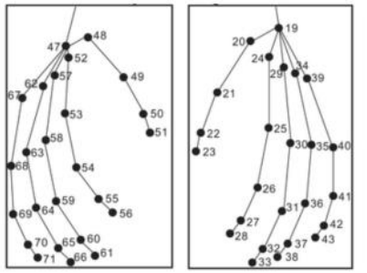
Key Challenges

Why Hands are Challenging ?

Compared to Body Motion	Hand Motion Requires
Fewer joints, sparse motion	Dense joints (~21 per hand) in small space
Broad global movements	Fine-grained articulations & subtle variations
Easier visibility	Frequent self-occlusion + rapid dynamics
Low anatomical constraints	Strong biomechanical limits & joint dependencies



Body skeleton



Hands skeleton

Key Challenges

AI can generate realistic faces, cities, even entire videos…

Yet it still struggles with hands ...

High dimensionality

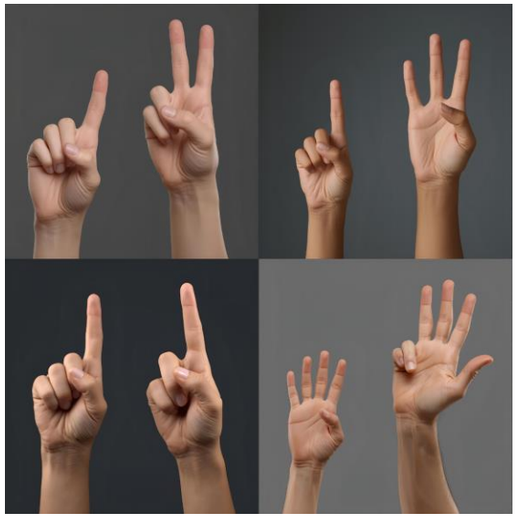
- 20+ articulated joints
- continuous rotations
- two hands → coordination problem
- not self-contained

Anatomy & Physical constraints

- Fixed number of fingers
- Constant bone lengths
- Limited degrees of freedom and finger configurations
- no penetration & correct contact & correct grasp
- plausible force/interaction
- temporal continuity

Precise 3D hand + object + language + temporal annotations are expensive.







Key Challenges

Hand Motion Generation: Key Aspects

1. WHAT TO MAKE ?

Gestures
Actions
Signs
Grasps
Interacts

2. HOW TO CONDITION ?

Text
Audio
Body
Object
Scene



3. IS IT PHYSICALLY VALID ?

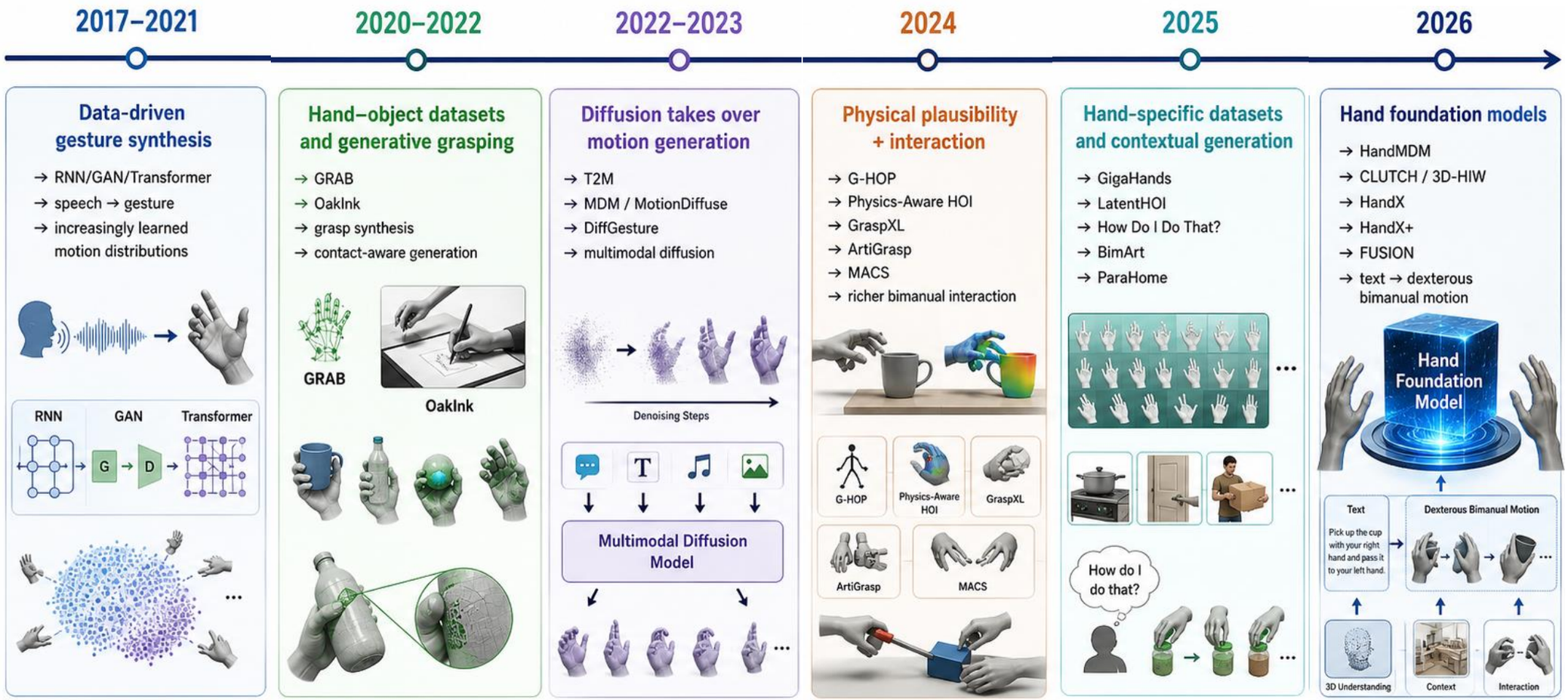
Contact
kinematics
Physics
Coordination
Latency

4. HOW DO WE KNOW?

Semantic evaluation
Diversity
Anatomy / Physics
User study

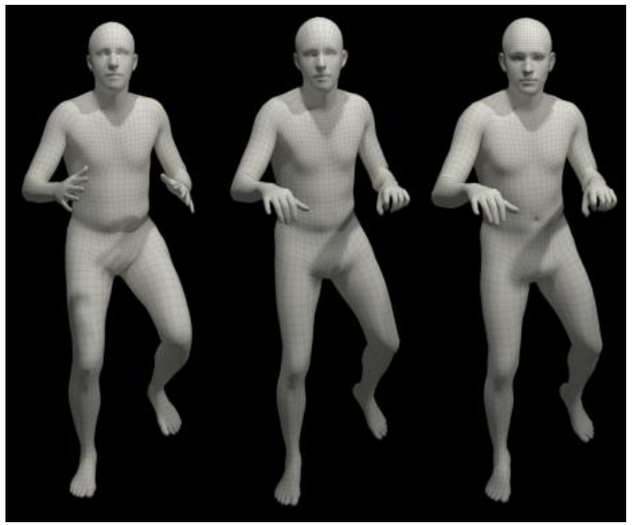
Timeline of Hand Motion Generation

- Learning Representations
- Object & Contact awareness
- Diffusion &Multi-modality
- Physical Plausibility &Interaction
- Context& Task Understanding
- Generalization &Foundation Models



Hand Motion Representation

How Motion Data is Represented ?

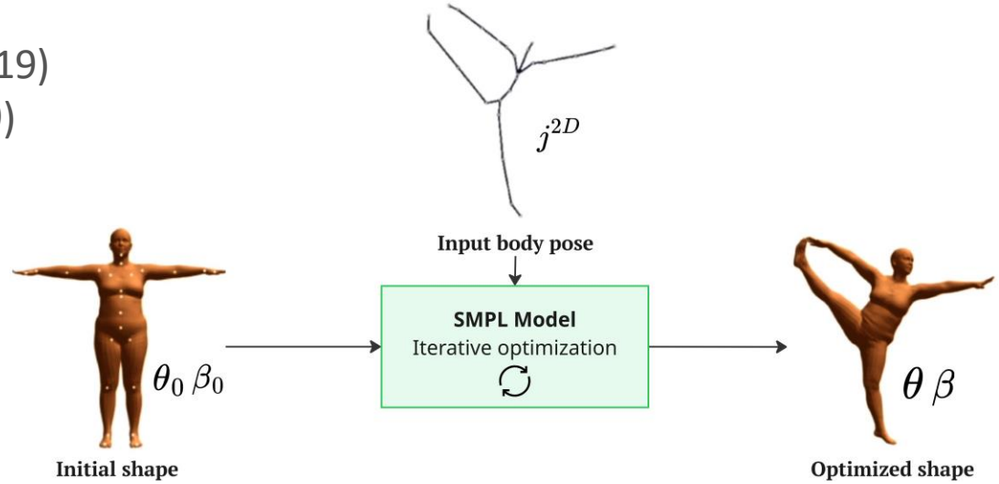


Skinned Multi-Person Linear (SMPL) model

- SMPL: body (Loper et al. 2015)
- SMPLH: body + hands (Romero et al. 2017)
- SMPLX: body + hands + expression (Pavlakos et al 2019)

Model fitting from recorded data

- SMPLify / SMPLify-X (Pavlakos et al. 2019)
- MoSh / MoSh++ (Mahmood et al. 2019)



- MANO
- Joint positions
- Joint rotations
- Mesh vertices
- Motion tokens
- contact/event representations

$$E(\theta, \beta, \mathbf{t}) = E_{\text{data}}(\theta, \beta, \mathbf{t}) + \lambda_{\text{pose}} E_{\text{pose}}(\theta) + \lambda_{\text{shape}} E_{\text{shape}}(\beta) + \lambda_{\text{prior}} E_{\text{prior}}(\theta)$$

Overview of Common Generation Pipelines

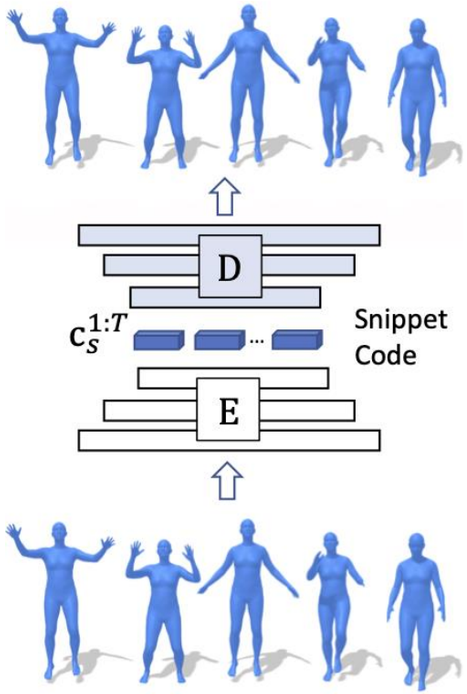
The generic human-motion revolution: First advanced step

T2M — CVPR 2022

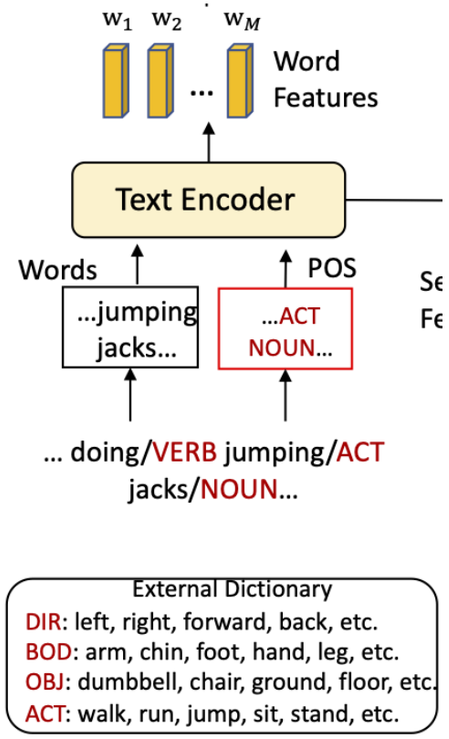
Introduced HumanML3D with **14,616 motions and 44,970 descriptions**, and used a temporal VAE / motion-snippet representation.



HumanML3D dataset



Learning Motion Representation



Language Encoding

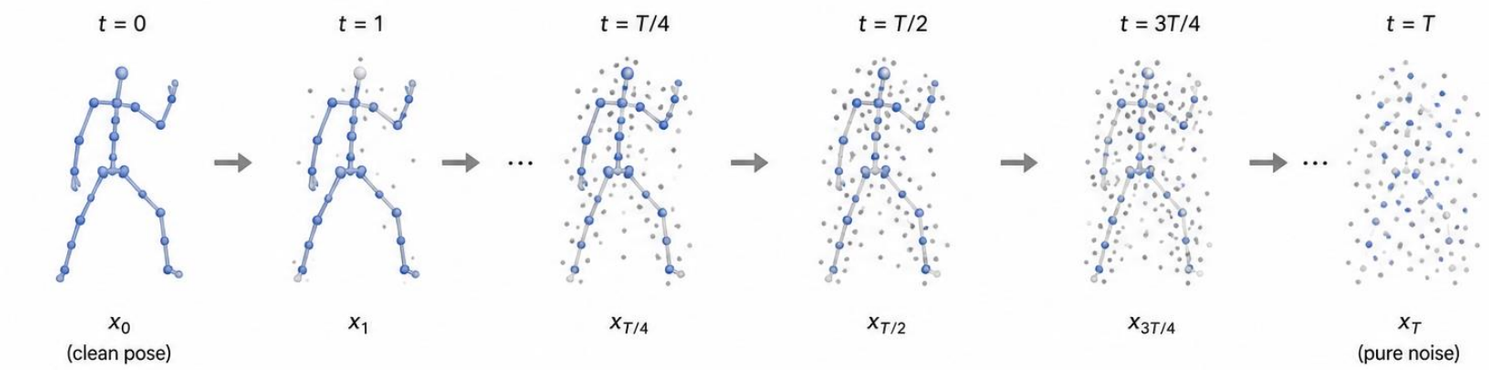
Guo, Chuan, et al. "Generating diverse and natural 3d human motions from text." *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2022.

Overview of Common Generation Pipelines

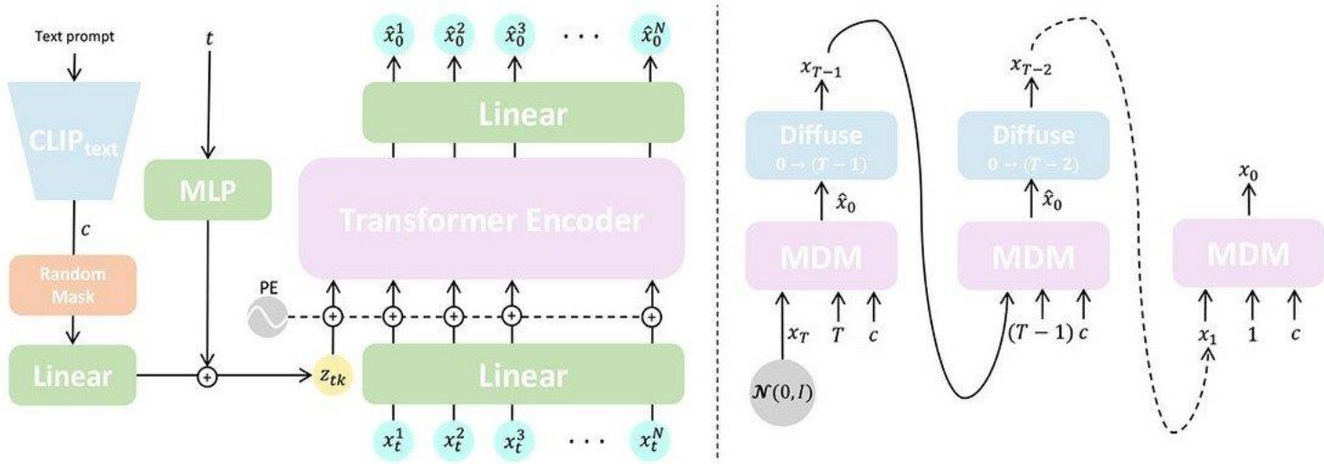
Diffusion-based Methods

Why diffusion changed motion generation

- Diverse trajectories
- Temporal coherence
- Conditional generation
- Controllable sampling



- **MDM** (Tevet et al., 2022): Transformer backbone with classifier-free guidance for controllability.
- **MotionDiffuse** (Zhang et al., 2022): DDPM with CLIP text embeddings and iterative denoising.
- **PhysDiff** (Yuan et al., 2023): Incorporates physics simulations via joint training.

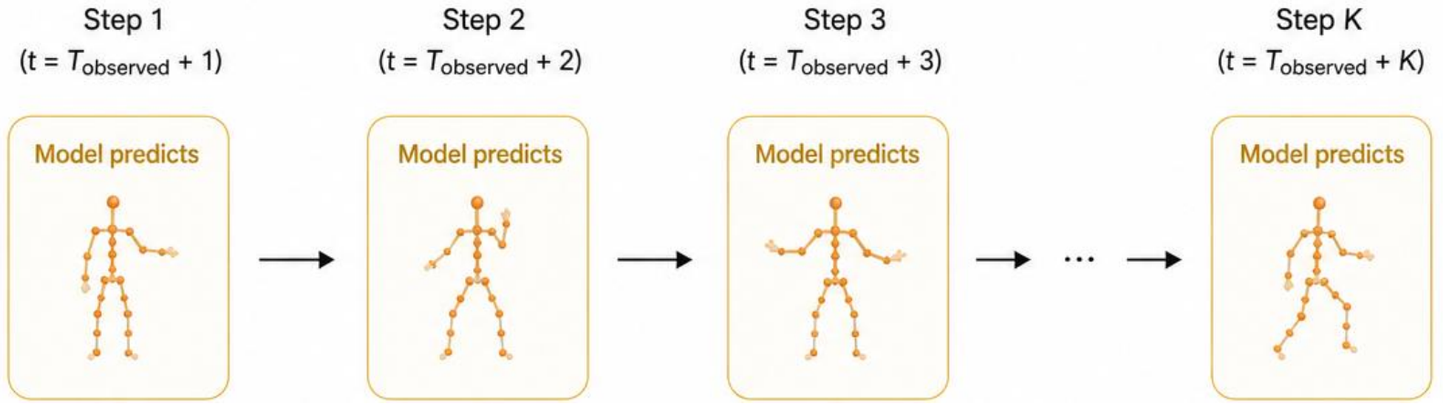


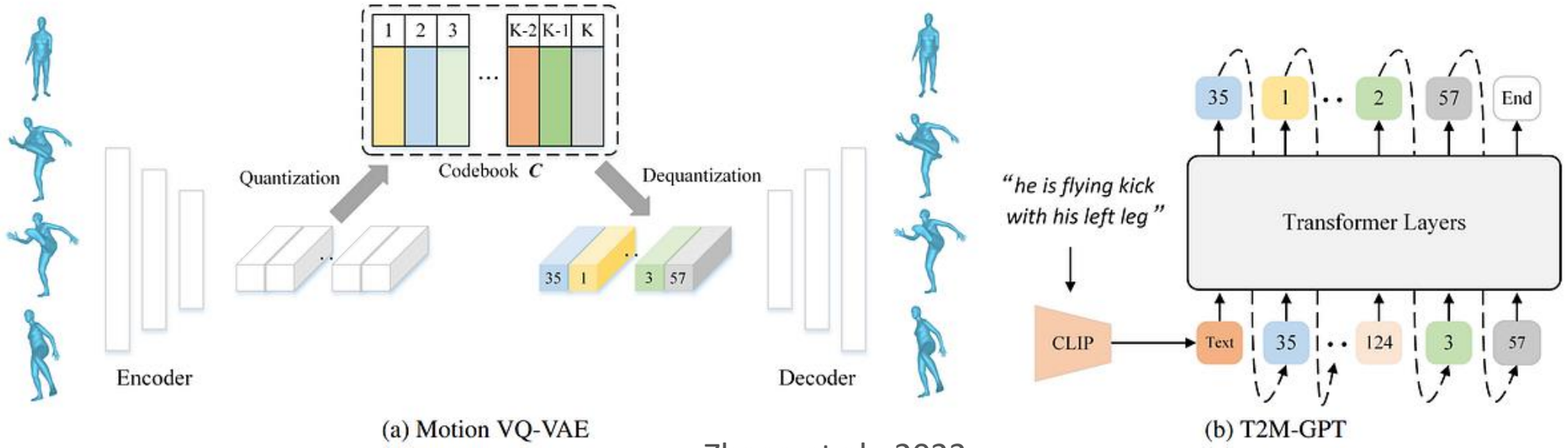
Tevet et al., 2022

Overview of Common Generation Pipelines

Auto-regressive

1.Action2Motion (Guo et al., 2020): RNN-VAE with Lie algebra for joint rotations.
2.TEMOS (Petrovich et al., 2022): Dual encoders for text (DistilBERT) and motion.
3.T2M-GPT (Zhang et al., 2023): VQ-VAE with GPT-style autoregressive generation of discrete codes.
4.MotionGPT (Jiang et al., 2023): Tokenizes motions using T5-like architecture.





Zhang et al., 2023

Overview of Common Generation Pipelines

What about hands ?

These models were mostly designed for whole-body motion, not detailed hand dynamics.

Whole-body motion generation does not automatically produce dexterous hand motion.

- Training data: Most classic motion datasets lack detailed fingers
- Representation: Hands are often treated as end-effectors
- Evaluation: Whole-body metrics can hide hand failures

But they establish the machinery that later becomes important for hands:

- Motion tokenization
- Text–motion embeddings
- Latent motion spaces
- Diffusion
- Multimodal conditioning



HumanML3D (Guo et al. 2022)



Motion-X(Lin et al. 2023)

(b) Multimodal Conditioning

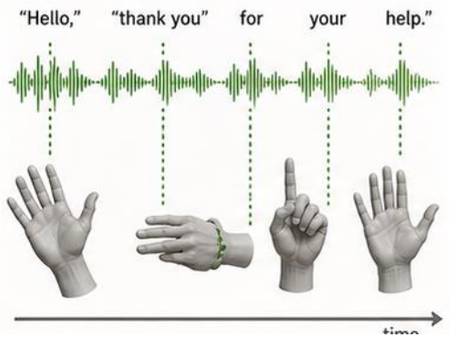
- Text-conditioned hand motion generation
- Audio-conditioned gesture generation during speech
- Body-conditioned and pose-conditioned hand motion
- Object-conditioned generation for grasping and manipulation

(b) Multimodal Conditioning

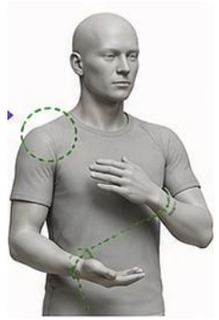
Text
What should the hand do?



Audio
When/how should it gesture?



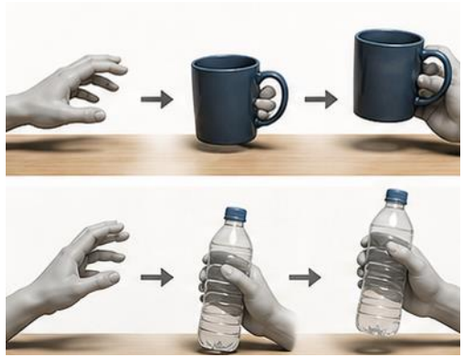
Body
How should hands fit the body?



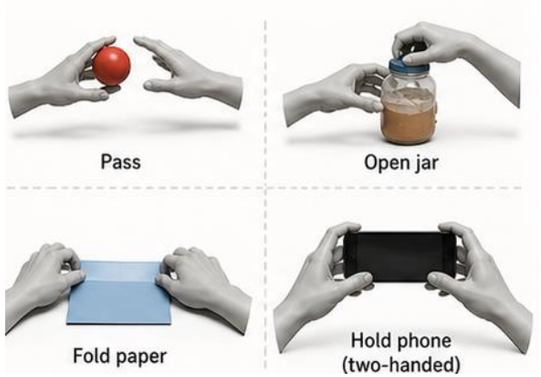
Scene
Where should interaction happen?



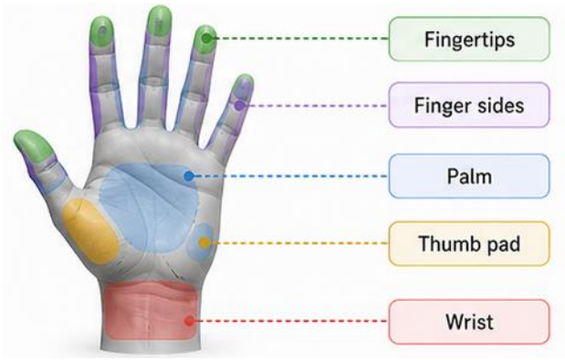
Object
How should it interact?



Other hand
How should hands coordinate?



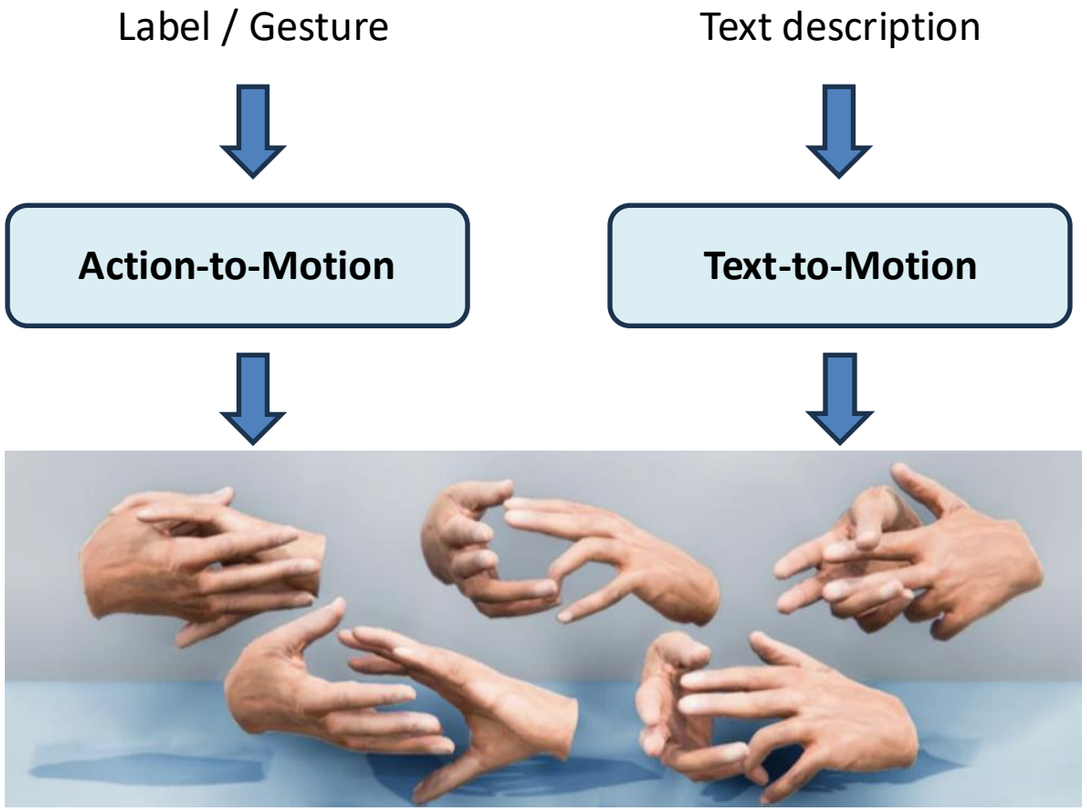
Contact
Where should the hand touch?



Text-conditioned Hand Motion Generation

Generic text-to-motion datasets describe **“walk”, “sit”, “dance”** much better than they describe **finger articulation, contact, handshape and wrist trajectories**.

- Gesture label
- Sentence
- Compositional description
- Fine-grained hand attributes



Text-conditioned Hand Motion Generation

Common benchmarks for text-conditioned hand motion generation

Dataset	Year	#motions	Vocabulary	Type	Main purpose / generation relevance
ASL-Text	2019	1,598	154	SL videos	Text→sign / zero-shot SL
MS-ZSSLR-W	2023	8,861	274	SL videos	Text/SL motion modeling
Motion-X	2023	81K	10,224	Mocap/video, body-focused	Large-scale whole-body expressive motion
SignAvatars	2024	70K	-	SL videos	3D sign-language motion production
BOTH57M	2024	1,815	612	Mocap, hand-focused	Text/body→two-hand motion
BOBSL3DT	2026	1,3M	1,500	SL videos	Large-scale text→3D hand motion
HandX	2026	84K	-	Mocap + curated HOI	Bimanual interaction generation
3D-HIW / CLUTCH	2026	32K	-	In the wild videos	Text2Motion & Motion2Text

2019 - 2022

sign labels

2022 - 2023

semantic text

2023- 2026

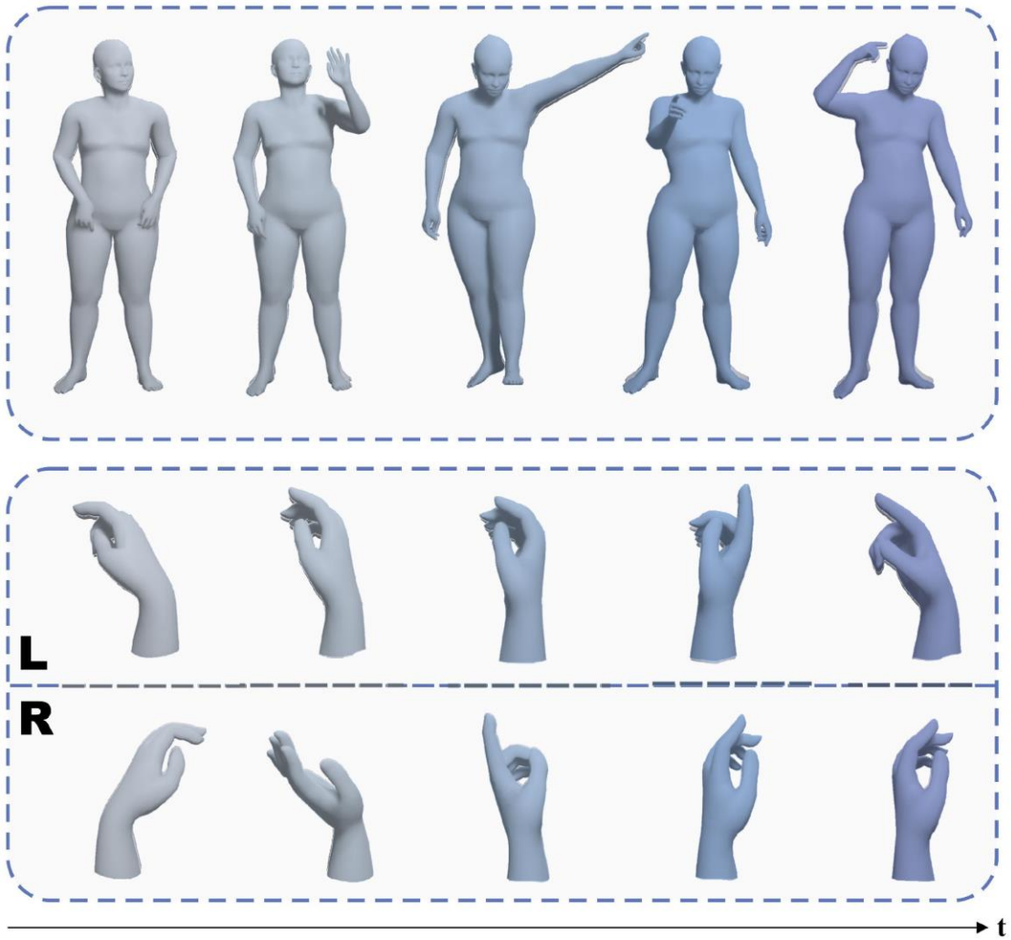
fine-grained / bimanual

Text-conditioned Hand Motion Generation

BOTH57M Dataset

BOTH2Hands introduces the dataset for generating two-hand motion.

- 57.8M total frames
- 8.31 hours
- 1,384 sequences
- mRGB fitting
  - 32 cameras
  - 60 FPS
- 23,477 text annotations
  - 3 / body
  - 3 / hands



One person extends their left index finger forward to indicate giving instructions to others.

A person extends their left hand forward, indicating command.

A person extends their left index finger and circles it around their temple to indicate the other person is a fool.

The left hand clenches into a fist and extends the index finger, while the right hand brings together all five fingers, then bends and straightens the index finger.

The right hand extends and closes the five fingers together, then the four fingers clasp with the index finger extended. The left hand bends the four fingers together while the index finger is extended.

The right hand clenches all five fingers together and then partially makes a fist with the index finger extended, while the left hand clenches all five fingers into a fist with the index finger extended.

Zhang, Wenqian, et al. "Both2hands: Inferring 3d hands from both text prompts and body dynamics." *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2024.

Text-conditioned Hand Motion Generation

HandMDM: Text-driven 3d hand motion generation from sign language data

It generates 3D hand motion conditioned on descriptions of:

- Handshape
- Location
- Finger movement
- Hand movement
- Arm movement



*The right hand, with the **index and little fingers extended** and other fingers curled, is held close to the **upper face**... **facing outwards**...*



*The right hand is held at **chest level** with the **index finger bent to touch the thumb tip**, forming a circle, while the **other fingers are spread**.*



*Hold your left hand **flat** in front of you, **palm up**... Shape your right hand by... **bending the middle finger at the lowest knuckle**...*



*The right hand, with the **index finger extended** from a fist and the palm facing sideways, moves towards the **upper face**, leading with its **radial side**.*



*The left hand is held... in a fist with **the little finger extended**... **palm facing inward**. The right hand starts **flat**... while **rotating the forearm.***

It constructs large-scale text–motion pairs from sign-language video and demonstrates generalization to unseen signs and non-sign hand movements.

Bensabath, Léore, Mathis Petrovich, and Gül Varol. "Text-driven 3d hand motion generation from sign language data." *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. CVPR 2026.

Text-conditioned Hand Motion Generation

HandX: Scaling Bimanual Motion and Interaction Generation

Left: Index finger transitions from partially bent to fully extended; wrist moves down and side-to-side.
Right: Index and middle fingers bend at the base while other fingers remain extended.
Relation: Left index tip contacts right palm, then right index tip contacts left palm as hands swap vertical positions.



Left: All fingers held straight; index, middle, and ring fingers spread apart from a closed position.
Right: Flat and open hand; thumb performs a single bending motion mid-sequence.
Relation: Palms stay near and aligned with repeated tapping contacts between corresponding fingertips.



Left: The ring and pinky fingers perform a slow bending motion while thumb and middle stay extended.
Right: The hand transitions from closed to open as fingers straighten out.
Relation: Right fingertips contact the left palm, then break apart rapidly at the end.



Zhang, Zimu, et al. "HandX: Scaling Bimanual Motion and Interaction Generation." *CVPR 2026*

Text-conditioned Hand Motion Generation

CLUTCH: Contextualized Language model for Unlocking Text-Conditioned Hand motion modelling in the wild.

From studio mocap To hand motion in the wild

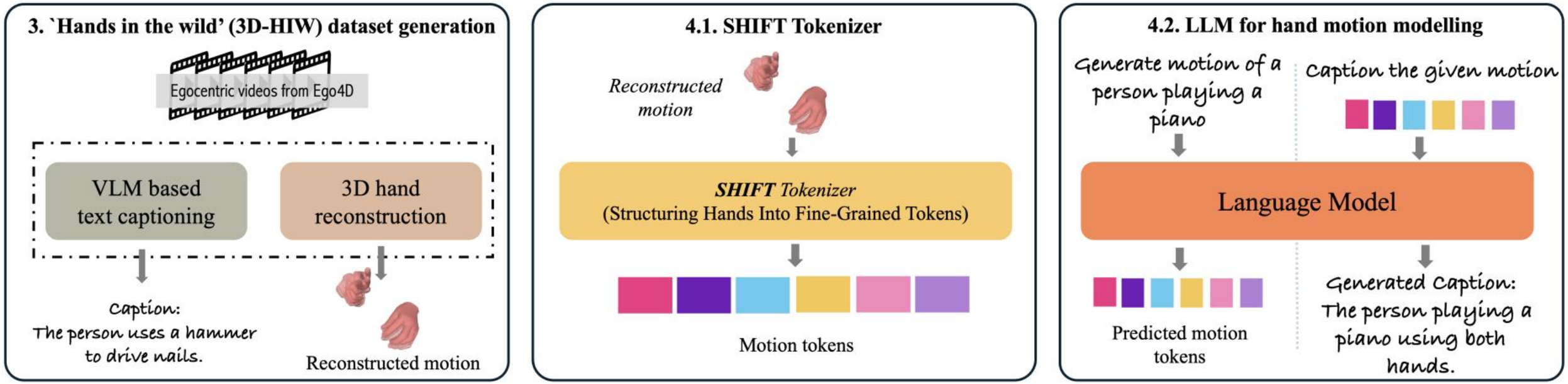


egocentric videos -> 3D hand tracking -> 3D-HIW 32K hand-motion sequences

Thambiraja, Balamurugan, et al. "CLUTCH: Contextualized Language model for Unlocking Text-Conditioned Hand motion modelling in the wild." *arXiv preprint arXiv:2602.17770* (2026).

Text-conditioned Hand Motion Generation

CLUTCH: Contextualized Language model for Unlocking Text-Conditioned Hand motion modelling in the wild.



SHIFT VQ-VAE -> motion tokens -> LLM -> hand motion -> geometric refinement

Thambiraja, Balamurugan, et al. "CLUTCH: Contextualized Language model for Unlocking Text-Conditioned Hand motion modelling in the wild." *arXiv preprint arXiv:2602.17770* (2026).

Audio-conditioned Generation During Speech

Speech does not specify only WHAT — it specifies WHEN

Audio is primarily a temporal constraint; text adds semantic constraint.



Audio gives strong information about:

- **when** a gesture should occur
- speech rhythm
- prosody
- emphasis
- pauses
- phonetic/acoustic structure

Speech →

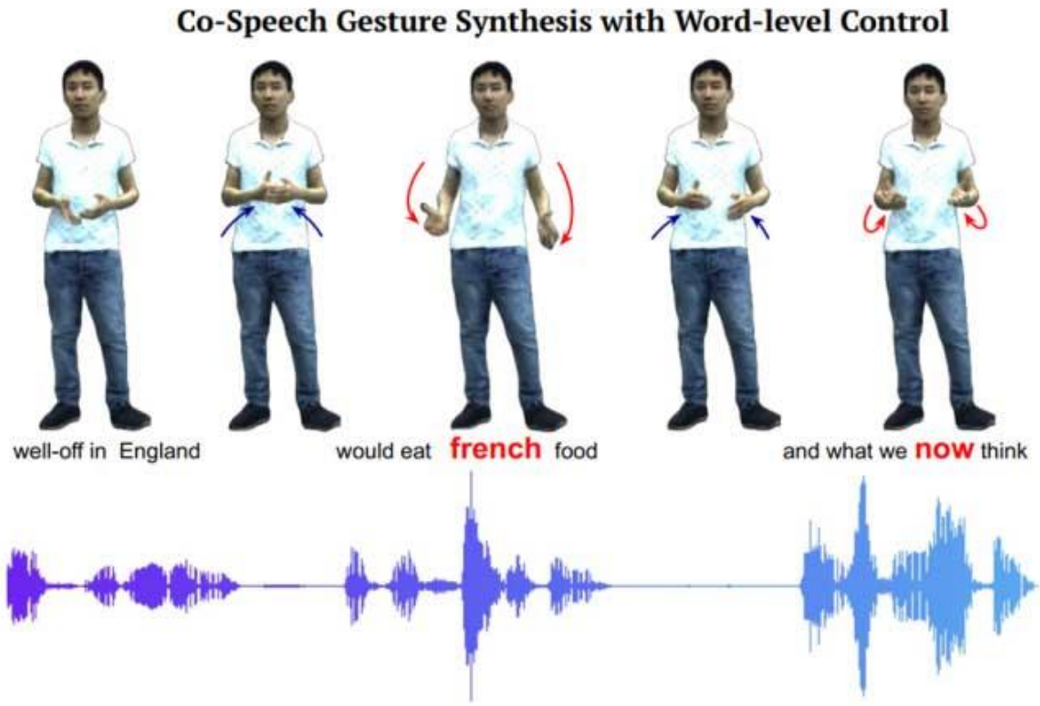
- lexical semantics
- prosody
- rhythm
- emphasis
- pauses
- turn-taking

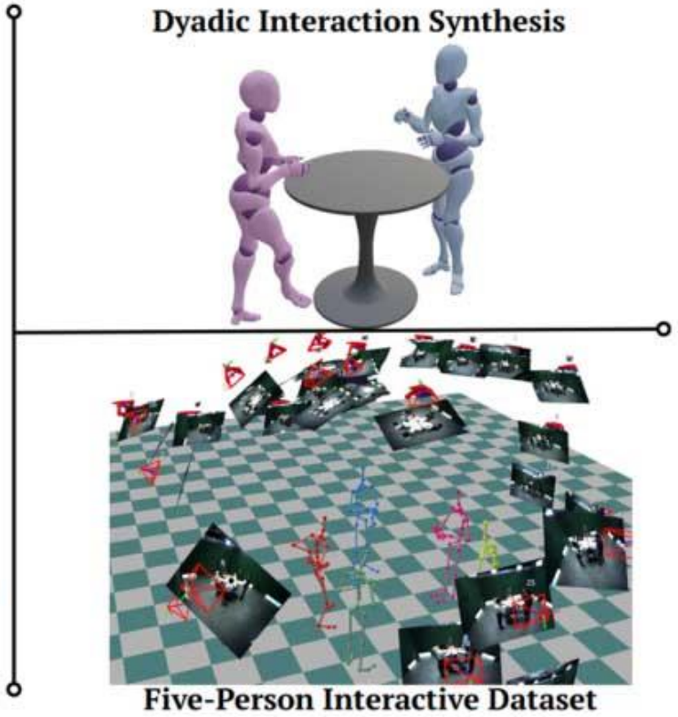
→ gesture

Audio-conditioned Generation During Speech

ConvoFusion: Multi-Modal Conversational Diffusion for Co-Speech Gesture Synthesis

Audio → when
Text → what
Word guidance → what to emphasize

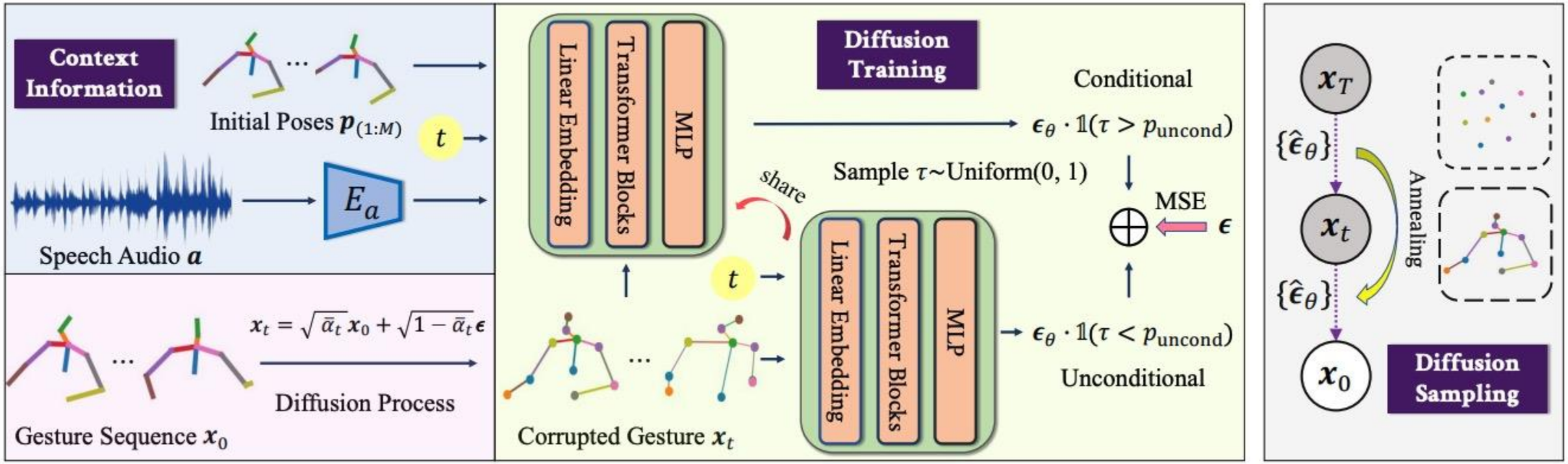




Mughal, Muhammad Hamza, et al. "Convofusion: Multi-modal conversational diffusion for co-speech gesture synthesis." *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2024.

Audio-conditioned Generation During Speech

DiffGesture: Bringing Diffusion to Co-Speech Gesture Generation
Audio-conditioned diffusion



Zhu, Lingting, et al. "Taming diffusion models for audio-driven co-speech gesture generation." *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2023.

Audio-conditioned Generation During Speech

But audio alone is often insufficient to determine:

- **what gesture**
- gesture semantics
- discourse function
- spatial meaning
- reaction to an interlocutor.

Body-conditioned Hand Motion Generation

Why Body Context Is Essential

- Body provides the reference frame, physical constraints, and motion rhythm
- Enables realistic, coherent, and purposeful hand gestures
- Essential for tasks where language and body jointly guide fine-grained hand motion

→ Hand motion generation is not just “more detailed body motion”

Why Hand Motion Generation Is Different ?

Hands Don’t Move Alone

- Hand motion depends on arm, shoulder, and torso dynamics
- Many gestures are defined relative to the body

Limitations of Hands-Only Datasets

- Capture detailed fingers but ignore global body context
- No spatial frame
- Cannot model coordination between both hands and the body



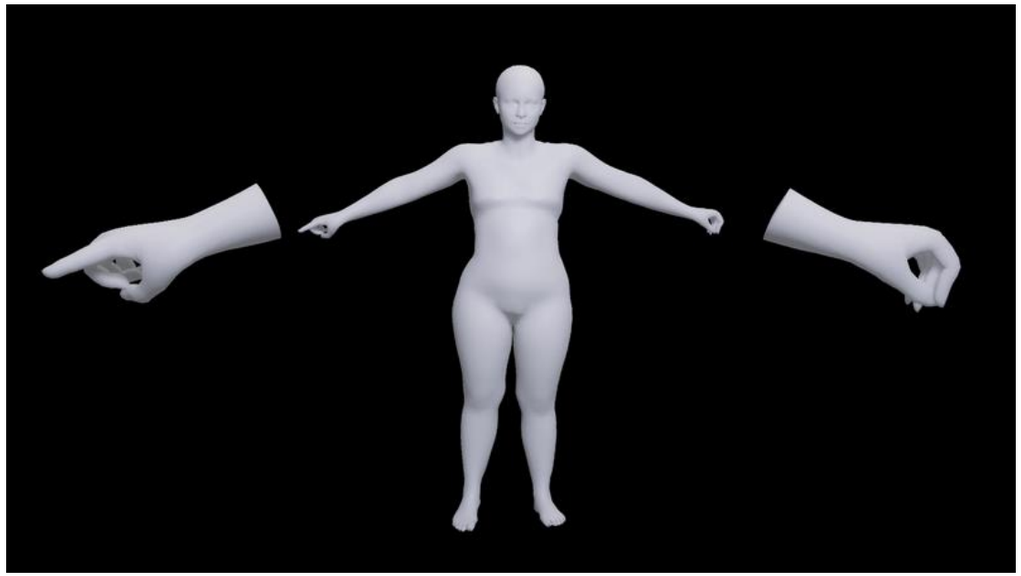
Body-conditioned Hand Motion Generation

The hand cannot be generated independently of the body

A full-body diffusion prior jointly models body and hands.
It demonstrates:

- detailed fingers
- object-conditioned motion
- self-interaction
- full-body coordination

and shows why body–hand joint modeling is preferable to treating hands as an isolated subsystem.



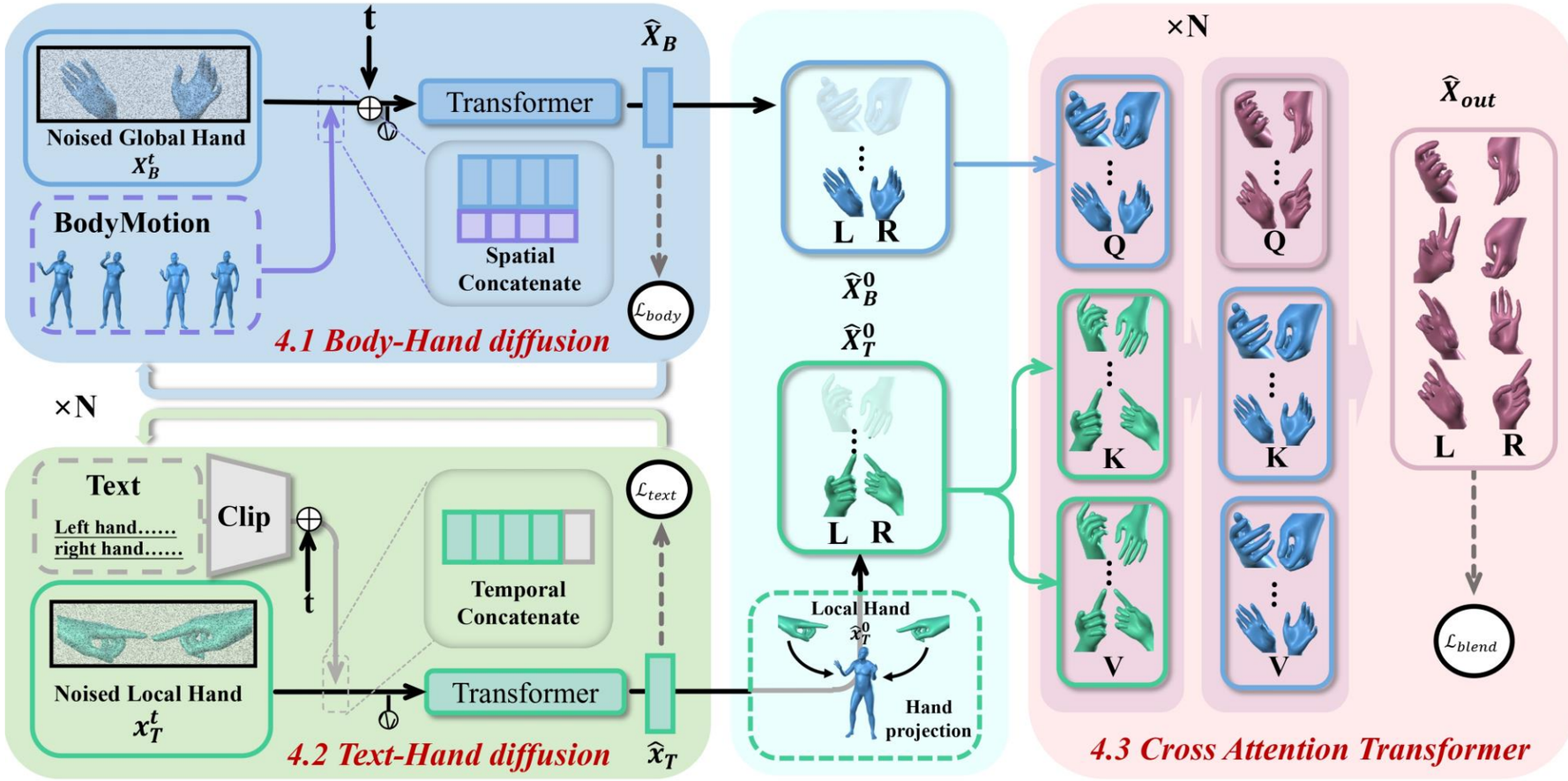
How can we generate detailed two-hand motion that is simultaneously consistent with the person's body movement and an explicit textual instruction?

Body-conditioned Hand Motion Generation

Both2hands: Inferring 3d hands from both text prompts and body dynamics.

The problem is that the two conditioning signals provide complementary information:

- **Body dynamics → implicit context:** where/how the hands should move given the body.
- **Text → explicit control:** what the hands should do.

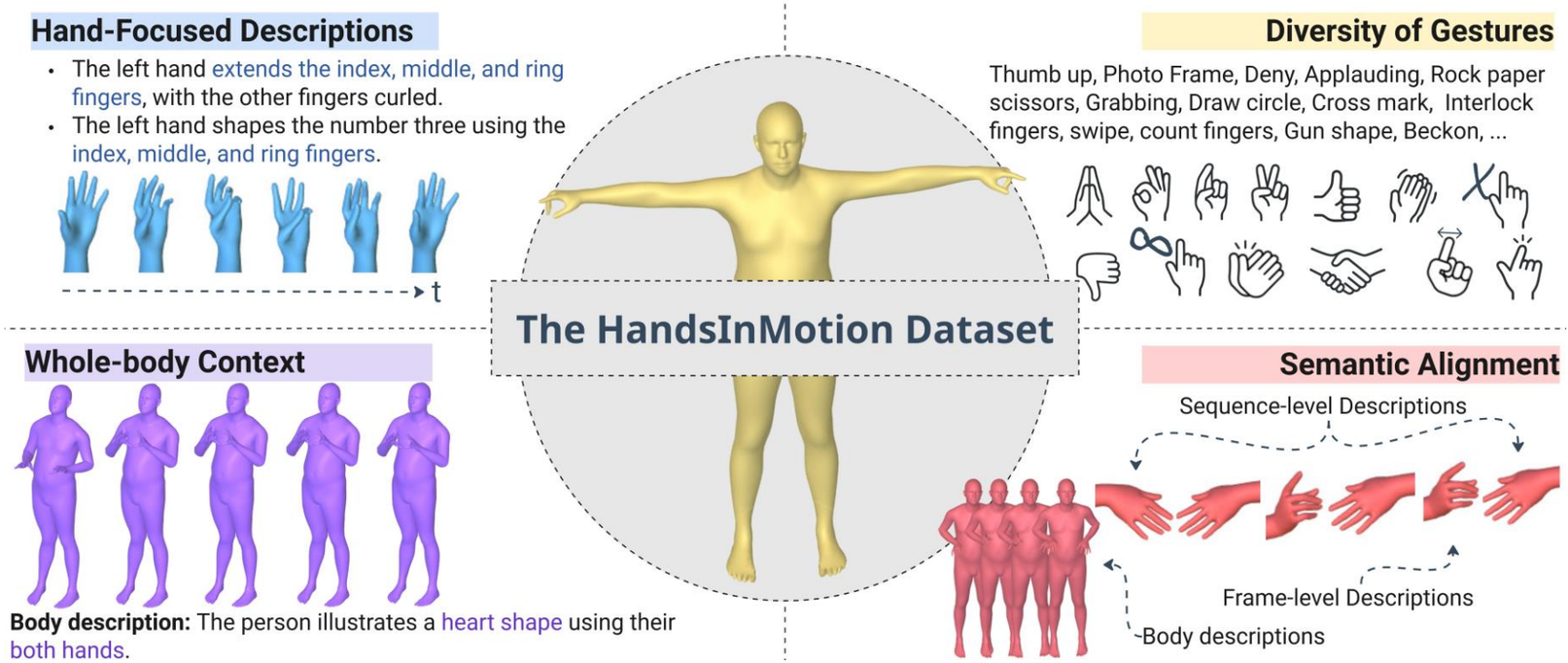


Zhang, Wenqian, et al. "Both2hands: Inferring 3d hands from both text prompts and body dynamics." *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2024.

Body-conditioned Hand Motion Generation

HandsInMotion Dataset

- 6610 Clips
- Mocap-based
- 84 gesture classes
- Sequence & frame level annotations
  - 3 high level
  - 3 finegrained descriptions / sequence



Body-conditioned Hand Motion Generation

HandsInMotion Dataset





Acquisition Setup

Mocap setup

- mRGB pipelines often struggle with occlusions & fine finger articulation
- For hand-centric motion modeling → accuracy matters
- We use Optical Motion Capture (Qualisys) for sub-millimeter precision



Marker Configuration

- 52 retroreflective markers per subject
- Hands → 30 markers → MCP, PIP, DIP
- Body → 22 markers following SMPL alignment

Gesture Recording Protocol

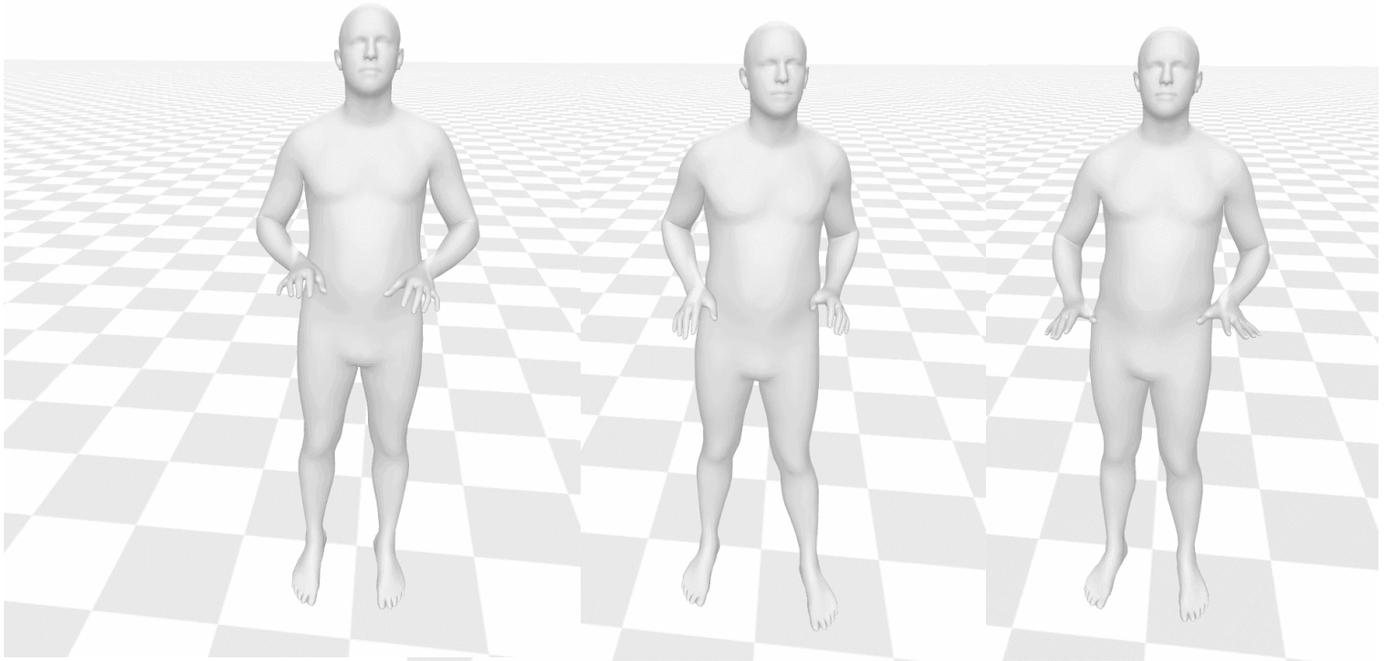
- 30 participants
- 84 gestures: performed with left, right & both hands
- Each gesture repeated 3 times & natural variations
- Sequence duration → 10–30s at 120 FPS

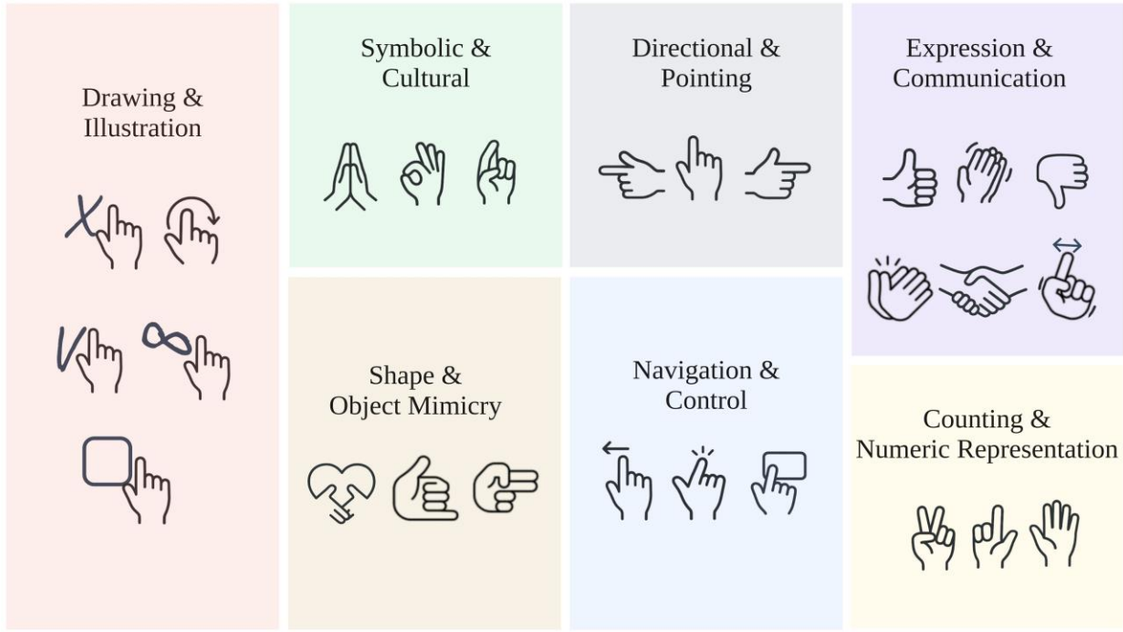
Body-conditioned Hand Motion Generation

HandsInMotion Dataset

Annotations / sequence

- 1 Gesture label
- 3 High-level descriptions
- 3 Fine-grained descriptions
- Frame-level annotation (1 / frame)





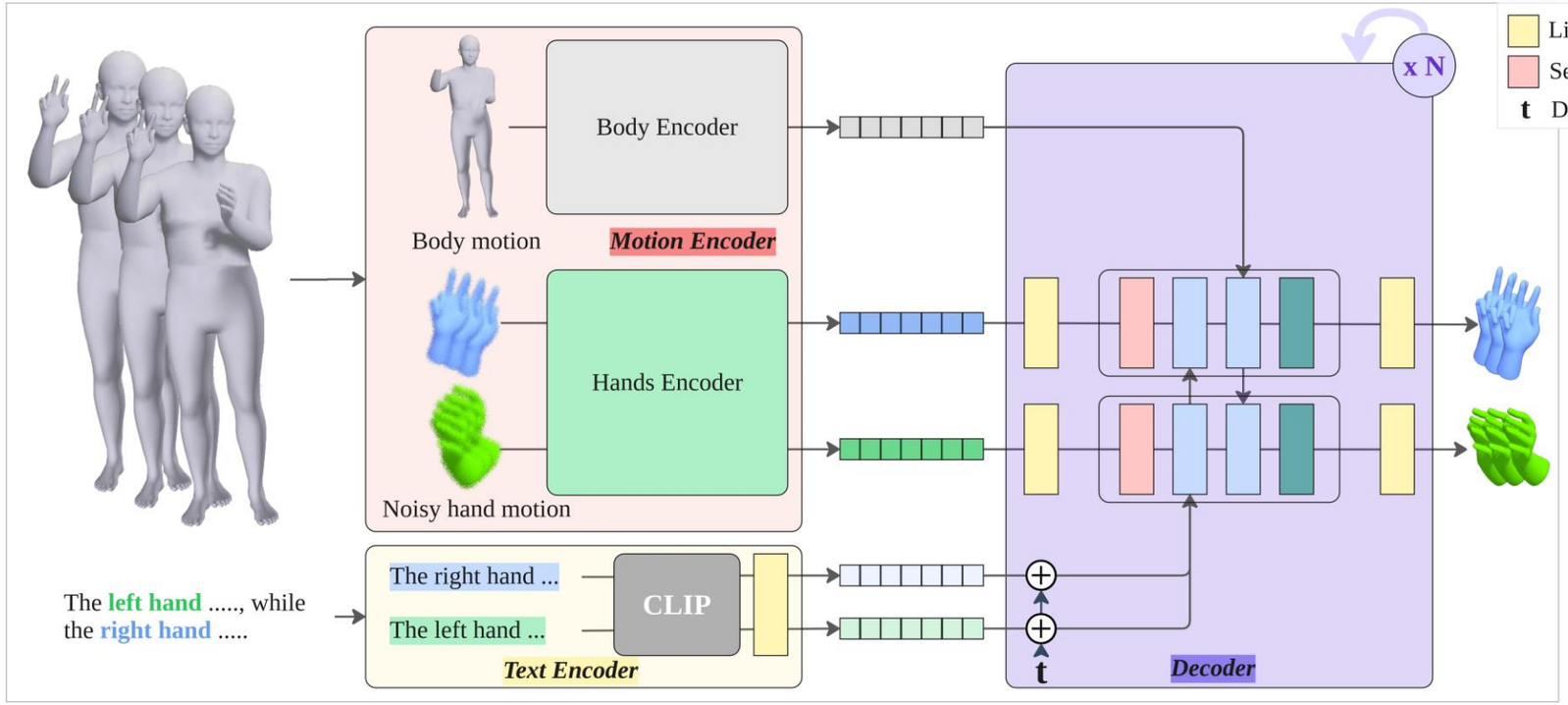
Body-conditioned Hand Motion Generation

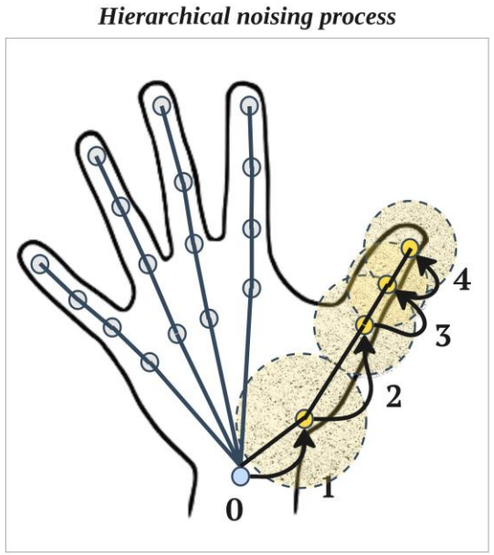
HandsInMotion Method

An anatomically guided diffusion framework that embeds hand-specific anatomical constraints directly into the noising process

•Hierarchical conditioned forward process

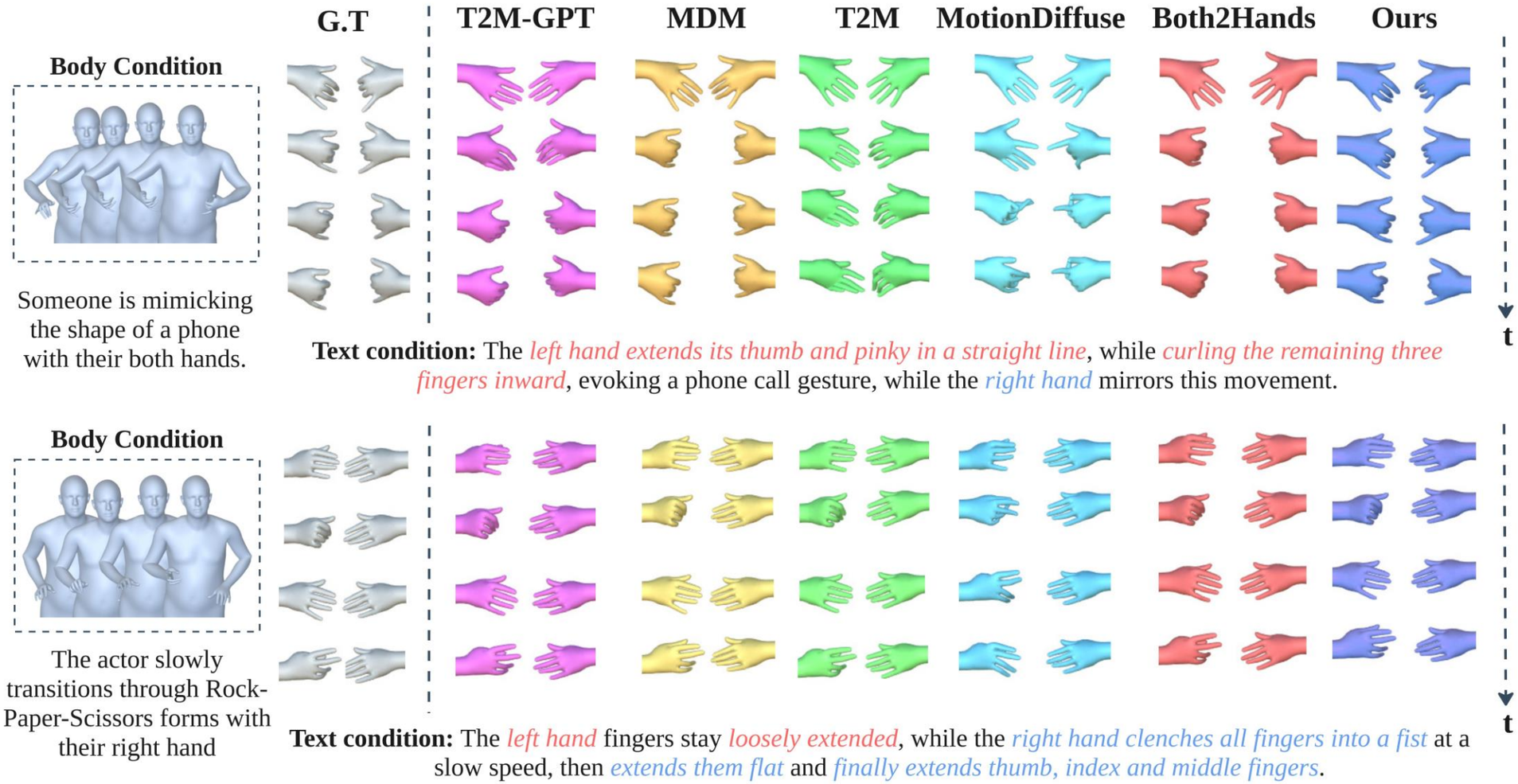
- Joint-specific anatomical constraints define noise ranges
- Timestep–dependent constraint relaxation to preserve motion diversity
- Anatomical priors grounded in dataset statistics:





Body-conditioned Hand Motion Generation

HandsInMotion - Qualitative Results

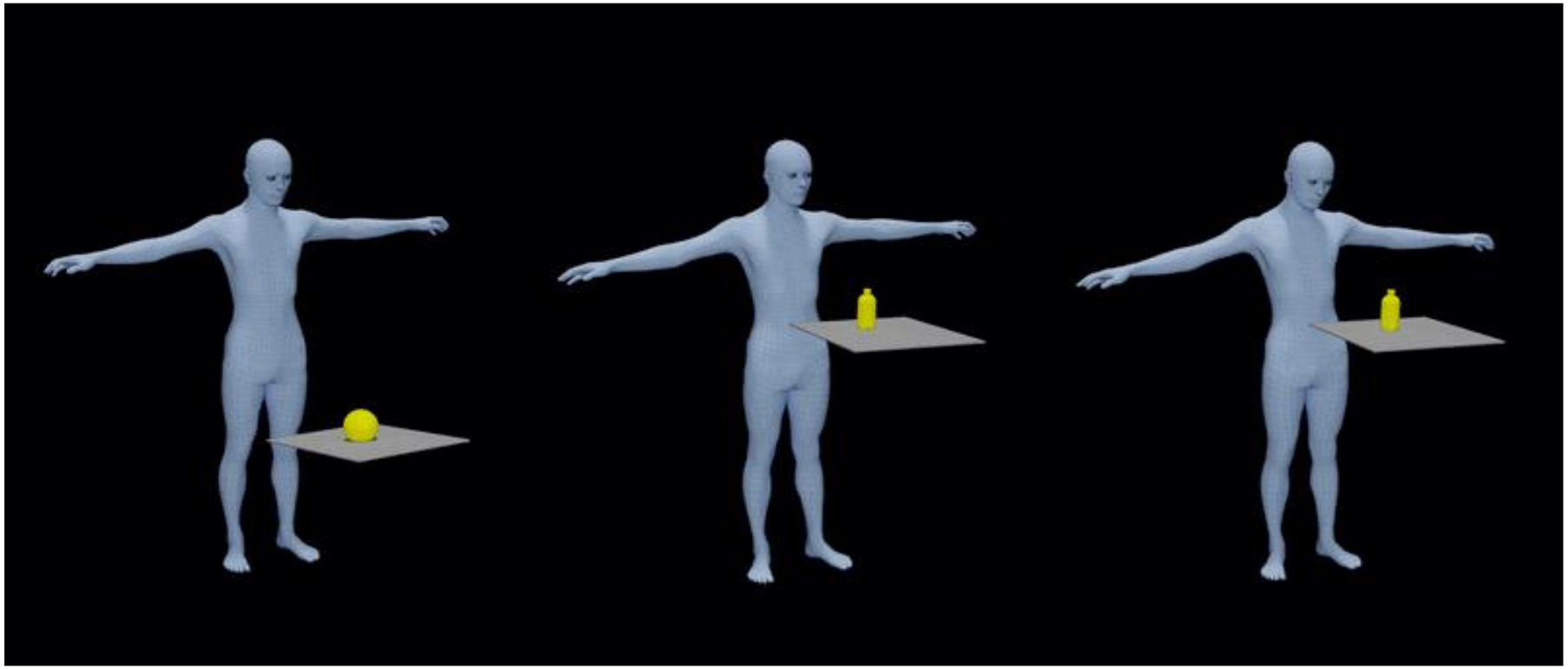


Object-conditioned Hand Motion Generation

Grasping Objects

GRAB dataset provided whole-body human grasping data with detailed 3D meshes/contact and introduced GrabNet for conditional grasp generation.

object
↓
affordance / geometry
↓
contact points
↓
wrist trajectory
↓
finger closure
↓
grasp

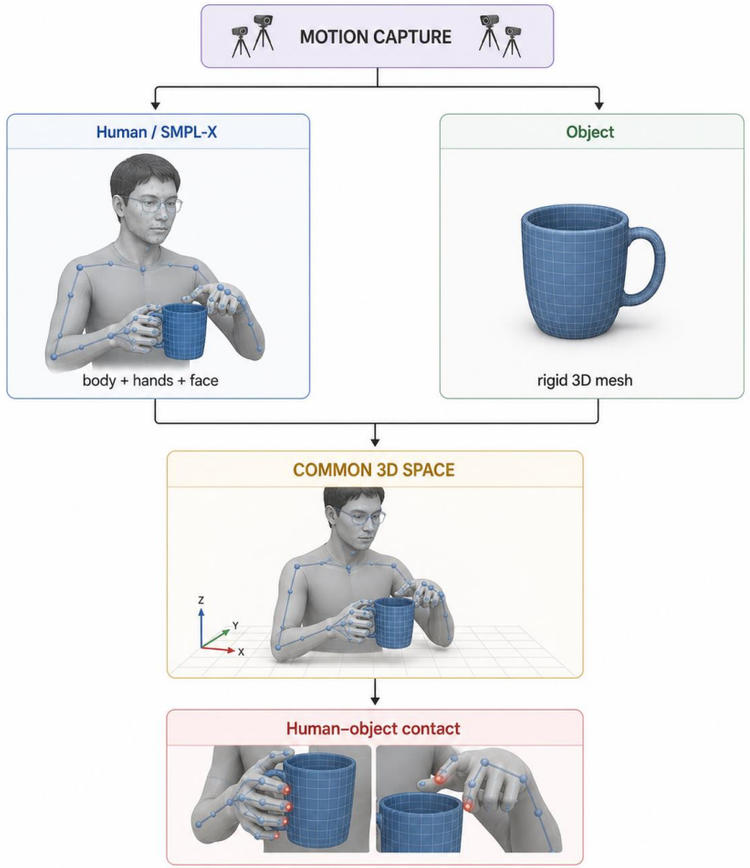


Taheri, Omid, et al. "GRAB: A dataset of whole-body human grasping of objects." *European conference on computer vision*. Cham: Springer International Publishing, 2020.

Object-conditioned Hand Motion Generation

Objects Tracking & Reconstruction

Affordance & Geometry





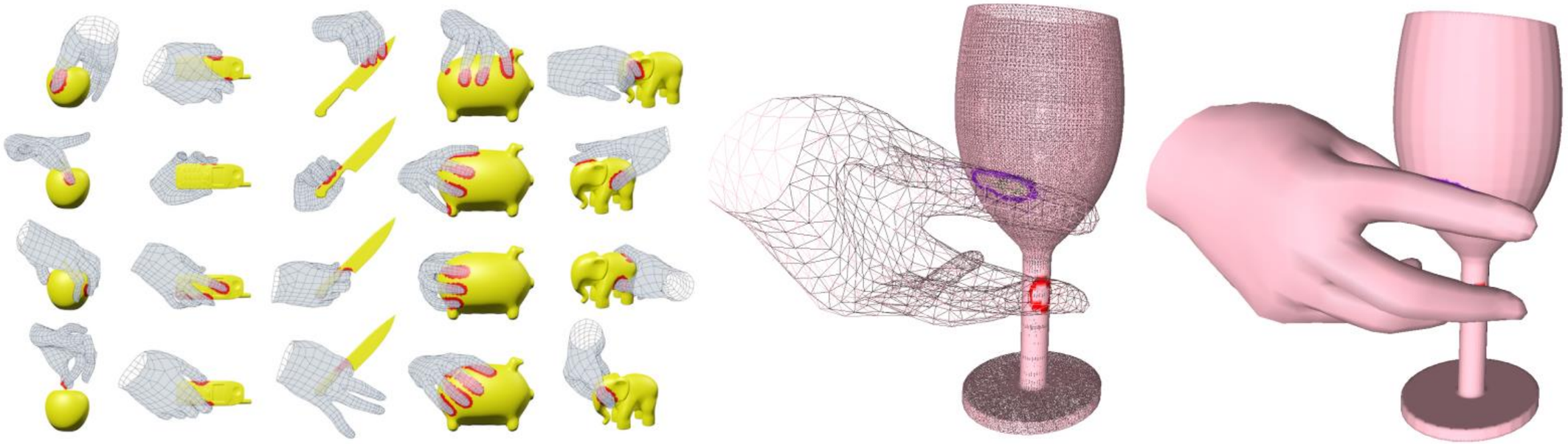
- human with **SMPL-X**, whose pose, shape and translation parameters give a moving 3D body mesh.
- Manipulated object is represented as a **rigid mesh** with its rotation and translation tracked throughout the sequence.

Taheri, Omid, et al. "GRAB: A dataset of whole-body human grasping of objects." *European conference on computer vision*. Cham: Springer International Publishing, 2020.

Object-conditioned Hand Motion Generation

Contact points / map

Object geometry + hand pose + contact → interaction

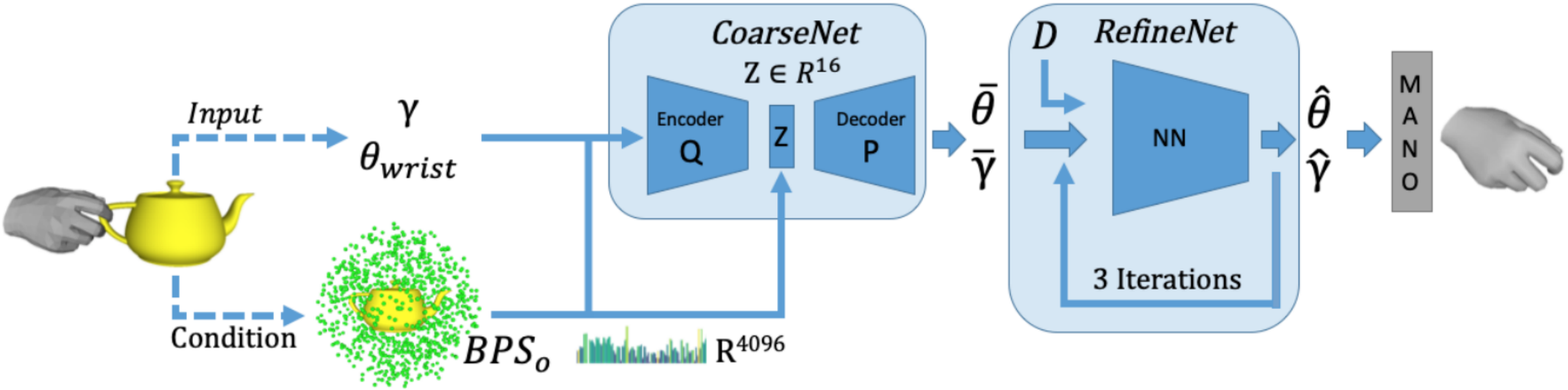


Contact in GRAB is inferred from geometric proximity and other criteria, rather than being directly measured by tactile sensors.

Taheri, Omid, et al. "GRAB: A dataset of whole-body human grasping of objects." *European conference on computer vision*. Cham: Springer International Publishing, 2020.

Object-conditioned Hand Motion Generation

GrabNet: object → hand grasp



- GRAB made human–object interaction geometrically observable
- GrabNet showed that this representation can support object-conditioned grasp generation.

Taheri, Omid, et al. "GRAB: A dataset of whole-body human grasping of objects." *European conference on computer vision*. Cham: Springer International Publishing, 2020.

Object-conditioned Hand Motion Generation

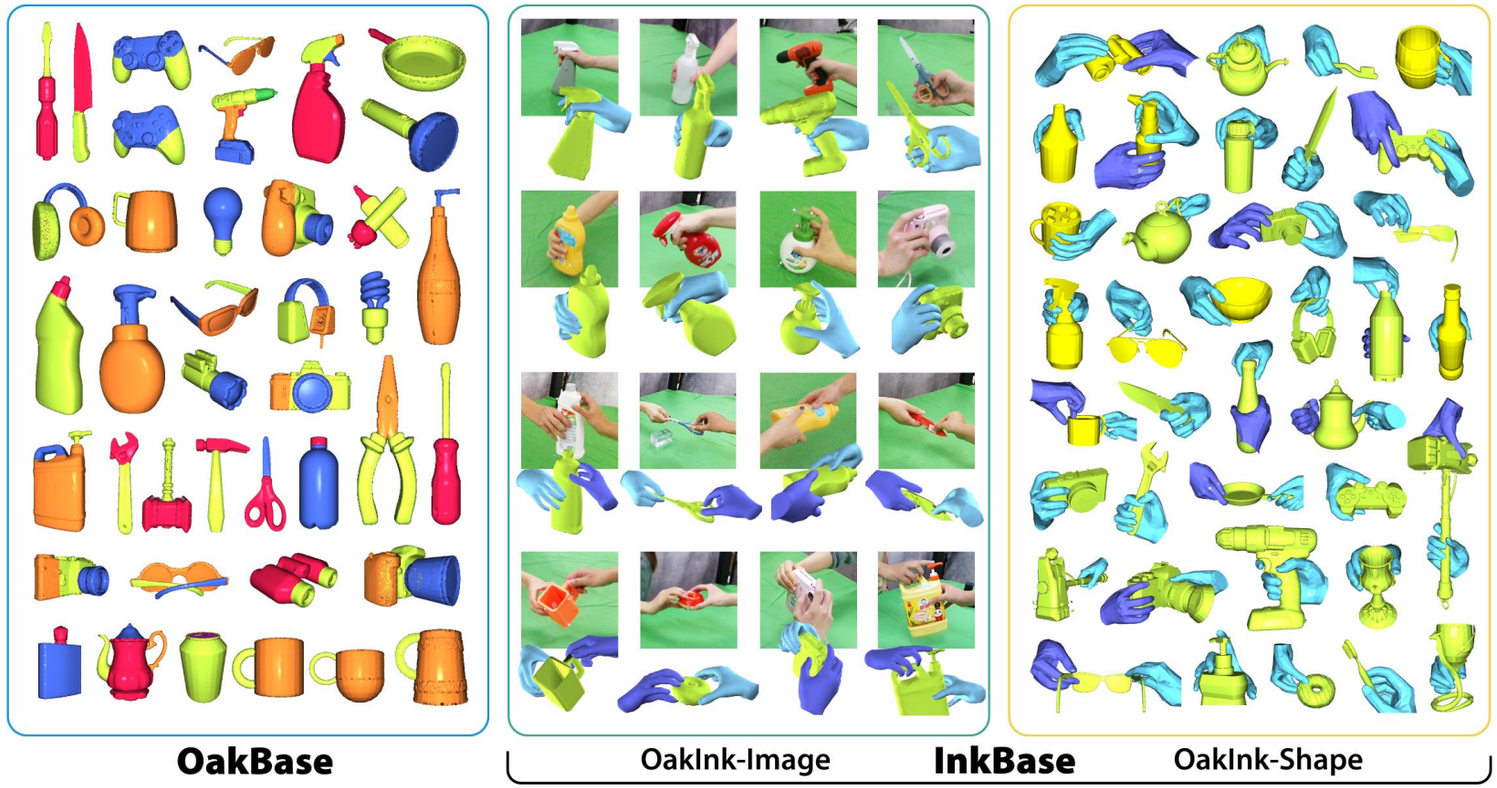
OakInk: A Large-scale Knowledge Repository for Understanding Hand-Object Interaction

GRAB tells us how humans grasp objects; OakInk adds knowledge about *what an object affords* and *why/how the human interacts with it*.

- Object affordances
- Hand interactions
- Dynamic grasping/handover
- Large-scale hand-object pairs

OakBase
- 1,800 common household objects
- 32 object categories
- part-level segmentation
- 30 semantic attribute phrases

The object is not treated simply as mug but rather as a structured object with meaningful parts and attributes.



Yang, Lixin, et al. "Oakink: A large-scale knowledge repository for understanding hand-object interaction." *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2022.

Object-conditioned Hand Motion Generation

OAKINK2: A Dataset of Bimanual Hands-Object Manipulation in Complex Task Completion

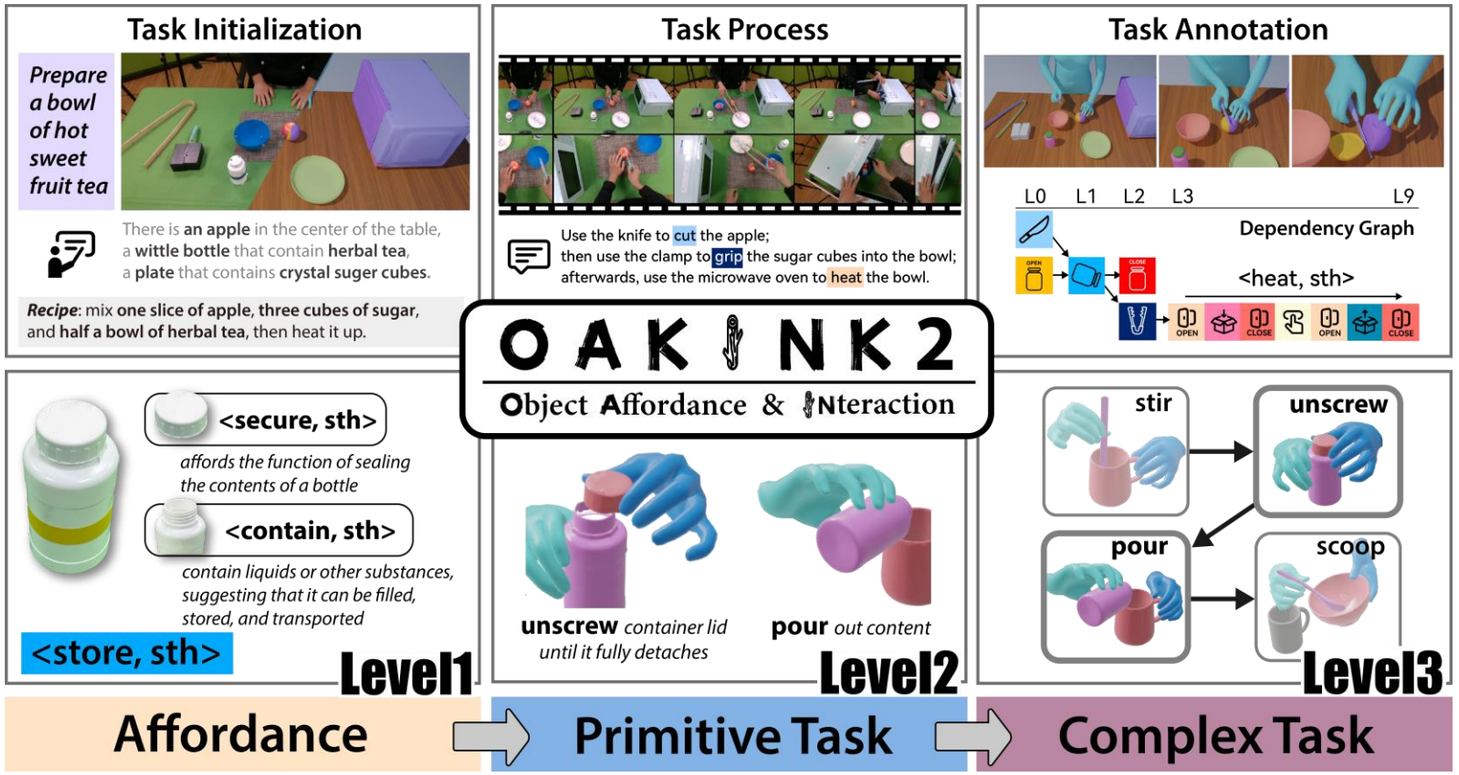
OakInk (2022)
- single-object,
- affordance + intent

OAKINK2 (2024)
- bimanual + multi-object
- complex tasks

COMPLEX TASK:
- "Prepare fruit tea"

PRIMITIVE TASKS
- cut apple
- grasp sugar
- put sugar in bowl
- heat bowl

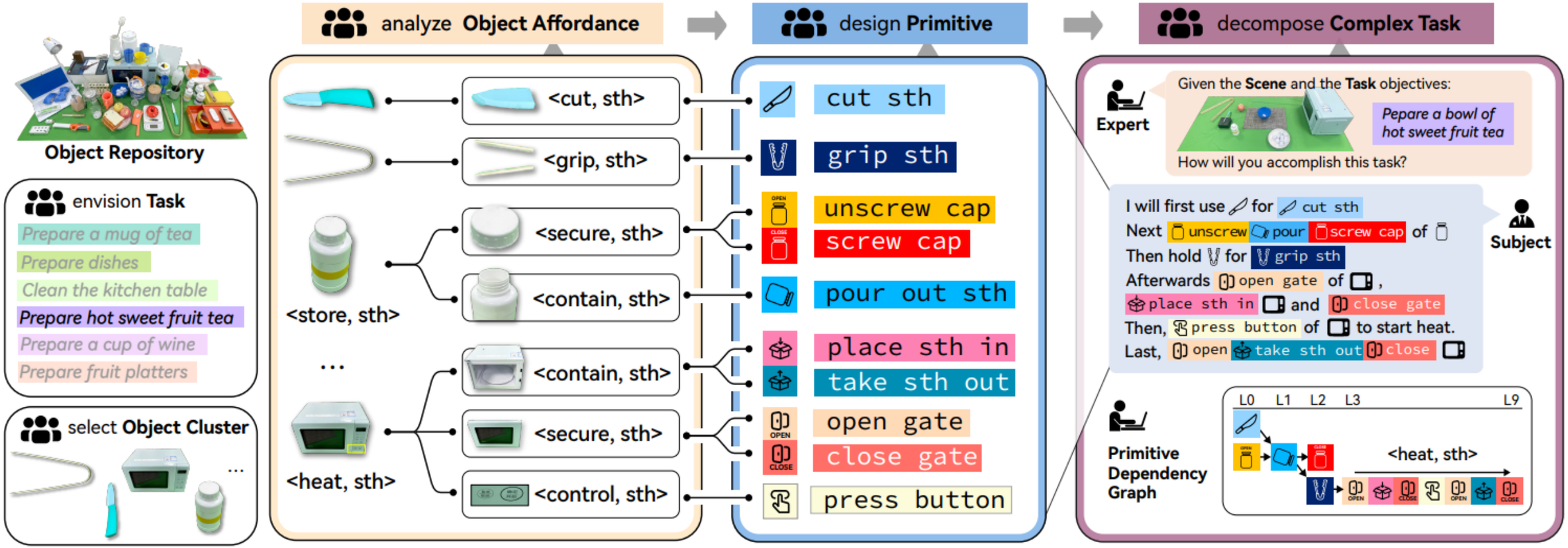
AFFORDANCES: cut / grasp / contain / heat



Zhan, Xinyu, et al. "Oakink2: A dataset of bimanual hands-object manipulation in complex task completion." *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2024

Object-conditioned Hand Motion Generation

OAKINK2: A Dataset of Bimanual Hands-Object Manipulation in Complex Task Completion

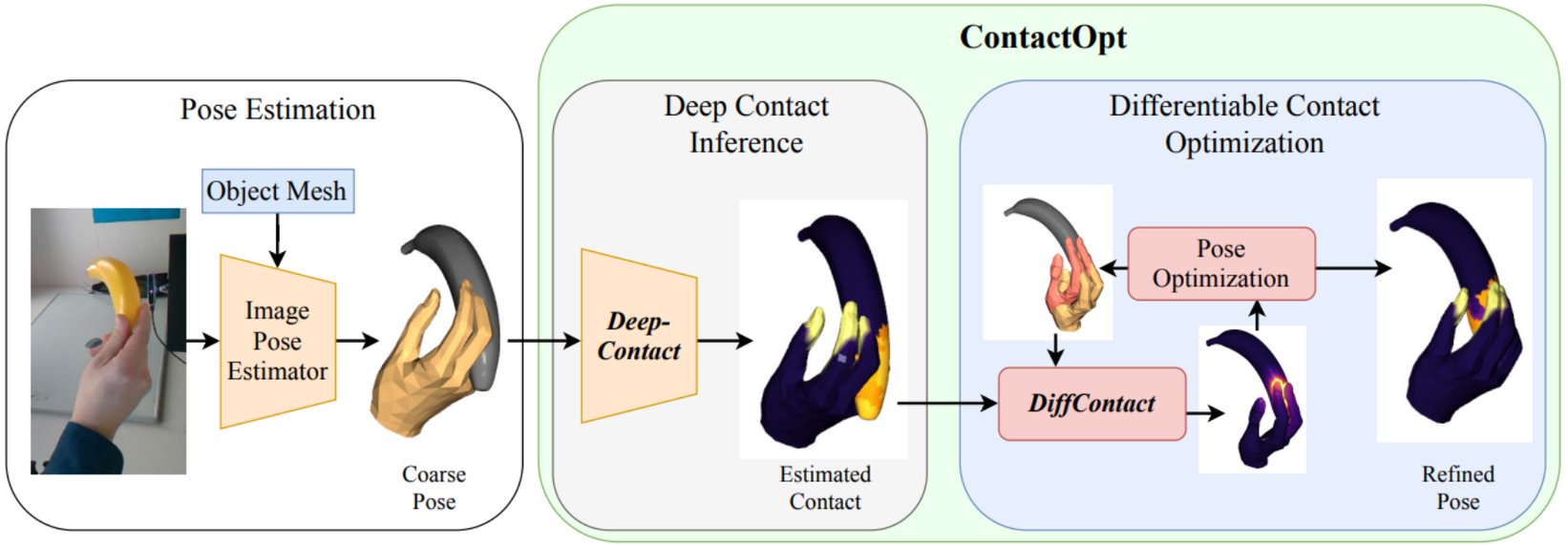


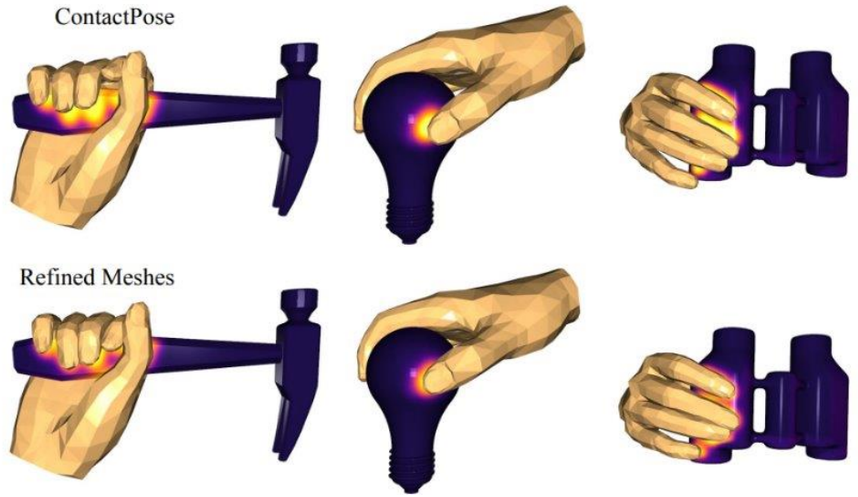
Zhan, Xinyu, et al. "Oakink2: A dataset of bimanual hands-object manipulation in complex task completion." *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2024

Object-conditioned Hand Motion Generation

From grasp pose to grasp motion: A grasp is not a pose — it is a trajectory

Contact is extremely sensitive: the paper notes that even sub-millimeter changes normal to an object surface can determine whether a fingertip is actually touching the object.



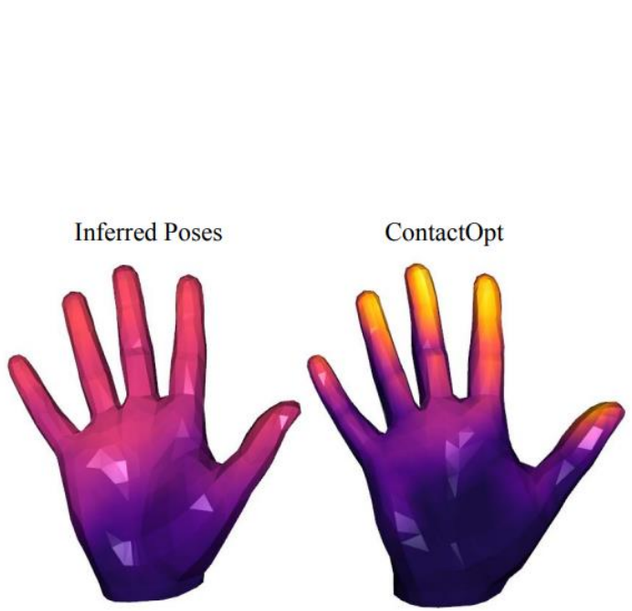


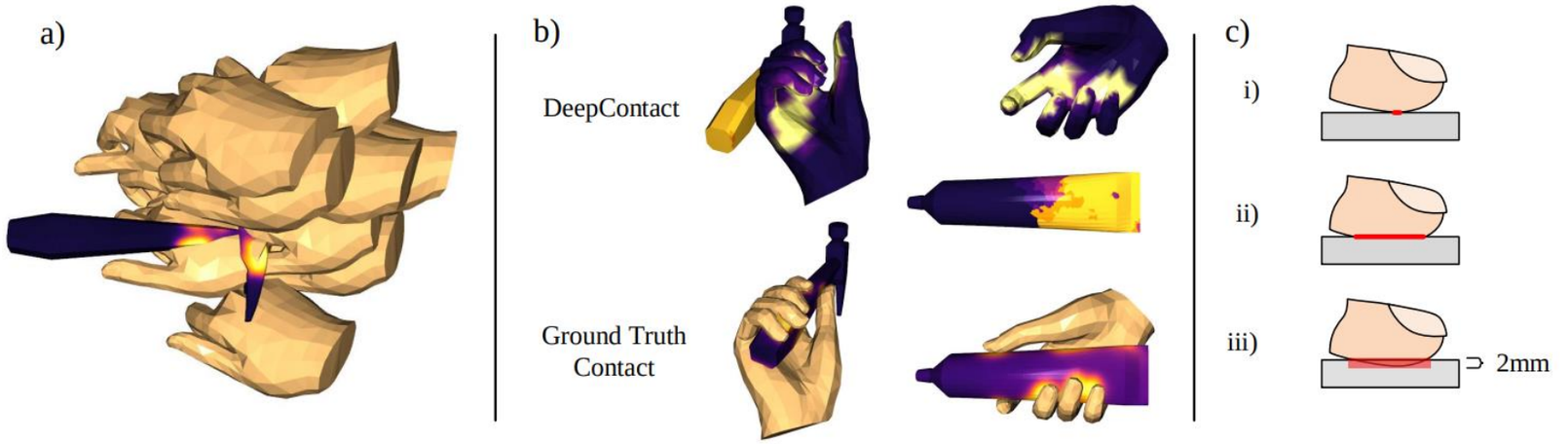
Given the hand and object meshes, **DeepContact predicts desirable contact across both surfaces**. Each mesh vertex receives a value between 0 and 1 indicating whether it should be in contact.

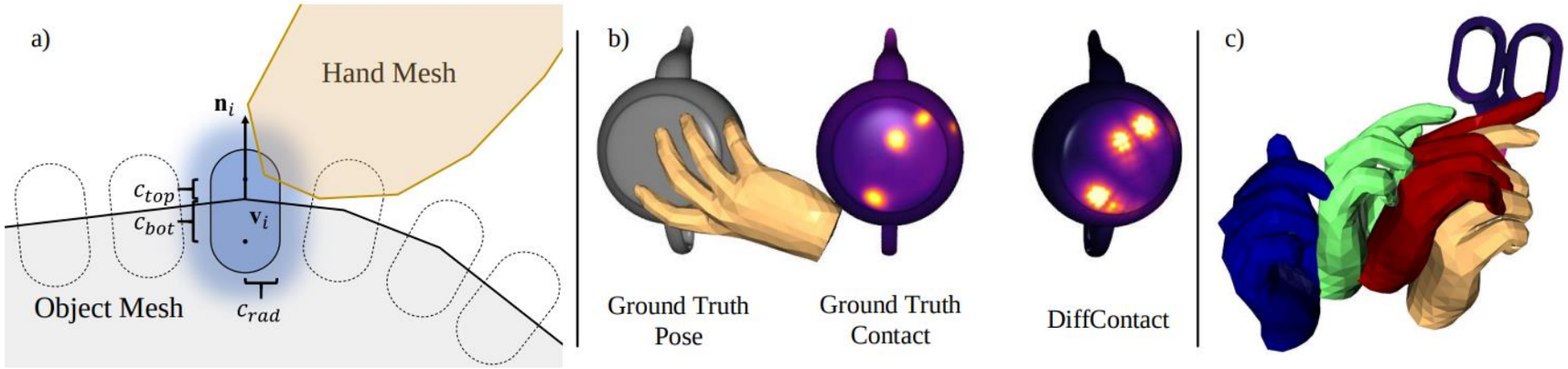
Grady, Patrick, et al. "Contactopt: Optimizing contact to improve grasps." *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2021.

Object-conditioned Hand Motion Generation

From grasp pose to grasp motion: A grasp is not a pose — it is a trajectory





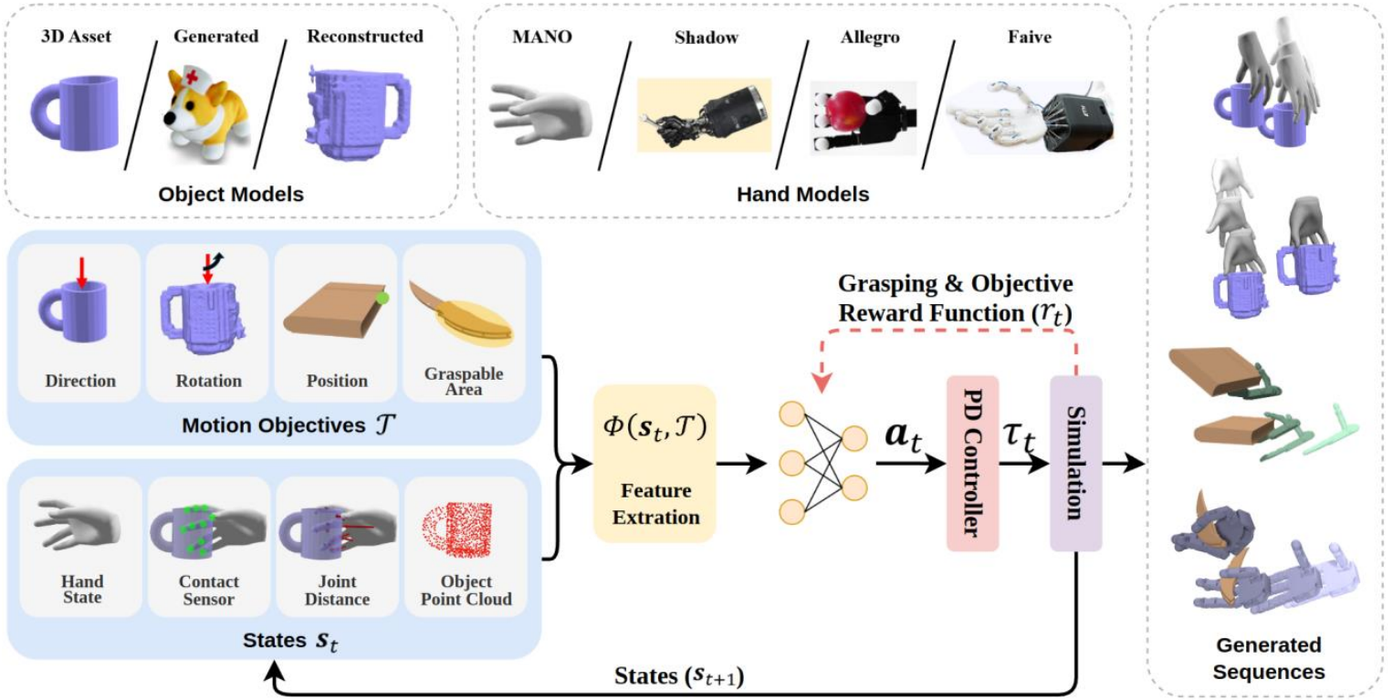


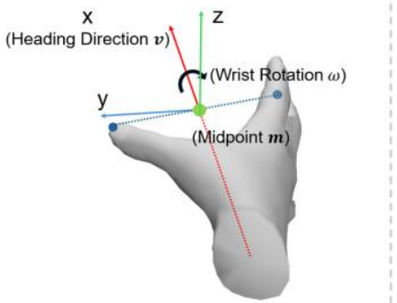
Grady, Patrick, et al. "Contactopt: Optimizing contact to improve grasps." *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2021.

Object-conditioned Hand Motion Generation

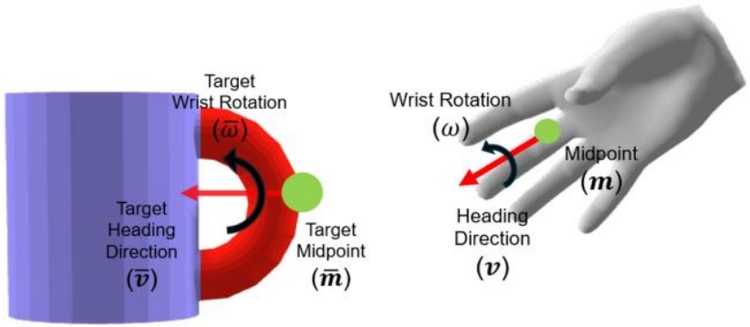
GraspXL: Generating Grasping Motions for Diverse Objects at Scale

Earlier approaches often focused on **one grasp objective** or required expensive hand–object interaction data. GraspXL instead asks:
Can we generate diverse grasping motions for arbitrary unseen objects, while explicitly controlling how the grasp happens?





(a) Hand local coordinate system



(b) Notations

Zhang, Hui, et al. "Graspxl: Generating grasping motions for diverse objects at scale." *European Conference on Computer Vision*. Cham: Springer Nature Switzerland, 2024

Object-conditioned Hand Motion Generation

GraspXL: Generating Grasping Motions for Diverse Objects at Scale

Synthesize grasping motions with objects generated from tests or reconstructed from in-the-wild videos using off-the-shelf methods.



DreamFusion [Poole et al, 2022]

Generated object



HOLD [Fan et al, 2024]

Reconstructed object

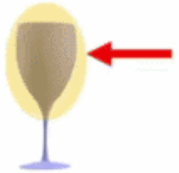








Generate grasping motions with the same objective for different hand models.





MANO Allegro Shadow

Zhang, Hui, et al. "Graspxl: Generating grasping motions for diverse objects at scale." *European Conference on Computer Vision*. Cham: Springer Nature Switzerland, 2024

Object-conditioned Hand Motion Generation

Physics enters the generator

Visual plausibility ≠ physical plausibility

Physics-Aware Hand-Object Interaction Denoising
Explicitly models:

- grasp credibility
- manipulation feasibility

as physical-plausibility constraints.



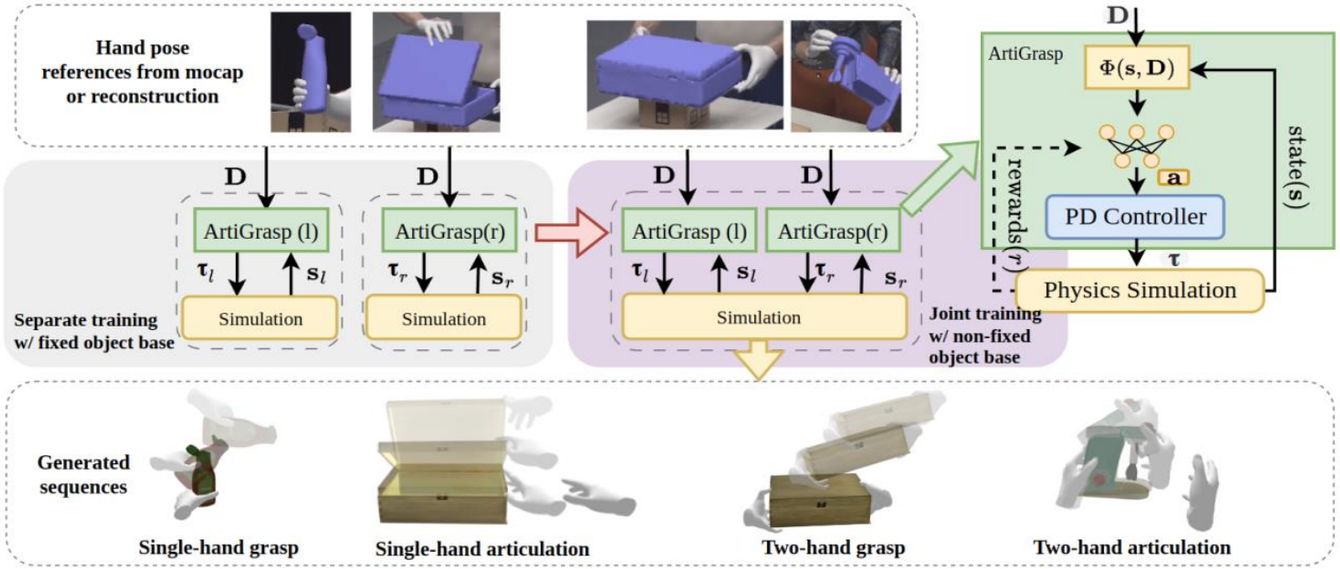
Luo, Haowen, Yunze Liu, and Li Yi. "Physics-aware hand-object interaction denoising." *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2024.

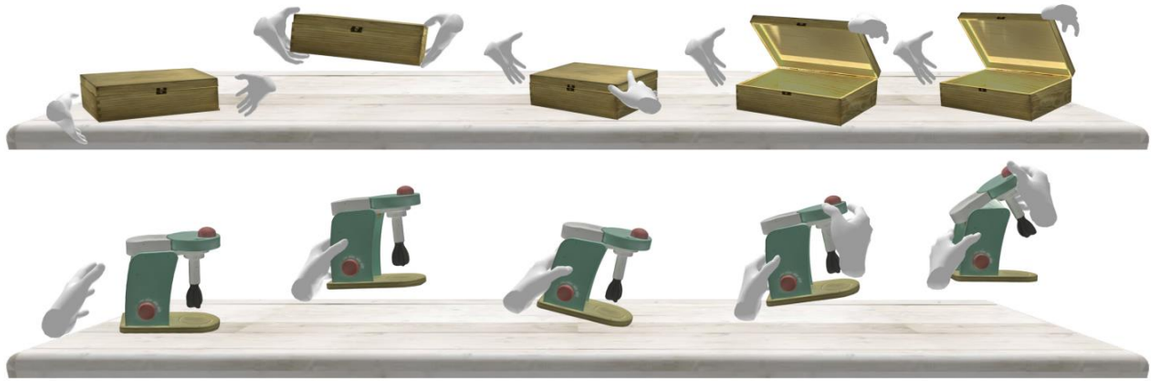
Object-conditioned Hand Motion Generation

ArtiGrasp: Physically Plausible Synthesis of Bi-Manual Dexterous Grasping and Articulation

Uses reinforcement learning + physics simulation to generate bimanual grasping and articulation.
Key message:

- The generator is increasingly becoming **a learned prior + a physical constraint solver**.





Zhang, Hui, et al. "Artigrasp: Physically plausible synthesis of bi-manual dexterous grasping and articulation." *2024 International Conference on 3D Vision (3DV)*. IEEE, 2024.

Object-conditioned Hand Motion Generation

Generalization to unseen objects

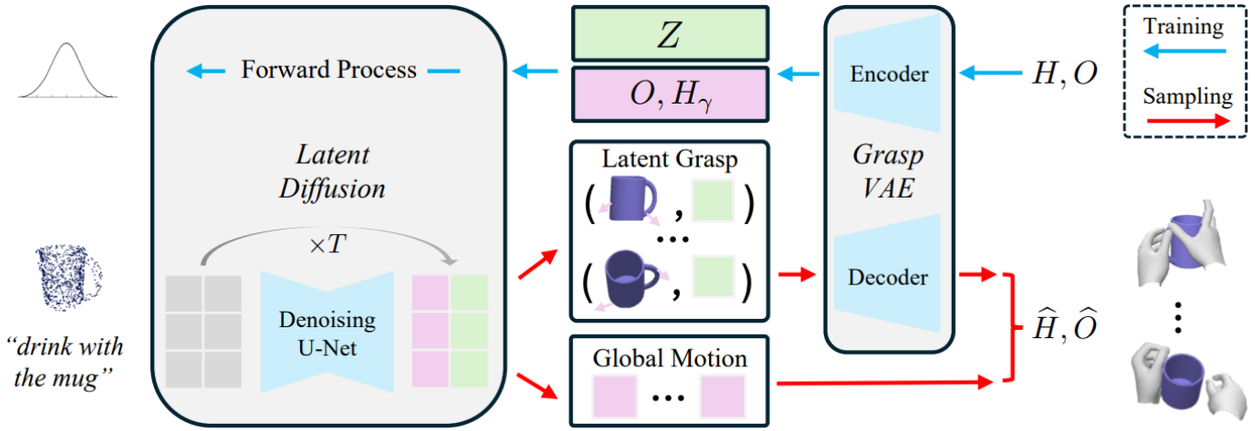
Can the hand interact with an object it has never seen?

LatentHOI: LatentHOI: On the Generalizable Hand Object Motion Generation with Latent Hand Diffusion.
The paper explicitly targets generalization to unseen objects.
Its central idea:

- separate high-level temporal motion from fine-grained spatial hand–object interaction in a latent diffusion model.
- It evaluates on **GRAB, DexYCB and OakInk**, including single-hand and bimanual motion.

training objects
↓
memorize grasps
X

training objects
↓
learn interaction prior
↓
generalize to newgeometry
✓



Zhang, Hui, et al. "Artigrasp: Physically plausible synthesis of bi-manual dexterous grasping and articulation." *2024 International Conference on 3D Vision (3DV)*. IEEE, 2024.

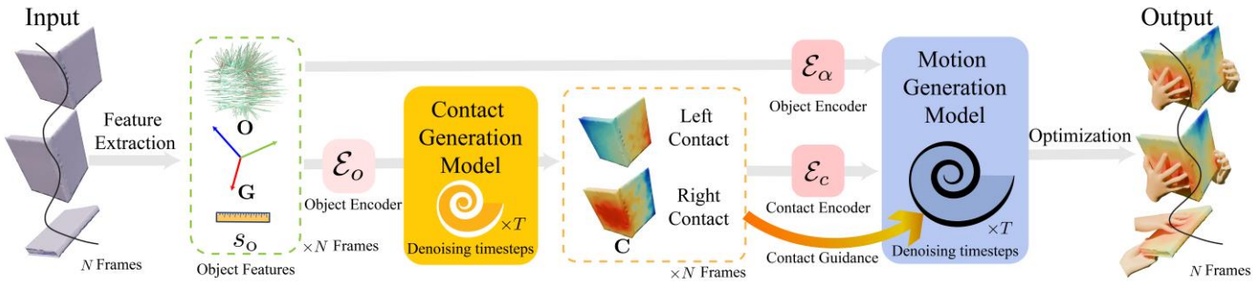
Object-conditioned Hand Motion Generation

Bimanual generation: Two hands ≠ two independent generators

Need to model:

- temporal synchronization
- complementary roles
- contact events
- relative geometry
- independent finger articulation





BimArt: A Unified Approach for the Synthesis of 3D Bimanual Interaction with Articulated Objects

Zhang, Wanyue, et al. "Bimart: A unified approach for the synthesis of 3d bimanual interaction with articulated objects." *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2025.

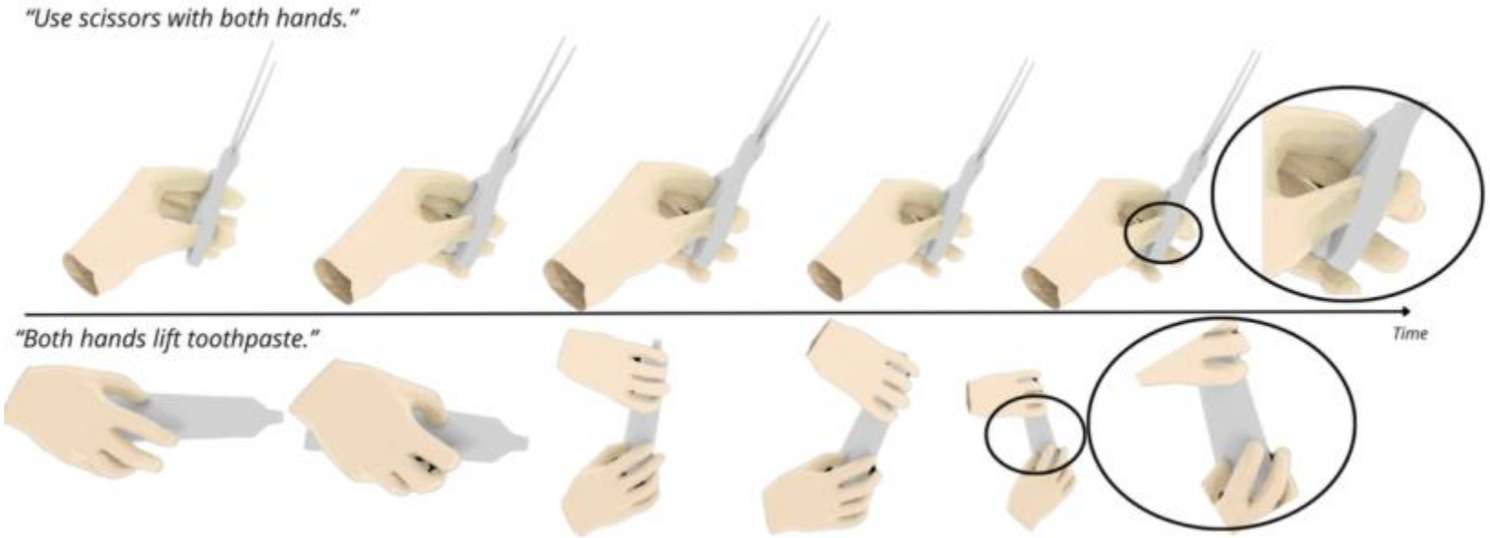
Object-conditioned Hand Motion Generation

When and Where to Touch

A Temporal Contact Prior for Text-Driven 3D Hand-Object Interaction

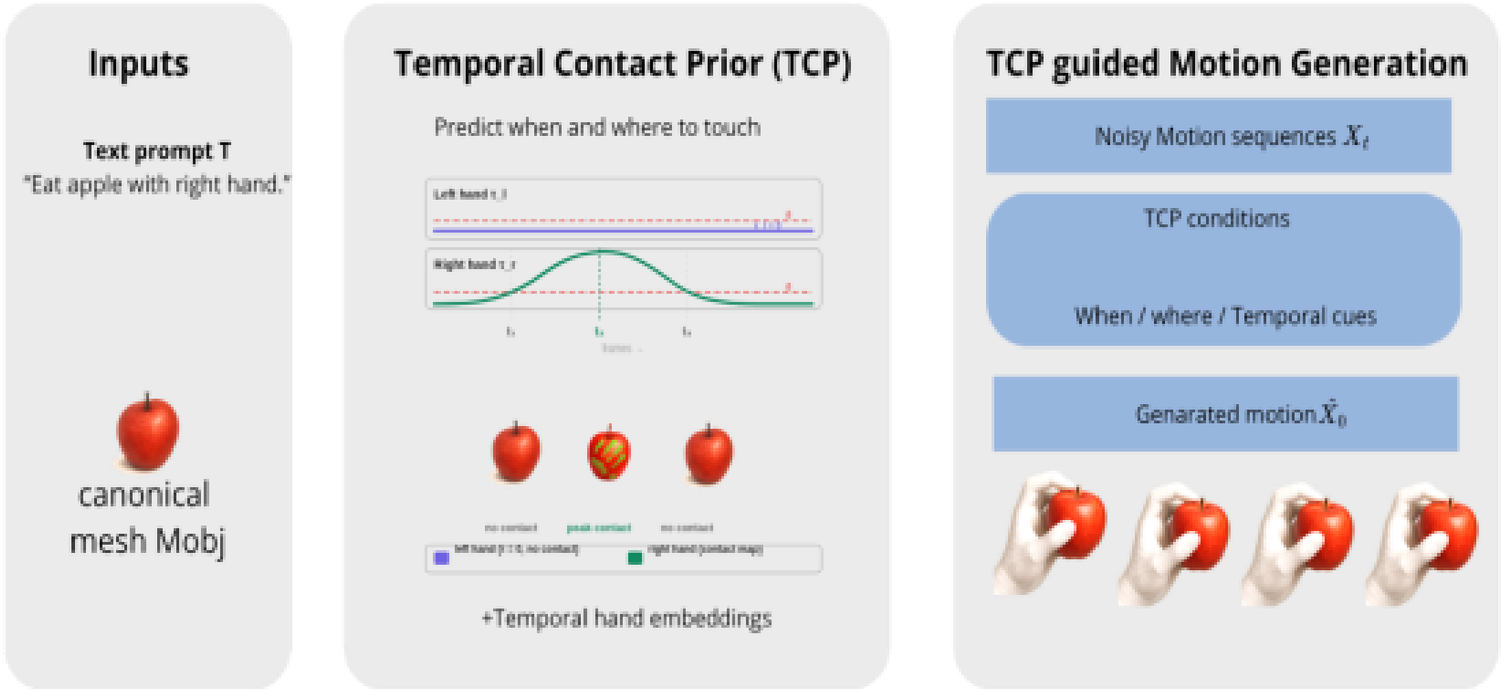
Text gives the intent, but contact makes the motion believable.

- Hand-object interaction needs millimeter-level precision under object-surface constraints.
- Real contact evolves: approach, touch, maintain, shift, release.
- Static contact cues miss this temporal structure and lead to missed grasps, sliding, or penetration.



Object-conditioned Hand Motion Generation

Temporal Contact Prior: what it predicts

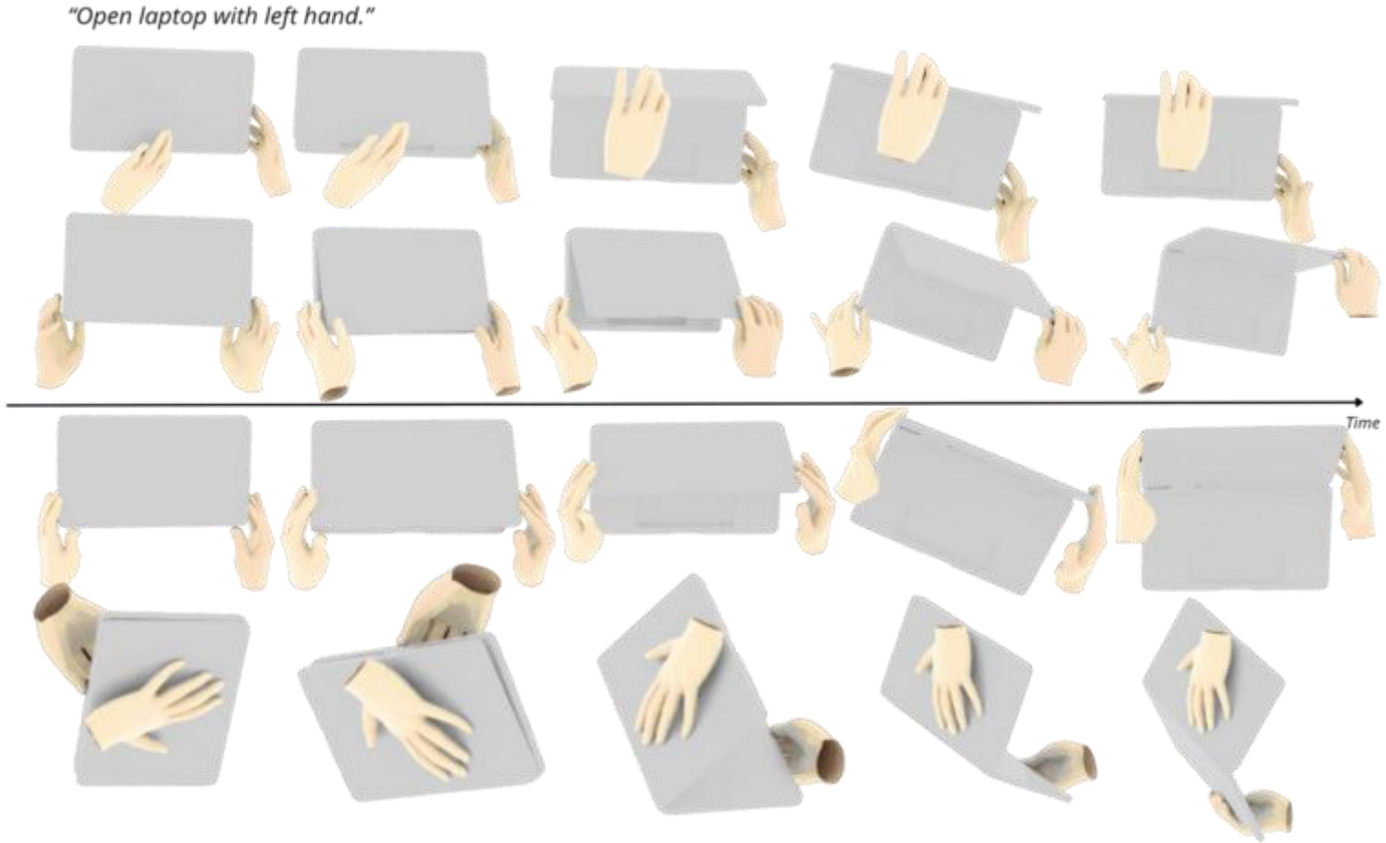


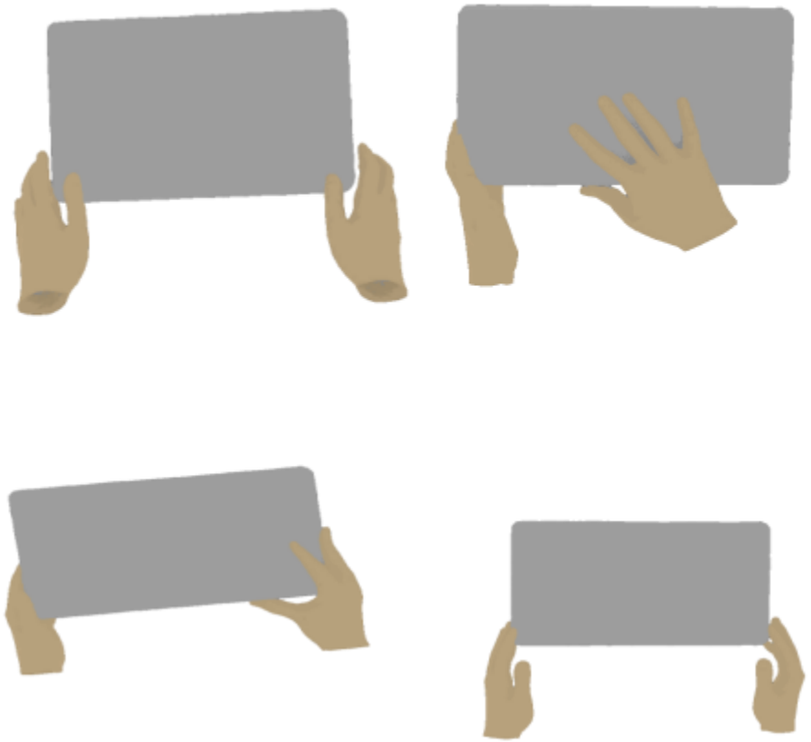
For each hand and frame, TCP outputs:

- Contact activation - when the hand should touch.
- Spatial attention - where on the object surface.
- Temporal embedding - a compact summary of interaction dynamics.

Object-conditioned Hand Motion Generation

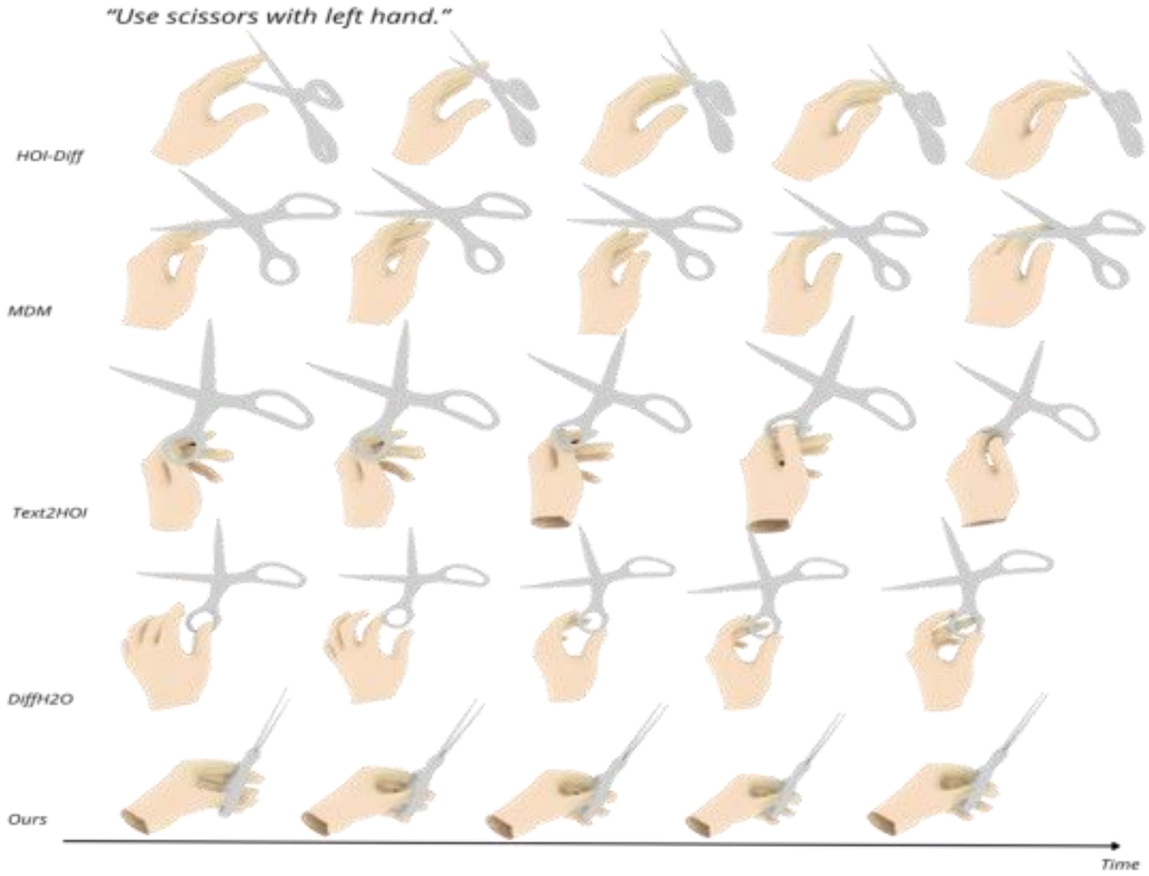
Qualitative results: coherent sequences and diversity





Object-conditioned Hand Motion Generation

Qualitative comparison: Comparison with prior methods



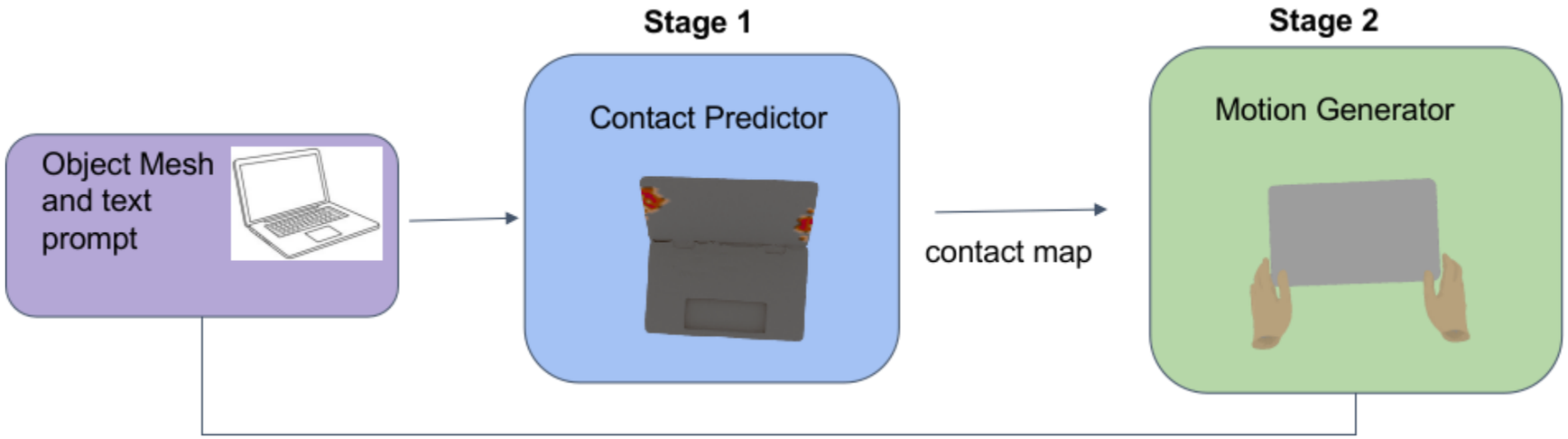
Baselines often miss the scissor rings or produce unstable contact. 2WT-HOI better aligns fingers with functional object regions. The improvement is visible where semantic intent requires precise contact.

Object-conditioned Hand Motion Generation

GCD: Geometry-Constrained Contact-Aware Diffusion for Text-Driven 3D Hand-Object Motion Synthesis

Key observations:

- Hands and objects play asymmetric roles
- The object follows text intent The hand adapts to geometry



Object-conditioned Hand Motion Generation

GCD: Geometry-Constrained Contact-Aware Diffusion for Text-Driven 3D Hand-Object Motion Synthesis

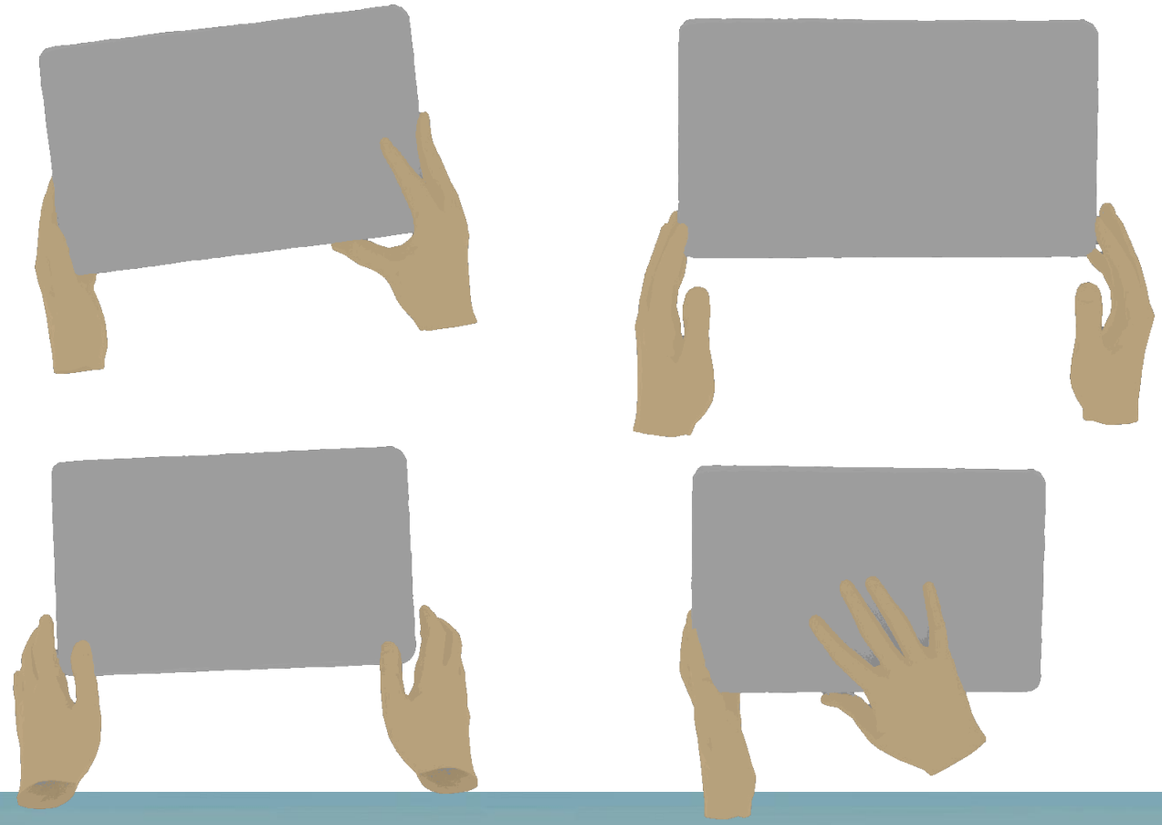
Inspect the Stanford Bunny with both hands.

Prompt (Grab dataset) : Inspect stanford bunny with both hands.



Lift the train with both hands.

Prompt (Grab dataset) : Lift train with both hands.



Ensuring Physical Plausibility

- Biomechanical constraints
- Joint limits and natural motion priors
- Body-hand coordination
- Contact consistency and object interaction
- Real-time generation and latency constraints

Ensuring Physical Plausibility

Kinematic validity	Dynamic validity	Interaction validity
• joint limits • anatomy • self-collision • DOF	• Smoothness • velocity • acceleration • forces	• contact • Penetration • grasp • object motion

Levels of plausibility

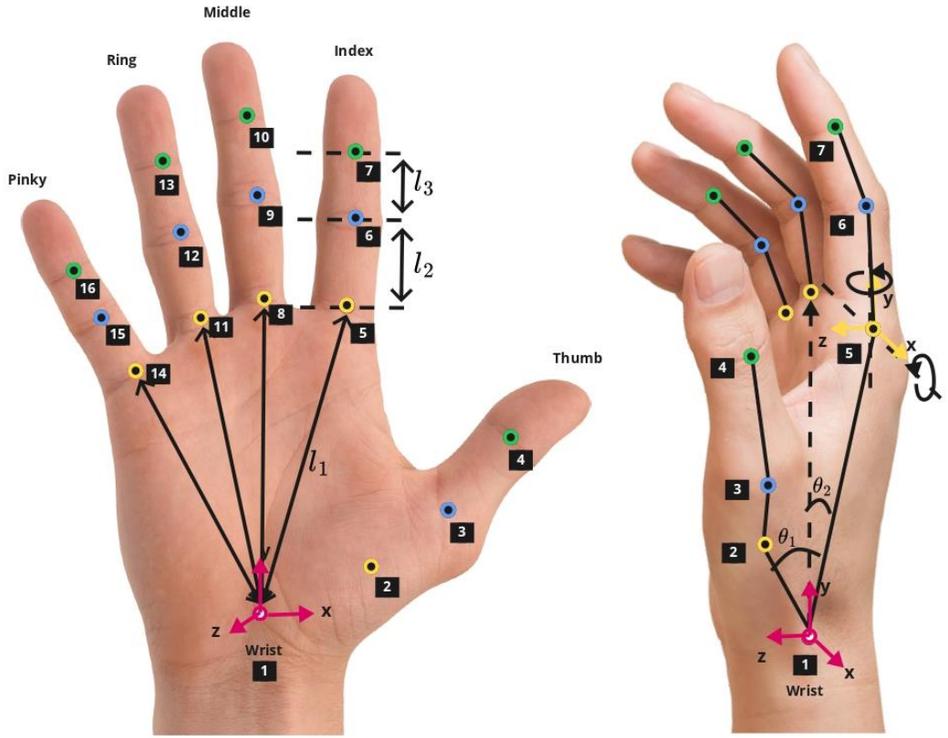
Level	Question	Typical failure
Geometric	Is the pose anatomically possible?	bent/overextended finger
Kinematic	Is the trajectory coherent?	jitter / discontinuity
Dynamic	Could a human physically execute it?	impossible acceleration/torque
Contact	Does interaction obey geometry/contact?	penetration / floating
Functional	Can the action actually work?	object slips / grasp fails

Ensuring Physical Plausibility

A hand can be kinematically valid but physically implausible

Example:
- Fingers within joint limits
- No self-collision
- Smooth trajectory

…but:
- Impossible acceleration
- Unrealistic tendon/muscle configuration
- Object penetrates the palm
- Grasp cannot support the object



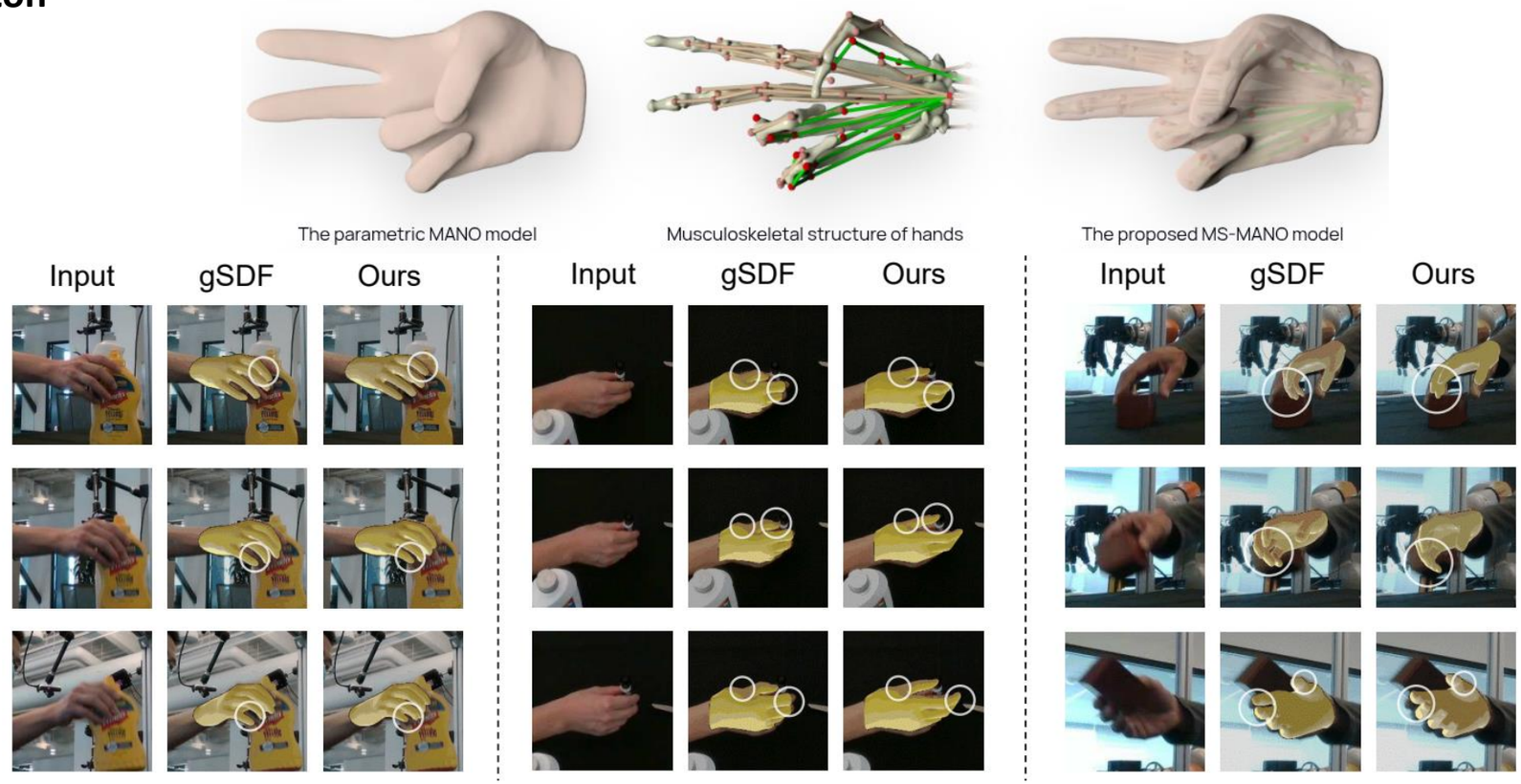
1. Wrist
2. Thumb CMC
3. Thumb MCP
4. Thumb IP
5. Index MCP
6. Index PIP
7. Index DIP
8. Middle MCP
9. Middle PIP
10. Middle DIP
11. Ring MCP
12. Ring PIP
13. Ring DIP
14. Pinky MCP
15. Pinky PIP
16. Pinky DIP

Ensuring Physical Plausibility

The hand is not an arbitrary articulated skeleton

MS-MANO explicitly argues that simplified joint-actuated hand models can generate unnatural motion and introduces a musculoskeletal model coupling muscles/tendons with MANO.

A pose prior constrains what the hand looks like. A biomechanical prior constrains how the hand can move.



Xie, Pengfei, et al. "Ms-mano: Enabling hand pose tracking with biomechanical constraints." *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2024.

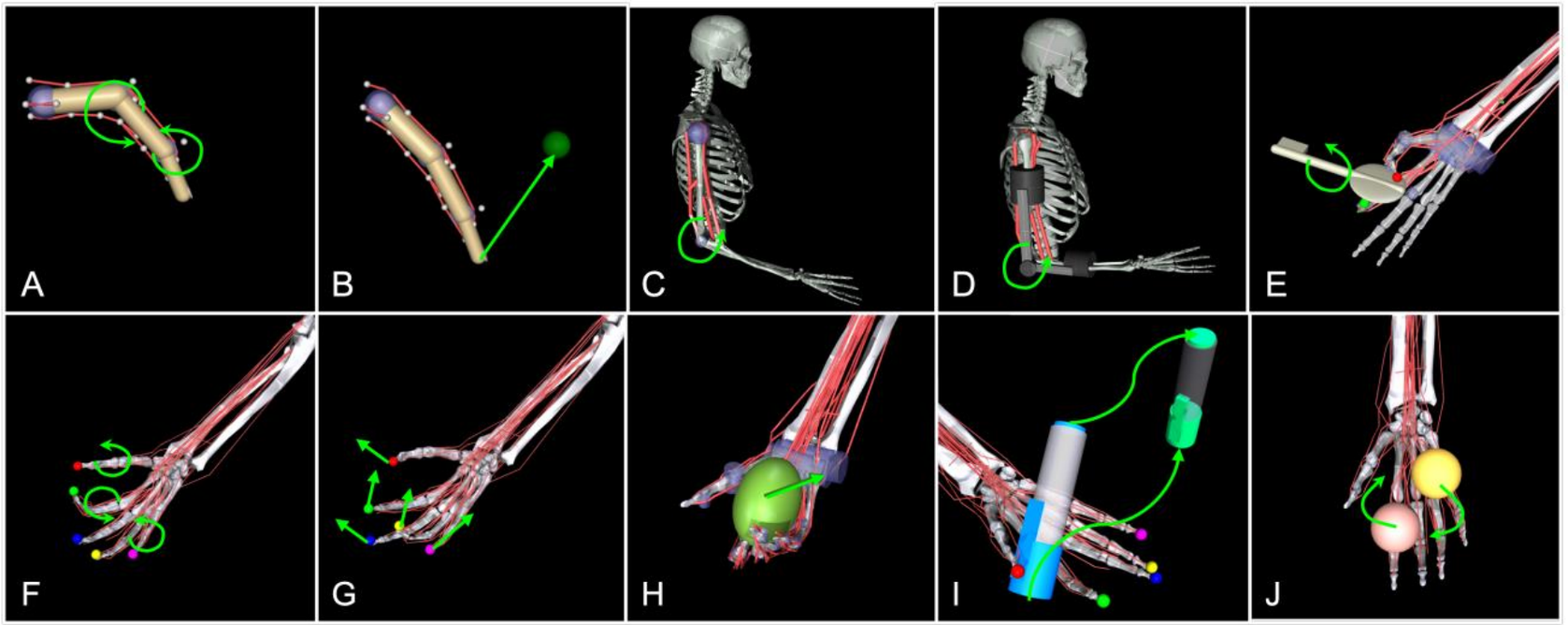
Ensuring Physical Plausibility

MyoSuite: physics as a generative environment

It provides physiologically detailed elbow, wrist and hand models with contact-rich tasks such as:

- turning a key
- pen twirling
- object rotation
- multi-object manipulation
- hand reaching

Could a motion generator produce trajectories **in simulation**, rather than merely fitting motion-capture trajectories?



Finger: joint pose (A), tip reach (B). Elbow: joint pose (C), exo-assisted (D). Hand: key turn (E), joint pose (F), tips reach (G), object reposition (H), pen-twirl (I), baoding balls (J)

Caggiano, Vittorio, et al. "MyoSuite--A contact-rich simulation suite for musculoskeletal motor control." *arXiv preprint arXiv:2205.13600* (2022)

Ensuring Physical Plausibility

Joint limits are necessary—but insufficient

“Within joint limits” does not mean “natural”

For interaction: joint configuration + contact configuration + object geometry must be compatible.

Natural motion priors: Instead of hard constraints, Learn“how humans naturally move”

Hard prior
- reject impossible states.

Learned prior
- assign low probability to unnatural states.

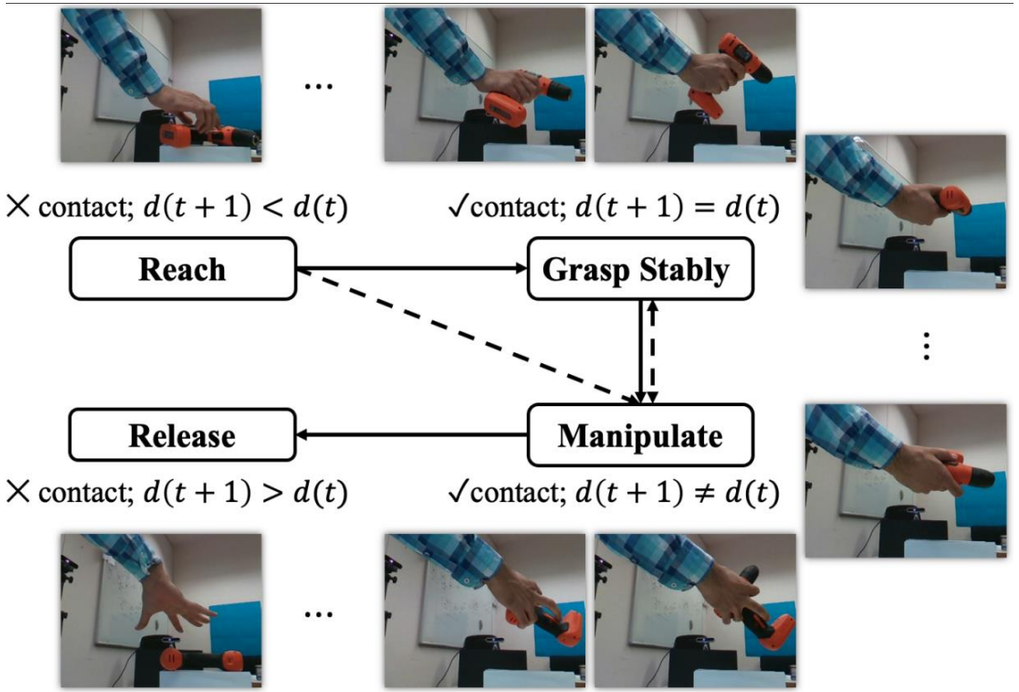
Hybrid
- learned prior + explicit physical constraints.

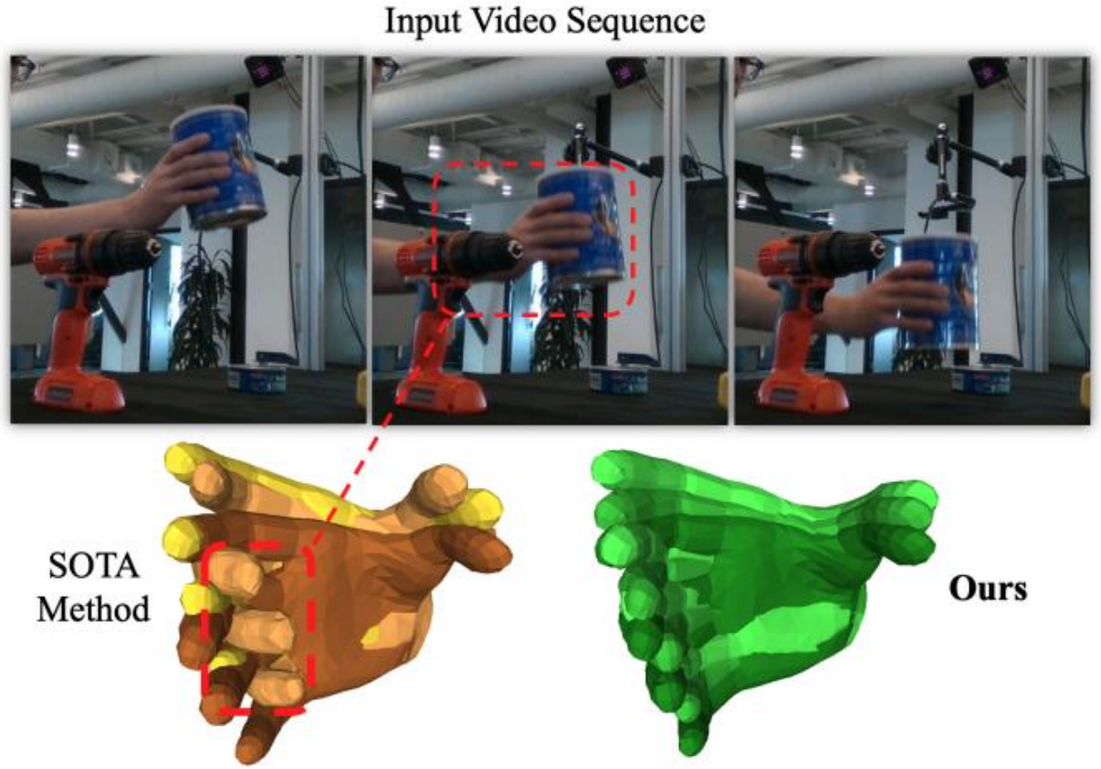
The recent literature is increasingly moving toward the third formulation.

Ensuring Physical Plausibility

Physics-aware diffusion - Can diffusion itself learn physical plausibility?

The model refines noisy hand motion through diffusion and explicitly introduces four interaction states:





Physics does not necessarily need to be a differentiable simulator inside every diffusion step. It can be encoded as structured motion knowledge

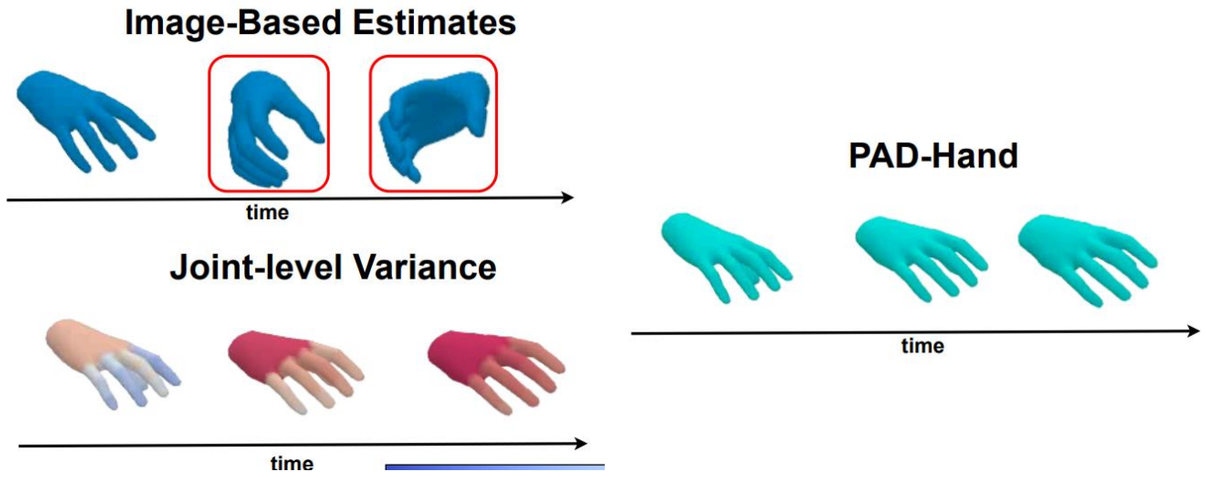
Zhang, Yufei, et al. "Diffusion-based 3d hand motion recovery with intuitive physics." *2025 IEEE/CVF International Conference on Computer Vision (ICCV)*. IEEE, 2025.

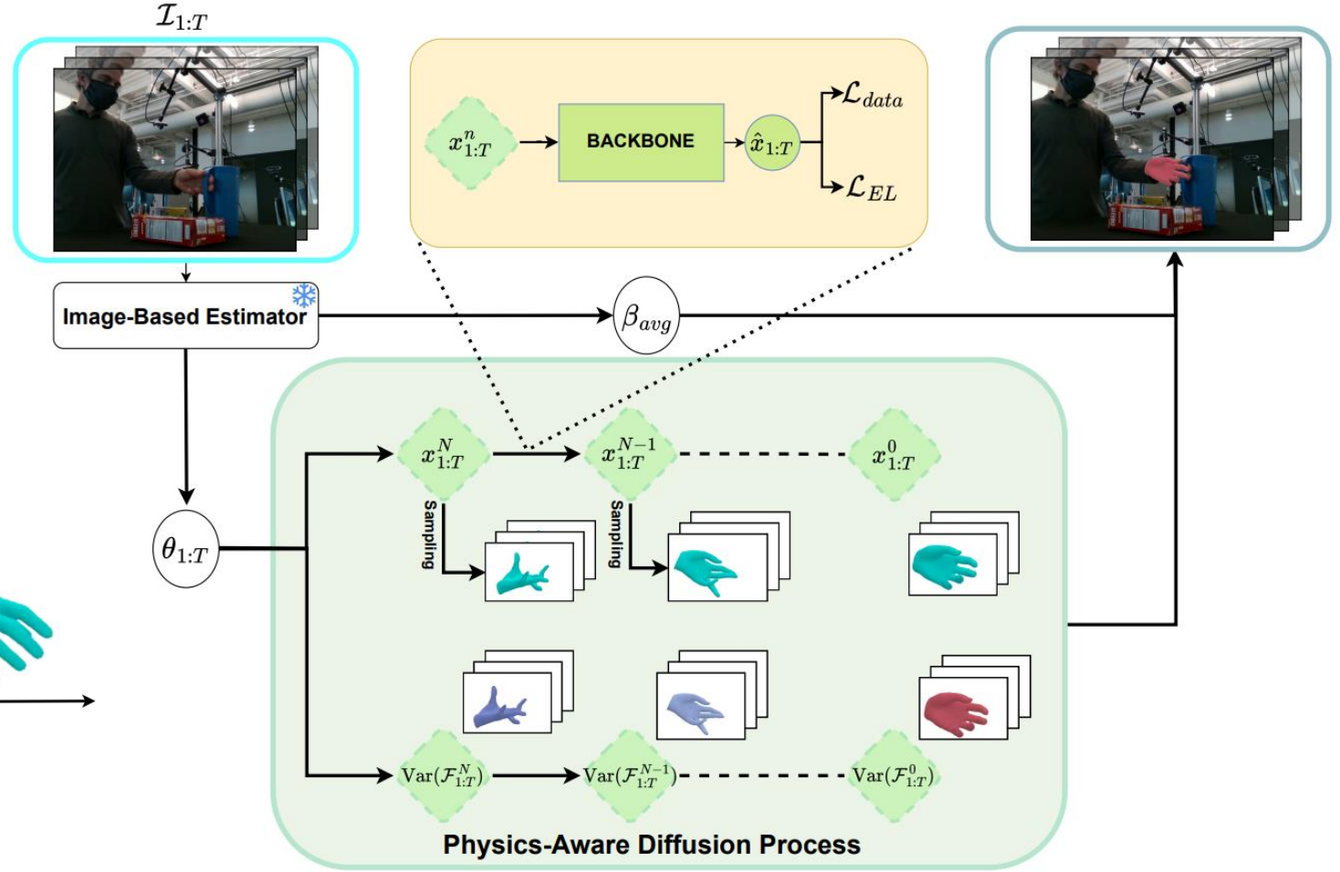
Ensuring Physical Plausibility

Physics becomes probabilistic - From “physics loss” to “physics uncertainty”

Instead of merely enforcing physical consistency, PAD-Hand:

- formulates Euler–Lagrange dynamics for articulated hands;
- treats dynamic residuals as virtual observations;
- integrates them into conditional diffusion;
- estimates **per-joint, per-time physics-consistency variance**.





Should a generative model tell us not only what motion is plausible, but where it is physically uncertain?

Ismayilzada, Elkhan, Yufei Zhang, and Zijun Cui. "PAD-Hand: Physics-Aware Diffusion for Hand Motion Recovery." *arXiv preprint arXiv:2603.26068* (2026).

Ensuring Physical Plausibility

Contact is the bridge between hand and world - Contact maps as generative conditions

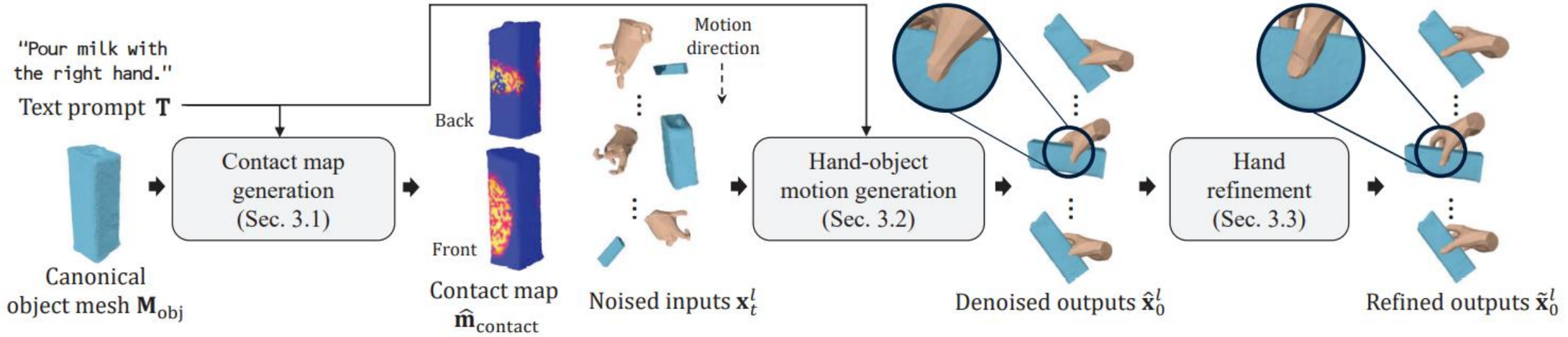


The interaction becomes a **coupled dynamical system**.

Text2HOI explicitly decomposes generation into:

1. hand-object contact generation
2. hand-object motion generation

and then uses a hand refiner to reduce penetration and improve temporal contact stability.



Contact is no longer merely something we measure after generation—it becomes a condition for generation.

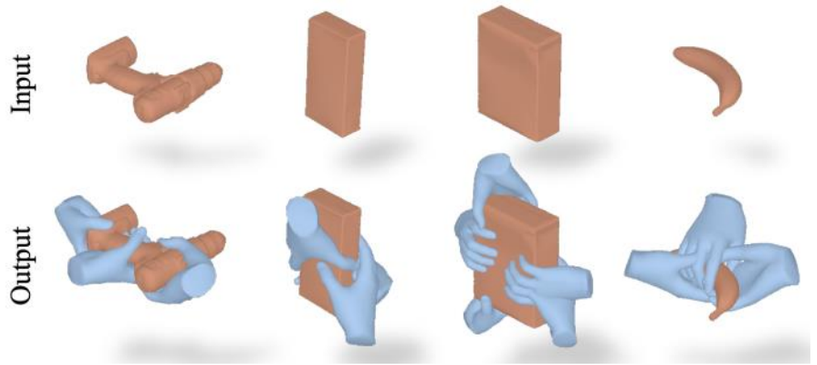
Cha, Junuk, et al. "Text2hoi: Text-guided 3d motion generation for hand-object interaction." *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2024.

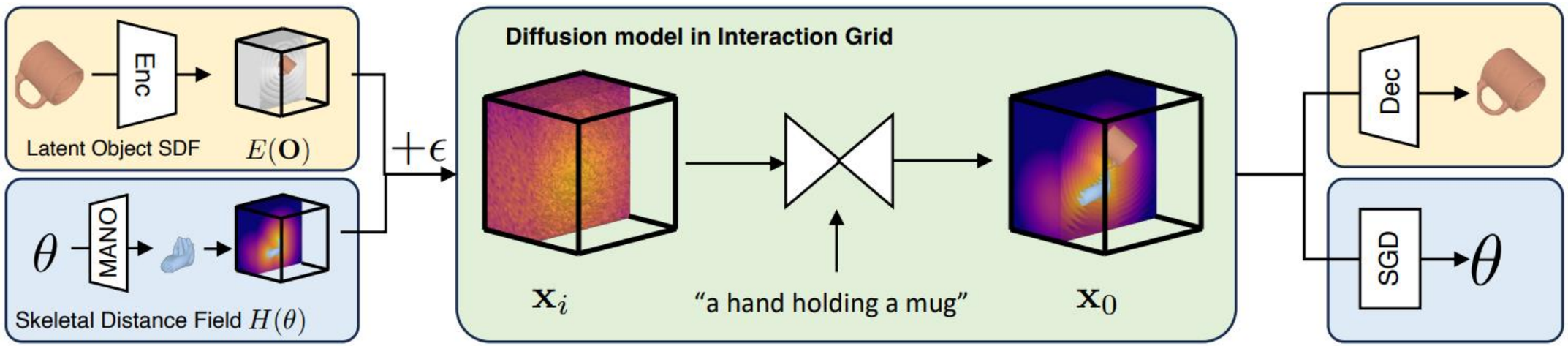
Ensuring Physical Plausibility

Joint hand-object priors - Learn the joint distribution

G-HOP learns a denoising diffusion prior over **both hand and object**.

Its representation uses a skeletal distance field for the hand and latent SDF-like representation for the object, allowing a joint spatial diffusion prior. It aggregates interaction data across **155 object categories.**





Prior, A. Hand-Object. "G-HOP: Generative Hand-Object Prior for Interaction Reconstruction and Grasp Synthesis.

Ensuring Physical Plausibility

Penetration is not the same as contact - The "contact paradox"

Goal:
contact ≈ YES
penetration ≈ NO

But naive optimization can produce:
more contact => more penetration

So we need multiple objectives
Good interaction=high contact−λpenetration

OOD Data	Model	*IV* [cm^3] ↓		*ID* [cm]↓		*CR* [%] ↑		*IVU*↓	*Phy* ↑	*SD* ↑	*OD* →
		Right	*Left*	*Right*	*Left*	*Right*	*Left*				
GRAB	Real Mocap	4.97	1.20	0.92	0.18	8.21	1.56	0.12	98.68	-	0.40
	IMoS [11]	10.38	-	1.25	-	4.61	-	0.53	84.88	0.00	0.46
	MDM [38]	9.12	2.61	1.24	0.51	8.21	1.29	0.19	89.81	0.18	0.29
	MLD [4]	9.62	3.14	1.06	0.49	10.23	0.87	0.24	85.68	**0.20**	0.31
	Ours	**6.38**	**1.66**	**0.77**	**0.29**	**11.94**	**1.11**	**0.10**	**96.16**	0.13	0.27
OAKINK	Text2HOI[2]	15.19	11.54	2.14	1.39	11.24	6.33	0.26	82.58	0.21	-
	MDM[38]	8.46	**2.47**	1.69	**0.34**	4.97	1.02	0.20	60.89	**0.23**	0.25
	MLD [4]	9.15	4.25	1.79	0.56	5.36	0.77	0.29	46.41	0.32	0.46
	Ours	**7.22**	3.11	**1.103**	0.369	**7.80**	**1.73**	**0.14**	**71.24**	0.22	0.32

LatentHOI explicitly evaluates:

- **Interpenetration Volume (IV)**
- **Interpenetration Depth (ID)**
- **Contact Ratio (CR)**
- **Interpenetration Volume per Contact Unit (IVU)**

and uses these to distinguish physically meaningful contact from simply intersecting meshes.

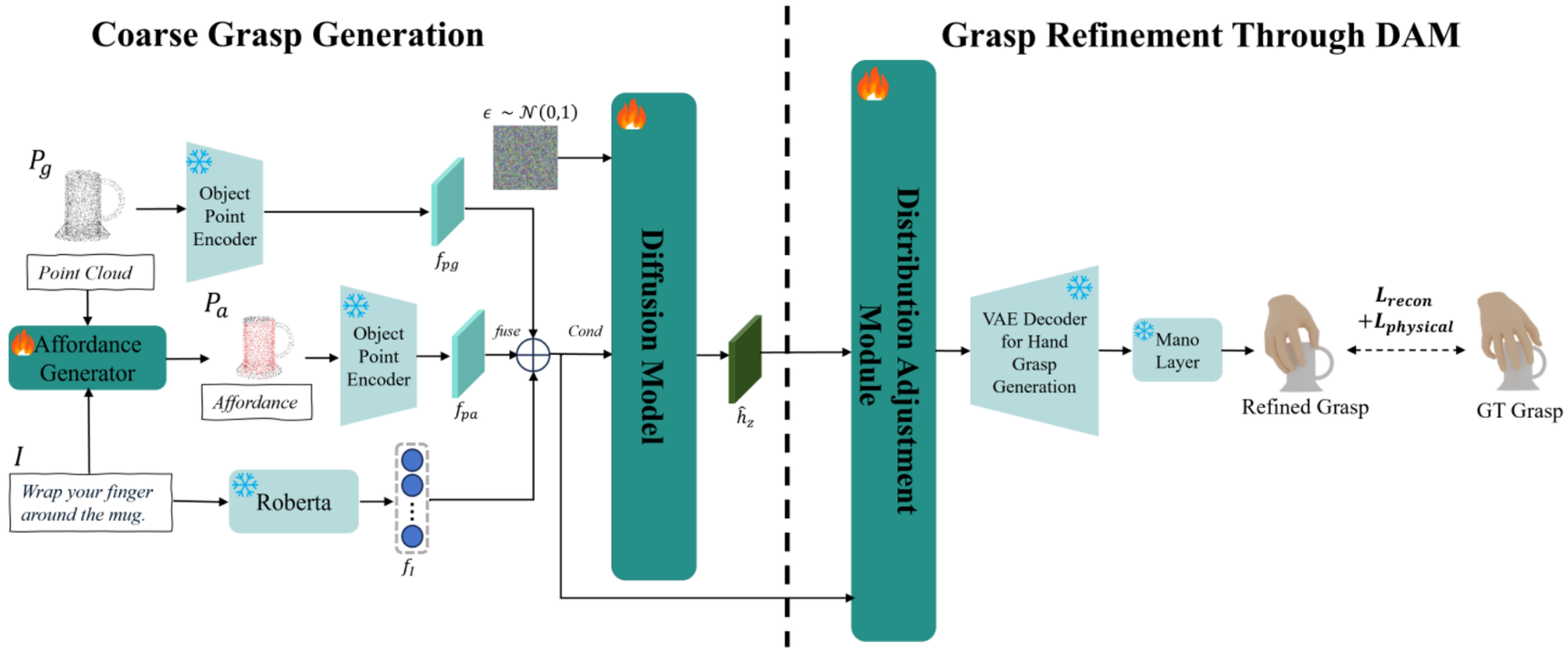
Li, Muchen, et al. "Latenthoi: On the generalizable hand object motion generation with latent hand diffusion." *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2025.

Ensuring Physical Plausibility

Affordance-aware physical plausibility - Geometry tells us where; affordance tells us why

AffordGrasp combines:

- object geometry
- spatial affordances
- instruction semantics
- latent hand representation
- diffusion



- Explicitly targets physically stable and semantically faithful grasps.
- It introduces distribution adjustment to enforce contact consistency and semantic alignment.

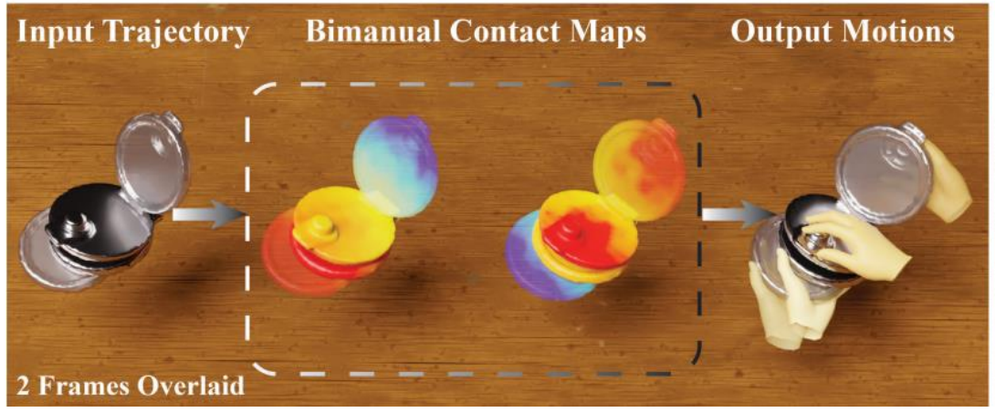
Wu, Xiaofei, et al. "AffordGrasp: Cross-Modal Diffusion for Affordance-Aware Grasp Synthesis." *arXiv preprint arXiv:2603.08021* (2026).

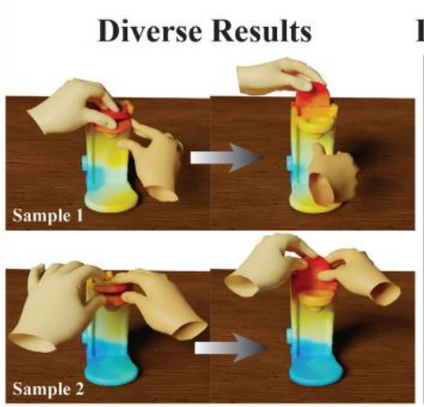
Ensuring Physical Plausibility

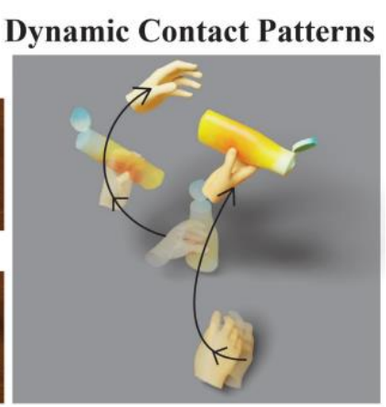
Bimanual physical plausibility - Two hands create new physical constraints

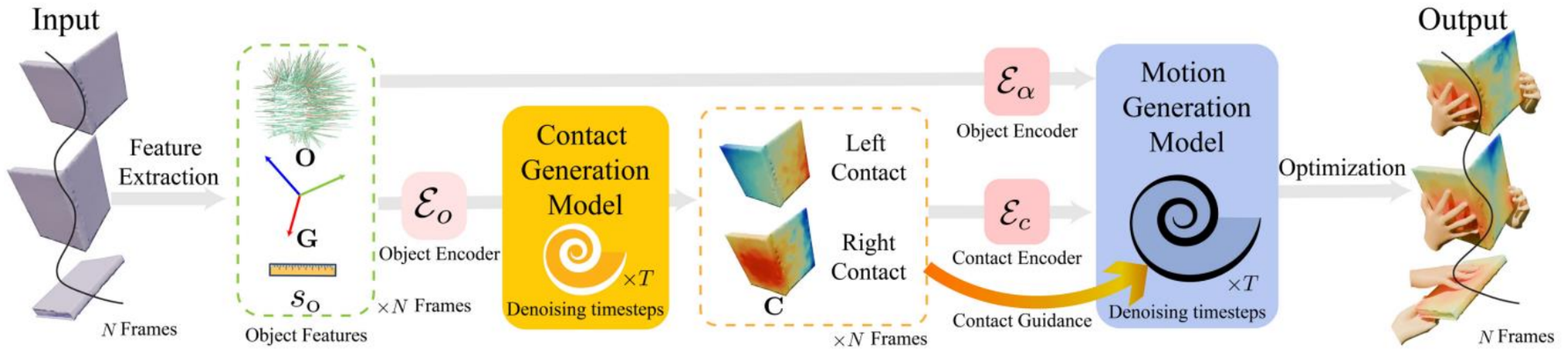
BimArt

- generates distance-based contact maps conditioned on object trajectories
- uses the learned contact prior to guide bimanual motion generation for articulated objects.









Zhang, Wanyue, et al. "Bimart: A unified approach for the synthesis of 3d bimanual interaction with articulated objects." *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2025.

Ensuring Physical Plausibility

Body–hand coordination - The hand is physically attached to the arm

EgoAllo

- jointly estimates body and hand motion from an egocentric device.
- Importantly, the authors report that body estimates provide kinematic and temporal constraints that reduce world-frame hand errors



Body motion is not just additional context—it can be a physical constraint on the hand.

Yi, Brent, et al. "Estimating body and hand motion in an ego-sensed world." *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2025.

Ensuring Physical Plausibility

The real-time problem - A physically perfect generator that takes 30 seconds is not an interactive system



Total latency:
T=Tsense +Tencode +Tgenerate +Tphysics +Trender

- Offline diffusion vs streaming generation
  - The architecture has to change for interaction
- Real-time under missing/noisy hand tracking
  - Real interaction is not clean motion capture
- Contact detection itself has a latency/accuracy problem
  - You cannot generate correct contact if you cannot detect contact

Ensuring Physical Plausibility

Where should physics enter the pipeline?

Location	Example
Data	physically filtered mocap
Representation	MANO / biomechanical model
Prior	learned motion distribution
Conditioning	contact / affordance
Training	physics losses
Sampling	physics-guided diffusion
Refinement	optimization / simulation
Inference	closed-loop correction
Evaluation	penetration/contact/task success

Evaluation and Common Pitfalls

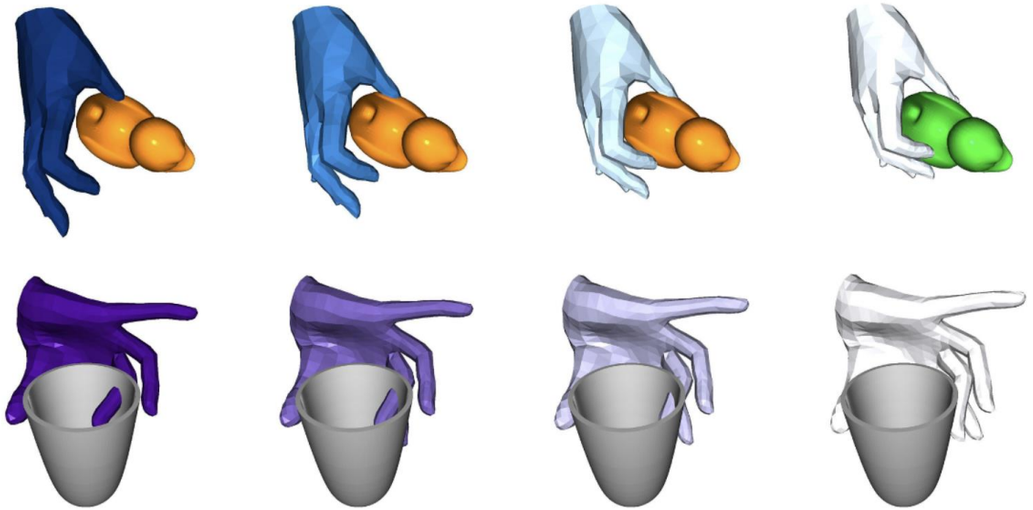
- Evaluation metrics: FID, diversity, realism, physical validity, and task success
- Common failures: floating hands, impossible grasps, jitter, lack of synchronization, and dataset bias
- Limitations of current evaluation protocols
- Open research challenges

Evaluation and Common Pitfalls

What Does "Good" Hand Motion Mean?

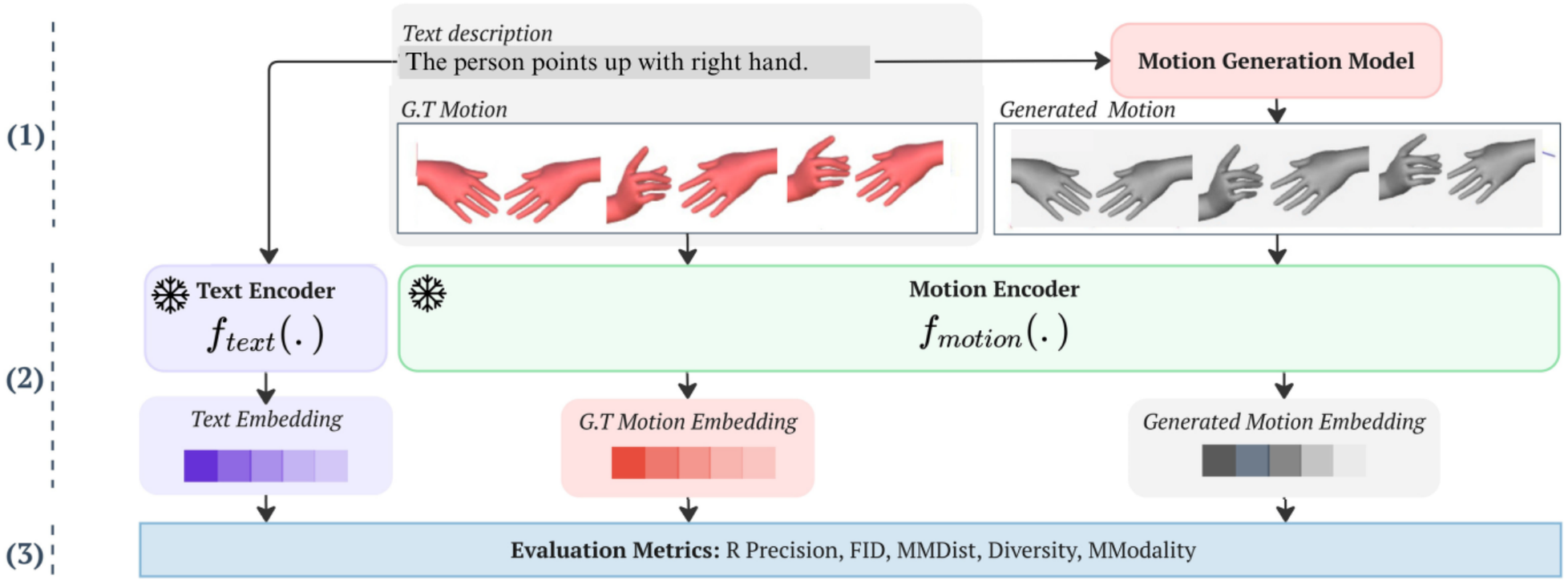
Evaluation is multi-dimensional

Question	What to measure
Does it look human?	realism / naturalness
Does it match the input?	text/audio/pose alignment
Is it diverse?	diversity / multimodality
Is it physically valid?	contact, penetration, joint limits
Does the interaction work?	grasp / manipulation success



Evaluation and Common Pitfalls

How motion data is evaluated in practice ?



- R Precision

$$R\text{-}Prec@k = \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}\left[z_i^{\text{text}} \in \text{Top-}k(z_i^{\text{motion}})\right],$$

- Fréchet Inception Distance (FID)

$$FID = \|\mu_r - \mu_g\|_2^2 + \text{Tr}(\Sigma_r + \Sigma_g - 2(\Sigma_r\Sigma_g)^{1/2}),$$

- Diversity

$$Diversity = \frac{1}{S_d}\sum_{i=1}^{S_d} \|v_i - v_i'\|_2,$$

- Multi-Modality

$$Multimodality = \frac{1}{C \times S_l}\sum_{c=1}^{C}\sum_{i=1}^{S_l} \|v_{c,i} - v_{c,i}'\|_2,$$

- Multi-Modal Distance

$$MM\text{-}Dist = \frac{1}{N}\sum_{i=1}^{N} \|z_i^{\text{motion}} - z_i^{\text{text}}\|_2,$$

Evaluation and Common Pitfalls

Dimension	Example metric
Motion distribution	FID / FGD
Text alignment	R-precision / matching
Pose accuracy	MPJPE
Diversity	Diversity / multimodality
Temporal quality	jerk / acceleration
Contact	contact precision/recall/F1
Physical plausibility	penetration/contact feasibility
Task success	grasp/manipulation success
Human perception	MOS / pairwise preference
Interaction	responsiveness / appropriateness

Evaluation and Common Pitfalls

Distributional Metrics: FID Is Not Enough

What it measures

- Does the distribution of generated motion resemble real motion?
- “does it look like the dataset?” — not “does it do the right thing?”

What it does NOT tell you

❌ Did the gesture match the text?
❌ Did it synchronize with speech?
❌ Did the hand actually touch the object?
❌ Was the grasp successful?
❌ Was the generated motion physically valid?

Diversity & Multimodality

What it measures

- **Diversity:** Different samples across different inputs
- **Multimodality:** Different valid motions for the same input

Why this matters for hands
A phrase such as: **“wave hello”**
doesn't specify:

- amplitude
- wrist orientation
- speed
- number of waves
- hand trajectory
- exact finger configuration

There are many valid solutions.

Evaluation and Common Pitfalls

Physical Validity: The Metrics That FID Cannot See

Metric	What it detects	Desired
Interpenetration volume	hand inside object	↓
Penetration depth	severity of collision	↓
Contact ratio	maintaining required contact	↑
Foot/hand skating	unrealistic sliding	↓

Contact ratio

For example:

$$CR = \frac{\#frames\ with\ valid\ contact}{\#interaction\ frames}$$

But there is a subtle problem:

High contact ratio alone does not imply a good grasp.

You can increase contact simply by pushing the hand *through* the object.

Therefore:

Good contact=high contact+low penetration

Evaluation and Common Pitfalls

Task Success: Does the Generated Hand Actually Work?

From:

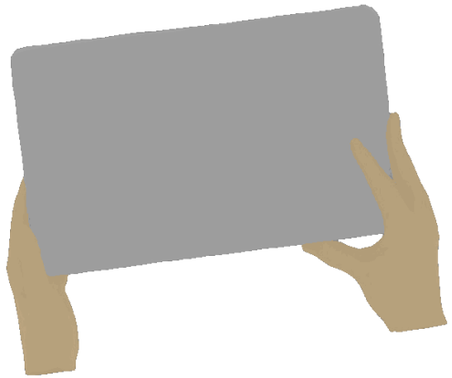
"Does the generated hand look realistic?"

to:

"Can the generated hand accomplish the intended interaction?"

Prompt (Grab dataset) : inspect stanford bunny with both hands.







Grasping
- object lifted
- object remains stable
- no excessive penetration
- desired contact maintained

Opening
- correct handle contact
- correct motion direction
- object articulation achieved

Pouring
- correct object orientation
- correct trajectory
- task endpoint reached

Handover
- donor maintains grasp
- receiver establishes grasp
- object remains stable during transfer

Evaluation and Common Pitfalls

Common Failures: What Metrics Miss

Floating hands



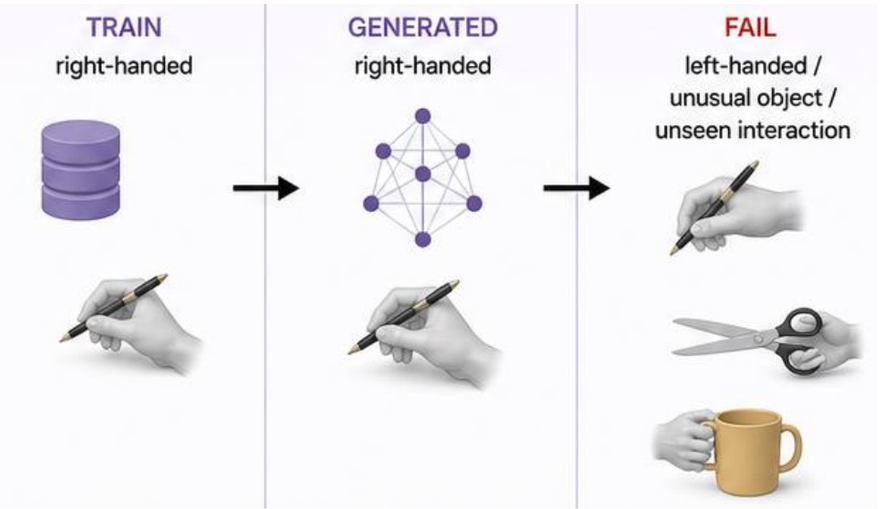
Hand visually approaches object but never contacts it.

Impossible grasps

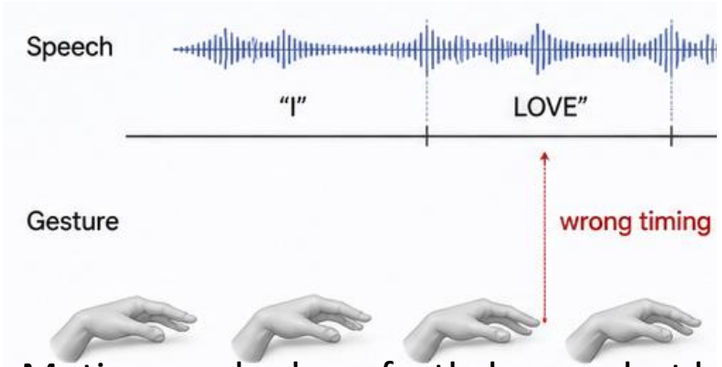


Fingers pass through the object or approach from impossible configurations.

Dataset bias

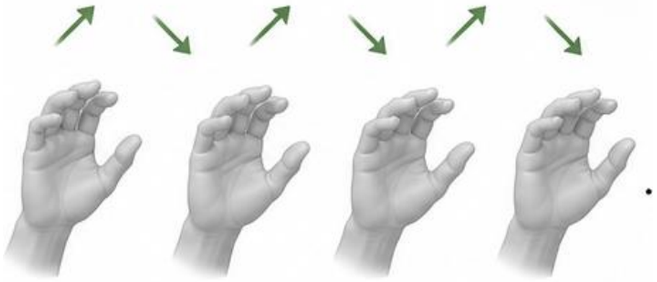


Desynchronization



Motion can look perfectly human but be inappropriate for the speech.

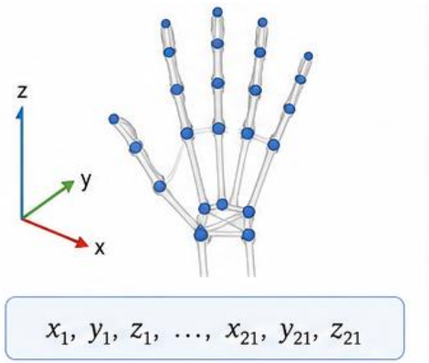
Jitter

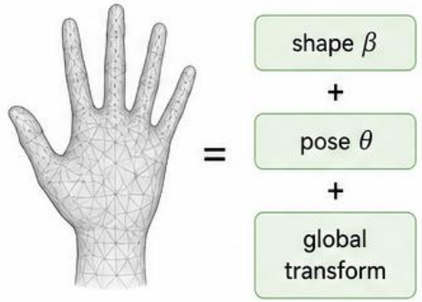


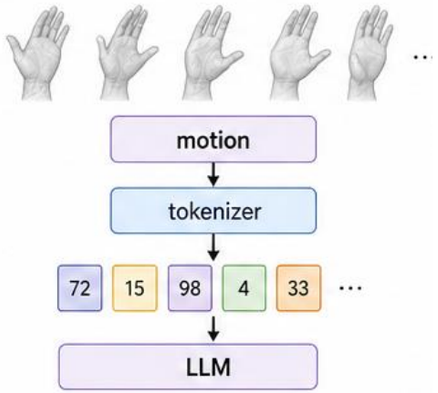
High-frequency oscillations that may not strongly affect distributional metrics.

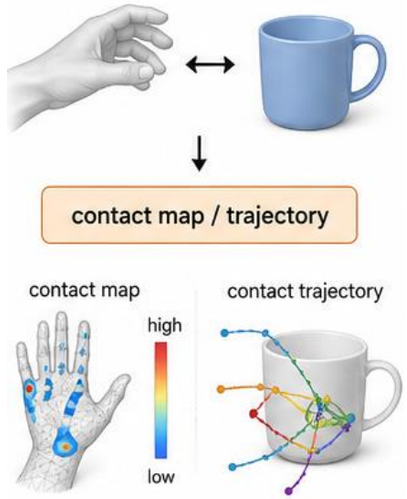
Open research challenges

What should a hand-motion representation contain?
Representation is still an open problem









Joint coordinates
Pros:
- simple
- directly interpretable

Cons:
- rotational ambiguities
- anatomical constraints
- scale/translation issues

MANO
Pros:
- anatomically structured

Cons:
- representation bottleneck
- fitting errors

Motion tokens
Pros:
- language-model compatible
- long sequences

Cons:
- quantization error
- fine finger details

Contact representations
Pros:
- interaction semantics

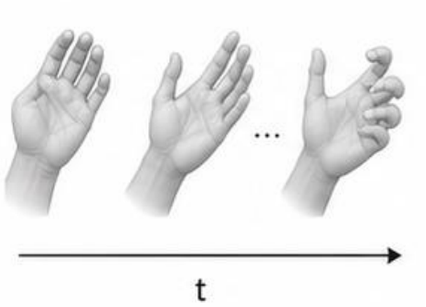
Cons:
- still difficult to model temporally

Open research challenges

Evaluation: the metric zoo

Automatic metrics are not enough !

Motion realism

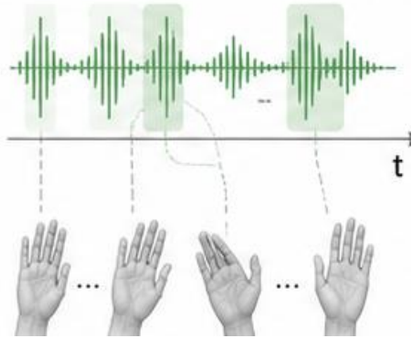


Hand Motion Evaluation

Diversity & Generalization

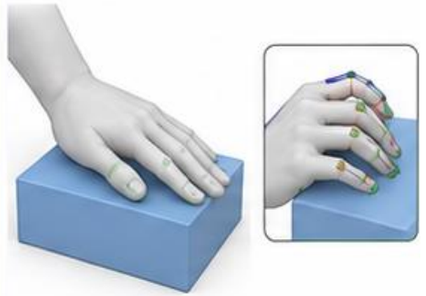


Semantic alignment: speech & text

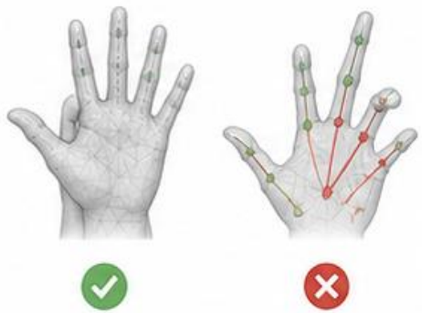


Physical Plausibility

Interaction



Anatomy

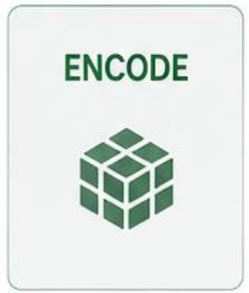


Open research challenges

Generation for interaction has a latency budget



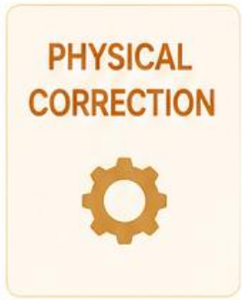
- Text
- Speech
- Video
- Audio
- …



- Text Encoder
- Speech Encoder
- Video Encoder
- Audio Encoder
- …



- Diffusion
- Autoregressive
- LLM
- …



- Physics
- IK
- Optimization
- Post-processing
- ….



- MANO
- …



- Act
- Interact
- …

quality ↑ sampling steps ↑ latency ↑

This motivates:

- latent diffusion
- distillation
- flow matching
- autoregressive token generation
- streaming generation

Open research challenges

What remains unsolved?

1. Semantic grounding
Can the model understand:
"Point to the object, then pinch it gently"?
rather than just generate a statistically similar gesture?

2. Contact
Correct *where*, *when* and *how* the hand touches.

3. Dexterity
Fine finger articulation remains much harder than wrist motion.

4. Bimanual coordination
Two hands have complementary—not independent—roles.

5. Body–hand coordination
Hands must make sense in the context of the whole body.

6. Object generalization
Generate interactions with unseen objects.

7. In-the-wild robustness
Noisy, occluded, diverse real-world motion.

8. Long-horizon generation
No drift, jitter or semantic degradation.

9. Real-time interaction
Generate and adapt continuously.

10. Evaluation
We still lack a universally accepted answer to:
What makes a generated hand motion good?

Conclusions

- Implementing effective, non trivial and reliable gesture-based interfaces is not as easy as it appears
  - Continuous task
  - Heterogeneous gestures with different time and spatial scales
  - Tracking failures
  - Non optimally designed dictionaries
- There's still need of
  - Better algorithms
  - New benchmarks
  - Dictionary design guidelines/tools

Reasonable ideas not investigated

- While many research works still performs small changes in networks and test results on old benchmarks (segmented), many simple research ideas seem not explored
  - Different kinds of gesture (static, dynamic coarse/fine) may require different recognizers, working at different time scales. Why not separate them?
  - Augmentation of nongestures covering uniformily trajectory and hand pose spaces may be feasible using physical constraints or data driven statistical models
  - Visual feedback may help the HMD user to perform the gestures in more recognisable ways
  - Simultaneously optimizing dictionary and recognizers
  - We are open for collaborations on these topics!

DISCUSSION

Discussion

- Lessons learned from recognition and generation pipelines
- Open problems in gesture-based methods for multimodal interaction
- Toward robust, adaptive, and user-centered gesture interfaces
- Questions and discussion with participants

Questions

- Multimodality with gestures: potential enhancements, disambiguation and other error sources

Questions

- Should users adapt to gesture interfaces, or should interfaces adapt to users?

Questions

- Should a system provide feedback that guides users toward more recognizable gestures?